# CTU: Mechanics Lectures Compilation

Iskender Yalcinkaya, PhD

October 21, 2026

Unrevised WIP document right now. More will be updated as our class materials gets updated.

# Contents

# Chapter 1

# Mathematical Tools needed for this lecture

Information about this chapter blah blah blah

## 1.1 Scalars

*Scalars* are quantities that are fully described by a magnitude (or numerical value) alone.

Examples of scalars in physics include mass, temperature, and energy:

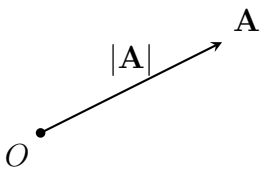
- 1 kilogram (mass)
- 20 degrees Celcius (temperature)
- 100 Joules (energgy)

Scalars can be *added, substracted, multiplied* and *divided* using the standard rules of arithmetic.

Scalars can only be added or substracted when they share the same physical dimension. For example, $1kg + 2kg = 3kg$ is valid but, $1kg + 20C$ is meaningless. However, scalars of different dimensions can be multiplied or divided, yielding a new unit.

## 1.2 Vectors

*Vectors* are quantities that have both a magnitude and a direction. They can be represented graphically as arrows, where the length of the arrow represents

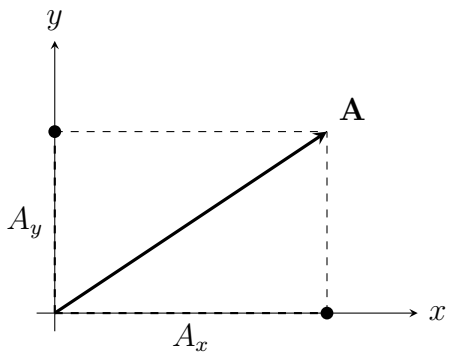



the magnitude and the direction of the arrow indicates the direction of the vector.

We will represent vectors using boldface letters in these notes, such as $\mathbf{A}$, or with an arrow above the letter on the blackboard, such as $\vec{A}$. The magnitude of a vector $\mathbf{A}$ is denoted by $|\mathbf{A}|$ or $A$.

### 1.2.1 Components of Vectors

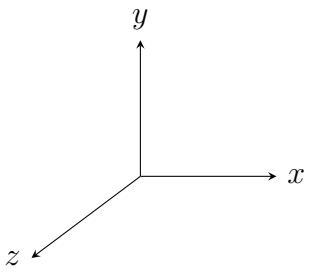
A vector can be broken down into its components along the coordinate axes. For example, a vector $\mathbf{A}$ in two-dimensional space can be expressed in terms of its components along the $x$-axis and $y$-axis by two numbers $A_x$ and $A_y$ as it can be seen in the figure below.




It is conventional to express a vector in terms of a column vector of its components along the coordinate axes. For example, a vector $\mathbf{A}$ in two-dimensional space can be expressed as:

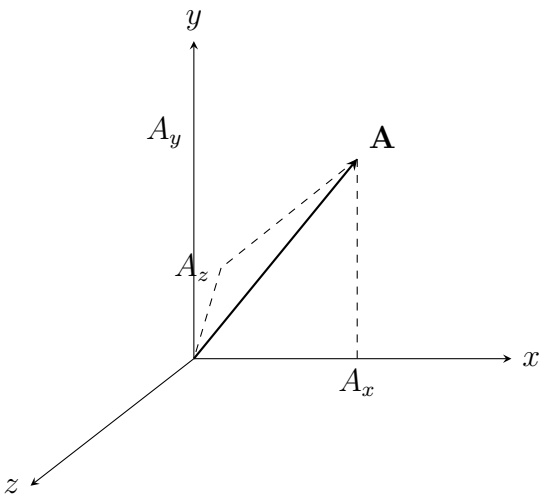
$$\mathbf{A} = \begin{bmatrix} A_x \\ A_y \end{bmatrix}.$$

The real world is three-dimensional, and when it comes to draw vectors in three-dimensional space on a paper, we need to use a three coordinate system. The way we draw a three-dimensional coordinate system on a paper is shown in the figure below:




One can think of these axes as the edges of a cube. The $x$-axis points to the right, the $y$-axis points up, and the $z$-axis points out of the page towards you. The positive direction of each axis is indicated by an arrowhead. A vector is in three-dimensional space then can be expressed as:

$$\mathbf{A} = \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}.$$




The magnitude of a vector $\mathbf{A}$ in three-dimensional space can therefore be calculated using the Pythagorean theorem:

$$|\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}.$$

### 1.2.2 Unit Vectors

A *unit vector* is a vector with a magnitude of 1. It is used to indicate direction without specifying magnitude. A unit vector can be obtained by dividing a

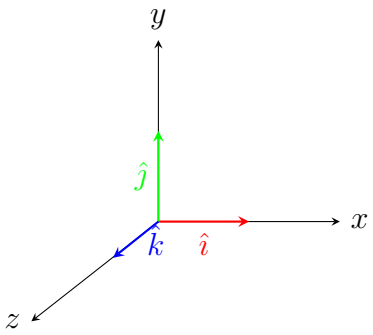
vector by its magnitude:

$$\mathbf{a} = \frac{\mathbf{A}}{|\mathbf{A}|} = \begin{bmatrix} \frac{A_x}{|\mathbf{A}|} \\ \frac{A_y}{|\mathbf{A}|} \\ \frac{A_z}{|\mathbf{A}|} \end{bmatrix}.$$

We can define *special unit vectors* along the coordinate axes. In three-dimensional space, we define the unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ as follows:

$$\mathbf{i} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{j} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

On the paper, we use hat notation for unit vectors, such as $\hat{\imath}$, $\hat{\jmath}$, and $\hat{k}$. The unit vectors are shown in the figure below:




Once we define the unit vectors, we can express any vector in three-dimensional space as a linear combination of the unit vectors:

$$\mathbf{A} = A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}.$$

## 1.3 Operations on Vectors

Like scalars, vectors can be added, substracted, multiplied, and divided.

### 1.3.1 Addition and Substraction

### 1.3.2 Scalar Multiplication

*Scalar multiplication* only changes the magnitude of a vector, not its direction. If the scalar is negative, it also reverses the direction of the vector:

$$c\mathbf{A} = cA_x\mathbf{i} + cA_y\mathbf{j} + cA_z\mathbf{k}.$$

### 1.3.3 Dot Product

The *dot product* of two vectors is defined as the product of their magnitudes and the cosine of the angle between them:

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos\theta = A_x B_x + A_y B_y + A_z B_z$$

The dot product is useful for finding the component of one vector along another and for calculating angles between vectors.

### 1.3.4 Cross Product

The *cross product* of two vectors produces a vector perpendicular to both:

$$\mathbf{A} \times \mathbf{B} = |\mathbf{A}||\mathbf{B}| \sin\theta\, \hat{\mathbf{n}},$$

where $\hat{\mathbf{n}}$ is determined by the right-hand rule. In three dimensions,

$$\begin{aligned}
\mathbf{A} \times \mathbf{B} &= (A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}) \times (B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k}) \\
&= A_x B_x(\mathbf{i} \times \mathbf{i}) + A_x B_y(\mathbf{i} \times \mathbf{j}) + A_x B_z(\mathbf{i} \times \mathbf{k}) \\
&\quad + A_y B_x(\mathbf{j} \times \mathbf{i}) + A_y B_y(\mathbf{j} \times \mathbf{j}) + A_y B_z(\mathbf{j} \times \mathbf{k}) \\
&\quad + A_z B_x(\mathbf{k} \times \mathbf{i}) + A_z B_y(\mathbf{k} \times \mathbf{j}) + A_z B_z(\mathbf{k} \times \mathbf{k}) \\
&= (A_y B_z - A_z B_y)\mathbf{i} + (A_z B_x - A_x B_z)\mathbf{j} + (A_x B_y - A_y B_x)\mathbf{k}.
\end{aligned}$$

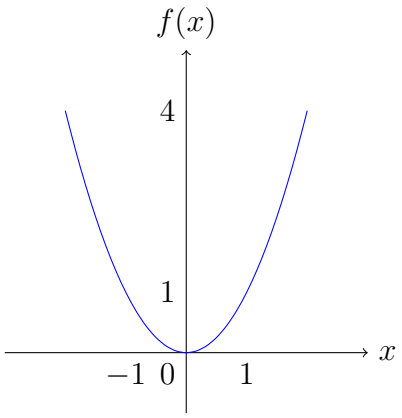
## 1.4 Basic Calculus

### 1.4.1 Function

A *function* is a rule that assigns a unique output to each input. For example, the function $f(x) = x^2$ assigns the square of $x$ to each value of $x$. The value $x$ is called the *parameter* of function $f$ and the value $f(x)$ is called the *output* of the function for that parameter. It is conventional to draw a figure to represent a function to have a better understanding of its behavior, where the horizontal axis represents the input values and the vertical axis represents the output values. For example, the function $f(x) = x^2$ can be represented as follows:

### 1.4.2 Derivative

For a function $f(x)$, its *derivative* describes the rate at which the output of the function changes with respect to its input. It is defined as the ratio of




the change in the output to the change in the input as the change in the input approaches zero. This is depicted in the following figure:

Figure 1.1: Geometrical interpretation of the derivative.

The derivative of a function $f(x)$ with respect to $x$ is denoted by $\frac{df}{dx}$ or $f'(x)$. It is defined as the limit of the average rate of change of the function as the interval approaches zero:

$$\frac{df}{dx} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

Geometrically, the derivative $f'(x)$ gives the slope of the tangent to the graph of $f(x)$ at a given point.

### 1.4.3 Different Notations for Derivatives

There are several common ways to denote the derivative of a function:

- $\frac{df}{dx}$: This is the *Leibniz notation*. It explicitly indicates that the derivative is taken with respect to $x$.
- $f'(x)$: This is the *Lagrange notation*, also called *prime notation*. It is a concise notation commonly used when working with functions.
- $\dot{f}$: This is the *Newton notation*, also called *dot notation*. It denotes differentiation with respect to time and is particularly common in physics and mechanics. For example,

$$\dot{f} = \frac{df}{dt}.$$

### 1.4.4 Derivatives of Common Functions

The derivatives of some common functions are as follows:

$$\begin{aligned}
\frac{d}{dx}(c) &= 0, \\
\frac{d}{dx}(x^n) &= nx^{n-1}, \\
\frac{d}{dx}(e^x) &= e^x, \\
\frac{d}{dx}(\ln x) &= \frac{1}{x}, \\
\frac{d}{dx}(\sin x) &= \cos x, \\
\frac{d}{dx}(\cos x) &= -\sin x, \\
\frac{d}{dx}(\tan x) &= \sec^2 x, \\
\frac{d}{dx}(\arcsin x) &= \frac{1}{\sqrt{1-x^2}}, \\
\frac{d}{dx}(\arccos x) &= -\frac{1}{\sqrt{1-x^2}}, \\
\frac{d}{dx}(\arctan x) &= \frac{1}{1+x^2}.
\end{aligned}$$

Some useful rules for differentiating functions $f(x)$ and $g(x)$ are

$$\begin{aligned}
\frac{d}{dx}[f(x)+g(x)] &= f'(x)+g'(x), \\
\frac{d}{dx}[cf(x)] &= cf'(x), \\
\frac{d}{dx}[f(x)g(x)] &= f'(x)g(x)+f(x)g'(x), \\
\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] &= \frac{f'(x)g(x)-f(x)g'(x)}{g(x)^2}.
\end{aligned}$$

For a composite function $f(g(x))$, the chain rule gives

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x).$$

### 1.4.5 Multivariable Functions and Partial Derivatives

$f = f(x, y, z, \ldots)$ is a *multivariable function* if it depends on more than one variable. For example, the surface $f(x, y) = x^2 + y^2$ is shown below.

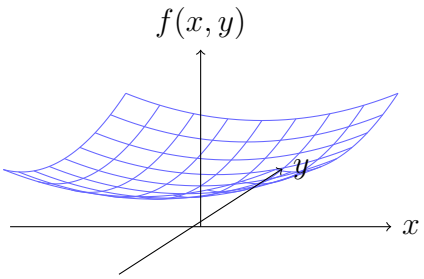



Figure 1.2: The surface $f(x, y) = x^2 + y^2$.

The *partial derivative* of a multivariable function with respect to one of its variables is the derivative of the function while treating all other variables as constants. For example, the partial derivative of $f(x, y)$ with respect to $x$ is denoted by $\frac{\partial f}{\partial x}$ and is defined as

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}.$$

For example, for the function $f(x, y) = x^2 + y^2$, the partial derivatives are

$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 2y.$$

For example, the partial derivative of $g(x, y) = x^2 + xy^2$ with respect to $x$ is

$$\frac{\partial g}{\partial x} = 2x + y^2,$$

and the partial derivative with respect to $y$ is

$$\frac{\partial g}{\partial y} = 2xy.$$

### 1.4.6 Differential and Total Differentials

The smallest change in the value of a function when we change all of its input variables is called the *differential* of the function. For a function $f = f(x, y)$, the differential $df$ is given by

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy,$$

where $dx$ and $dy$ are the small changes in the input variables $x$ and $y$, respectively. The differential provides an approximation of the change in the function's value based on the changes in its input variables. In general, for a function $f = f(x_1, x_2, \ldots, x_n)$, the differential is given by

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \cdots + \frac{\partial f}{\partial x_n} dx_n.$$

In a compact form, the differential can be expressed as

$$df = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} dx_i.$$

### 1.4.7 Total derivative

The *total derivative* of a function $f = f(x_1, x_2, \ldots, x_n, t)$ and $x_i(t) = x_i$ where $i = 1, 2, \ldots, n$ is given by

$$\frac{df}{dt} = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} \frac{dx_i}{dt} + \frac{\partial f}{\partial t}.$$

### 1.4.8 Maxima and Minima of Functions

If we have a function $f(x)$, we can find its maxima and minima by taking its derivative and setting it equal to zero:

$$f'(x) = 0.$$

We solve this equation for $x$.

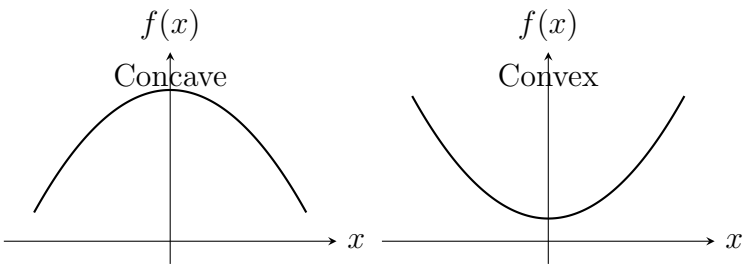
### 1.4.9 Concave and Convex Functions

A function $f(x)$ is *concave* if its second derivative is negative for all $x$ in its domain:

$$f''(x) < 0.$$

A function $f(x)$ is *convex* if its second derivative is positive for all $x$ in its domain:

$$f''(x) > 0.$$




### 1.4.10 Antiderivatives

The *antiderivative* of a function $f(x)$ is a function $F(x)$ whose derivative is $f(x)$:

$$F'(x) = f(x).$$

For example, since

$$\frac{d}{dx}(x^3) = 3x^2,$$

$x^3$ is an antiderivative of $3x^2$.

However, $x^3+C$ is also an antiderivative, where $C$ is an arbitrary constant, because

$$\frac{d}{dx}(x^3 + C) = 3x^2.$$

Therefore, an antiderivative is not unique: any two antiderivatives of the same function differ by a constant.

### 1.4.11 Indefinite Integrals

The *indefinite integral* represents the family of all antiderivatives of a function:

$$\int f(x)\,dx = F(x) + C,$$

where

$$F'(x) = f(x).$$

For example,

$$\int 3x^2\,dx = x^3 + C.$$

Some common indefinite integrals are

$$\begin{aligned}
\int c\,dx &= cx + C, \\
\int x^n\,dx &= \frac{x^{n+1}}{n+1} + C, \qquad n \neq -1, \\
\int e^x\,dx &= e^x + C, \\
\int \frac{1}{x}\,dx &= \ln|x| + C, \\
\int \cos x\,dx &= \sin x + C, \\
\int \sin x\,dx &= -\cos x + C.
\end{aligned}$$

The constant $C$ appears because the derivative of a constant is zero.

### 1.4.12 Definite Integrals

A *definite integral* has specified lower and upper limits:

$$\int_a^b f(x)\,dx.$$

Unlike an indefinite integral, a definite integral has a numerical value. It represents the signed area between the graph of $f(x)$ and the $x$-axis from $x = a$ to $x = b$. Regions where $f(x) < 0$ contribute negatively to the integral.

If $F(x)$ is an antiderivative of $f(x)$, then the Fundamental Theorem of Calculus gives

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

For example,

$$\int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3}.$$

Thus, differentiation and integration are inverse operations. Differentiation describes the instantaneous rate of change of a function, while integration can be used to accumulate a quantity or calculate the signed area under a curve.

## 1.5 Taylor Series

The *Taylor series* of a function $f(x)$ about a point $x_0$ is an infinite sum of terms that approximates the function near that point. The reason we use Taylor series is that it allows us to approximate complicated functions with polynomials, which are easier to work with.

**Theorem:** An *analytic function* can be written as a convergent power series. An analytic function is a function that can be infinitely differentiable. The examples of analytic functions include polynomials, exponential functions, and trigonometric functions. Therefore, if $f(x)$ is analytic about $x_0$, then

$$f(x) = c_0 + c_1(x - x_0) + c_2(x - x_0)^2 + c_3(x - x_0)^3 + \cdots$$

To determine the coefficients $c_n$, let us consider the derivatives of $f(x)$:

$$\begin{aligned}
f'(x) &= c_1 + 2c_2(x - x_0) + 3c_3(x - x_0)^2 + 4c_4(x - x_0)^3 + \cdots \\
f''(x) &= 2c_2 + 3 \cdot 2c_3(x - x_0) + 4 \cdot 3c_4(x - x_0)^2 + \cdots \\
f'''(x) &= 3 \cdot 2c_3 + 4 \cdot 3 \cdot 2c_4(x - x_0) + \cdots \\
f^{(4)}(x) &= 4 \cdot 3 \cdot 2c_4 + \cdots
\end{aligned}$$

Therefore, by evaluating these derivatives at $x = x_0$, we can find the coefficients $c_n$:

$$\begin{aligned}
f(x_0) = c_0 &\implies c_0 = f(x_0) \\
f'(x_0) = c_1 &\implies c_1 = f'(x_0) \\
f''(x_0) = 2c_2 &\implies c_2 = \frac{f''(x_0)}{2} \\
f'''(x_0) = 3 \cdot 2c_3 &\implies c_3 = \frac{f'''(x_0)}{3 \cdot 2} \\
f^{(4)}(x_0) = 4 \cdot 3 \cdot 2c_4 &\implies c_4 = \frac{f^{(4)}(x_0)}{4 \cdot 3 \cdot 2}
\end{aligned}$$

$$f(x) = c_0 + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \frac{f'''(x_0)}{3 \cdot 2}(x - x_0)^3 + \frac{f^{(4)}(x_0)}{4 \cdot 3 \cdot 2}(x - x_0)^4 + \cdots$$

This is called the Taylor series of $f(x)$ about $x_0$. The Taylor series can be expressed in a more compact form as follows:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

## 1.6 Worked Examples

### 1.6.1 Example 1: Scalars, Units, and the Anatomy of a Vector

**(a)** A student writes $m = 2\,\text{kg} + 20\,^\circ\text{C}$. Why is this meaningless? Then, for a metal block of mass $m = 2.7\,\text{kg}$ and volume $V = 1.0 \times 10^{-3}\,\text{m}^3$, compute the density.

**(b)** Show by dimensional analysis that $T = 2\pi\sqrt{L/g}$ can be a time.

**(c)** For $\mathbf{A} = 2\hat{\imath} + 3\hat{\jmath} + 6\hat{k}$, find $|\mathbf{A}|$, the unit vector $\hat{a}$, and the angle $\alpha$ between $\mathbf{A}$ and the $x$-axis.

**Solution**

**(a)** Addition requires the same physical dimension; mass and temperature have different dimensions, so the sum has no meaning. Multiplication and division are always allowed and create a new unit:

$$\rho = \frac{m}{V} = \frac{2.7\,\text{kg}}{1.0 \times 10^{-3}\,\text{m}^3} = 2700\ \text{kg/m}^3$$

**(b)**

$$\left[\sqrt{L/g}\right] = \sqrt{\frac{\text{m}}{\text{m/s}^2}} = \sqrt{\text{s}^2} = \text{s},$$

and $2\pi$ is a dimensionless scalar.

**(c)**

$$|\mathbf{A}| = \sqrt{2^2 + 3^2 + 6^2} = \sqrt{49} = 7,$$

$$\hat{a} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{2}{7}\hat{\imath} + \frac{3}{7}\hat{\jmath} + \frac{6}{7}\hat{k}, \qquad |\hat{a}| = \frac{1}{7}\sqrt{4 + 9 + 36} = 1$$

The direction cosine with the $x$-axis is

$$\cos\alpha = \frac{A_x}{|\mathbf{A}|} = \frac{2}{7} \quad \Longrightarrow \quad \alpha \approx 73.4^\circ$$

Note also that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{4+9+36}{49} = 1$.

*Board tip: draw the three-dimensional axes and the projections $A_x$, $A_y$, $A_z$ while doing part (c).*

### 1.6.2 Example 2: Vector Algebra and Force Equilibrium

Let $\mathbf{A} = 3\hat{\imath} - \hat{\jmath} + 2\hat{k}$ and $\mathbf{B} = -\hat{\imath} + 4\hat{\jmath} + 2\hat{k}$.

**(a)** Find $\mathbf{A} + \mathbf{B}$, $\mathbf{A} - \mathbf{B}$, $|\mathbf{A} + \mathbf{B}|$ and $2\mathbf{A} - 3\mathbf{B}$.

**(b)** A hiker walks 4 km east, then 3 km north. Find the displacement (magnitude and direction) and compare it with the distance travelled, which is a scalar.

**(c)** Two forces act on a particle: $\mathbf{F}_1 = (5\hat{\imath} + 2\hat{\jmath})$ N and $\mathbf{F}_2 = (-3\hat{\imath} + 6\hat{\jmath})$ N. What third force $\mathbf{F}_3$ keeps the particle in equilibrium?

**Solution**

**(a)**

$$\mathbf{A} + \mathbf{B} = 2\hat{\imath} + 3\hat{\jmath} + 4\hat{k}, \qquad |\mathbf{A} + \mathbf{B}| = \sqrt{4 + 9 + 16} = \sqrt{29} \approx 5.39,$$

$$\mathbf{A} - \mathbf{B} = 4\hat{\imath} - 5\hat{\jmath} + 0\hat{k},$$

$$2\mathbf{A} = 6\hat{\imath} - 2\hat{\jmath} + 4\hat{k}, \qquad 3\mathbf{B} = -3\hat{\imath} + 12\hat{\jmath} + 6\hat{k},$$

$$2\mathbf{A} - 3\mathbf{B} = 9\hat{\imath} - 14\hat{\jmath} - 2\hat{k}.$$

Note that $|\mathbf{A} + \mathbf{B}| \neq |\mathbf{A}| + |\mathbf{B}|$, since $\sqrt{29} \approx 5.39$ while $\sqrt{14} + \sqrt{21} \approx 8.3$.

**(b)** The displacement is $\mathbf{d} = 4\hat{\imath} + 3\hat{\jmath}$ km, so

$$|\mathbf{d}| = \sqrt{16 + 9} = 5 \text{ km}, \qquad \theta = \arctan\frac{3}{4} = 37^\circ \text{ north of east.}$$

The distance travelled is 7 km. It is a scalar, it has no direction, and it is different from the magnitude of the displacement.

**(c)** Equilibrium requires $\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = 0$, so

$$\mathbf{F}_3 = -(\mathbf{F}_1 + \mathbf{F}_2) = -(2\hat{\imath} + 8\hat{\jmath}) = (-2\hat{\imath} - 8\hat{\jmath}) \text{ N},$$

$$|\mathbf{F}_3| = \sqrt{4 + 64} = 2\sqrt{17} \approx 8.2 \text{ N},$$

pointing $76^\circ$ below the $-x$ axis, i.e. in the third quadrant.

*Board tip: do part (b) graphically first with the triangle or parallelogram rule, then analytically.*

### 1.6.3 Example 3: Dot Product, Angle, Projection and Work

Let $\mathbf{A} = 2\hat{\imath} + 3\hat{\jmath} - \hat{k}$ and $\mathbf{B} = \hat{\imath} - 2\hat{\jmath} + 2\hat{k}$.

**(a)** Find $\mathbf{A} \cdot \mathbf{B}$ and the angle $\theta$ between them.

**(b)** Find the scalar projection of $\mathbf{A}$ on $\mathbf{B}$, and the vector projection.

**(c)** Find $c$ such that $\mathbf{A}$ is perpendicular to $\mathbf{C} = \hat{\imath} + c\,\hat{\jmath}$.

**(d)** A force $\mathbf{F} = (3\hat{\imath}+4\hat{\jmath})$ N acts while the object undergoes a displacement $\mathbf{d} = (2\hat{\imath} - \hat{\jmath})$ m. Find the work $W = \mathbf{F} \cdot \mathbf{d}$.

**(e)** Derive the law of cosines from $|\mathbf{A} - \mathbf{B}|^2$.

**Solution**

**(a)**

$$\mathbf{A} \cdot \mathbf{B} = (2)(1) + (3)(-2) + (-1)(2) = 2 - 6 - 2 = -6,$$

$$|\mathbf{A}| = \sqrt{4 + 9 + 1} = \sqrt{14}, \qquad |\mathbf{B}| = \sqrt{1 + 4 + 4} = 3,$$

$$\cos\theta = \frac{-6}{3\sqrt{14}} = \frac{-2}{\sqrt{14}} \approx -0.535 \quad \Longrightarrow \quad \theta \approx 122^\circ$$

A negative dot product means that the angle is obtuse.

**(b)** The scalar projection is

$$A_{\parallel} = \mathbf{A} \cdot \hat{b} = \frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{B}|} = \frac{-6}{3} = -2,$$

and the vector projection is

$$A_{\parallel}\hat{b} = -2 \cdot \frac{1}{3}\left(\hat{\imath} - 2\hat{\jmath} + 2\hat{k}\right) = -\frac{2}{3}\hat{\imath} + \frac{4}{3}\hat{\jmath} - \frac{4}{3}\hat{k}.$$

**(c)**

$$\mathbf{A} \cdot \mathbf{C} = 2 + 3c = 0 \quad \Longrightarrow \quad c = -\frac{2}{3}.$$

**(d)**

$$W = (3)(2) + (4)(-1) = 6 - 4 = 2 \text{ J}.$$

Only the component of $\mathbf{F}$ along $\mathbf{d}$ does work.

**(e)**

$$|\mathbf{A} - \mathbf{B}|^2 = (\mathbf{A} - \mathbf{B}) \cdot (\mathbf{A} - \mathbf{B}) = A^2 + B^2 - 2\,\mathbf{A} \cdot \mathbf{B} = A^2 + B^2 - 2AB\cos\theta$$

### 1.6.4 Example 4: Cross Product, Area, Torque and Volume

Let $\mathbf{A} = 2\hat{\imath} + \hat{\jmath} - \hat{k}$ and $\mathbf{B} = \hat{\imath} + 2\hat{\jmath} + \hat{k}$.

**(a)** Compute $\mathbf{A} \times \mathbf{B}$ and verify that $\mathbf{B} \times \mathbf{A} = -\mathbf{A} \times \mathbf{B}$.

**(b)** Verify that $\mathbf{A} \cdot (\mathbf{A} \times \mathbf{B}) = 0$ and find the unit normal $\hat{n}$ to the plane of $\mathbf{A}$ and $\mathbf{B}$.

**(c)** Find the area of the parallelogram and of the triangle spanned by $\mathbf{A}$ and $\mathbf{B}$. Obtain $\theta$ from $|\mathbf{A} \times \mathbf{B}|$ and check it with the dot product.

**(d)** For $\mathbf{r} = (2\hat{\imath} + \hat{\jmath})$ m and $\mathbf{F} = 3\hat{\jmath}$ N, find the torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$.

**(e)** Find the volume of the parallelepiped with edges $\mathbf{a} = \hat{\imath} + 2\hat{\jmath}$, $\mathbf{b} = 3\hat{\jmath}$ and $\mathbf{c} = 4\hat{k}$.

**Solution**

**(a)**

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix} = \hat{\imath}(1+2) - \hat{\jmath}(2+1) + \hat{k}(4-1) = 3\hat{\imath} - 3\hat{\jmath} + 3\hat{k}.$$

Exchanging the two rows changes the sign, so

$$\mathbf{B} \times \mathbf{A} = -3\hat{\imath} + 3\hat{\jmath} - 3\hat{k}.$$

The cross product is anticommutative.

**(b)**

$$\mathbf{A} \cdot (\mathbf{A} \times \mathbf{B}) = 6 - 3 - 3 = 0,$$

so the result is perpendicular to both vectors. Since $|\mathbf{A} \times \mathbf{B}| = 3\sqrt{3}$,

$$\hat{n} = \frac{1}{\sqrt{3}} \left( \hat{\imath} - \hat{\jmath} + \hat{k} \right).$$

**(c)** The parallelogram area is $|\mathbf{A} \times \mathbf{B}| = 3\sqrt{3} \approx 5.20$ and the triangle area is $\frac{3\sqrt{3}}{2} \approx 2.60$. With $|\mathbf{A}| = |\mathbf{B}| = \sqrt{6}$,

$$\sin\theta = \frac{3\sqrt{3}}{6} = \frac{\sqrt{3}}{2}.$$

Checking with the dot product, $\mathbf{A} \cdot \mathbf{B} = 2 + 2 - 1 = 3$, so $\cos\theta = \frac{3}{6} = \frac{1}{2}$. Both give $\theta = 60^\circ$.

**(d)**

$$\boldsymbol{\tau} = (2\hat{\imath} + \hat{\jmath}) \times (3\hat{\jmath}) = 6(\hat{\imath} \times \hat{\jmath}) + 3(\hat{\jmath} \times \hat{\jmath}) = 6\hat{k} \text{ N} \cdot \text{m},$$

which points out of the page by the right-hand rule. The part of $\mathbf{r}$ parallel to $\mathbf{F}$ contributes nothing.

**(e)**

$$V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})| = \left\| \begin{vmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{vmatrix} \right\| = 12.$$

### 1.6.5 Example 5: Derivatives, Kinematics, Maxima and Concavity

**(a)** Differentiate

$$f(x) = 3x^4 - 5x^2 + 7, \quad g(x) = x^2 \sin x, \quad h(x) = \sin(3x^2 + 1), \quad p(x) = \frac{x^2+1}{x-1}, \quad q(x) = e^{-x^2}.$$

**(b)** A ball is thrown so that $y(t) = 5 + 20t - 5t^2$, with $y$ in metres and $t$ in seconds. Find $v(t)$ and $a(t)$, the time and value of the maximum height, and confirm with the second derivative that it is a maximum.

**(c)** For simple harmonic motion $x(t) = A\sin(\omega t + \varphi)$, show that $a = -\omega^2 x$.

**(d)** For damped motion $x(t) = Ae^{-bt}\cos\omega t$, find $\dot{x}$.

**Solution**

**(a)** Using the power and sum rules, the product rule, the chain rule and the quotient rule,

$$\begin{aligned} f'(x) &= 12x^3 - 10x, \\ g'(x) &= 2x\sin x + x^2\cos x, \\ h'(x) &= 6x\cos(3x^2+1), \\ p'(x) &= \frac{2x(x-1) - (x^2+1)}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}, \\ q'(x) &= -2x\,e^{-x^2}. \end{aligned}$$

**(b)**

$$v = \dot{y} = 20 - 10t, \qquad a = \dot{v} = \ddot{y} = -10 \text{ m/s}^2.$$

The maximum occurs when $v = 0$, that is at $t = 2$ s, where

$$y(2) = 5 + 40 - 20 = 25 \text{ m}.$$

Since $y''(t) = -10 < 0$ everywhere, the curve is concave, so $t = 2$ s is indeed a maximum.

**(c)**

$$v = A\omega\cos(\omega t+\varphi), \qquad a = -A\omega^2\sin(\omega t+\varphi) = -\omega^2 x.$$

This is exactly the equation of motion produced by the spring force $F = -kx$.

**(d)** By the product and chain rules,

$$\dot{x} = A\left[-b\,e^{-bt}\cos\omega t - \omega\,e^{-bt}\sin\omega t\right] = -A\,e^{-bt}\left(b\cos\omega t + \omega\sin\omega t\right).$$

### 1.6.6 Example 6: Partial Derivatives, Total Differential and Total Derivative

**(a)** For $f(x,y) = x^2y + 3xy^3 - y$, find $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$, and show that

$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x}.$$

**(b)** The volume of a cylinder is $V = \pi r^2 h$. With $r = 2.0$ cm, $h = 10.0$ cm, $dr = 0.1$ cm and $dh = 0.2$ cm, estimate $dV$ and the relative error $dV/V$.

**(c)** The temperature in a room is $T(x,y,t) = xy+5t$. A bug moves along $x(t) = t^2$ and $y(t) = 2t$. Find $dT/dt$ in two different ways.

**Solution**

**(a)** Treating the other variable as a constant,

$$\frac{\partial f}{\partial x} = 2xy + 3y^3, \qquad \frac{\partial f}{\partial y} = x^2 + 9xy^2 - 1.$$

Then

$$\frac{\partial}{\partial y}\left(2xy+3y^3\right) = 2x + 9y^2, \qquad \frac{\partial}{\partial x}\left(x^2+9xy^2-1\right) = 2x + 9y^2,$$

so the mixed partial derivatives are equal.

**(b)**

$$dV = \frac{\partial V}{\partial r}dr + \frac{\partial V}{\partial h}dh = 2\pi rh\,dr + \pi r^2\,dh = 2\pi(2)(10)(0.1) + \pi(4)(0.2) = 4.8\pi \approx 15\ \text{cm}^3.$$

Since $V = \pi(4)(10) = 40\pi \approx 126\ \text{cm}^3$,

$$\frac{dV}{V} = \frac{4.8\pi}{40\pi} = 0.12 = 12\%, \qquad \text{or equivalently} \qquad \frac{dV}{V} = 2\frac{dr}{r} + \frac{dh}{h} = 2(0.05) + 0.02.$$

The exponent multiplies the relative error.

**(c)** Using the total derivative,

$$\frac{dT}{dt} = \frac{\partial T}{\partial x}\frac{dx}{dt} + \frac{\partial T}{\partial y}\frac{dy}{dt} + \frac{\partial T}{\partial t} = y(2t) + x(2) + 5 = (2t)(2t) + 2t^2 + 5 = 6t^2 + 5.$$

By direct substitution, $T = (t^2)(2t) + 5t = 2t^3 + 5t$, so

$$\frac{dT}{dt} = 6t^2 + 5,$$

which agrees. Physically, the temperature the bug feels changes both because it moves and because the room heats up, the latter being the term $\partial T/\partial t = 5$.

### 1.6.7 Example 7: Indefinite and Definite Integrals in Physics

**(a)** Evaluate

$$\int \left(3x^2 - 4x + 5\right) dx, \qquad \int \left(\frac{2}{x} + e^x\right) dx, \qquad \int \sin(2x)\, dx.$$

**(b)** Evaluate $\int_0^3 (2x + 1)\, dx$ and check the answer geometrically using the area of a trapezoid. Then evaluate $\int_0^{\pi} \sin x\, dx$ and $\int_{-1}^{1} x^3\, dx$, and explain the second result.

**(c)** A particle has $a(t) = 6t$ m/s$^2$, with $v(0) = 2$ m/s and $x(0) = 0$. Find $v(t)$, $x(t)$, and the displacement during the first 2 s.

**(d)** Find the work done by a spring force $F = kx$ in stretching from 0 to $x_0$, with $k = 200$ N/m and $x_0 = 0.1$ m.

**Solution**

**(a)**

$$\int \left(3x^2 - 4x + 5\right) dx = x^3 - 2x^2 + 5x + C,$$

$$\int \left(\frac{2}{x} + e^x\right) dx = 2\ln|x| + e^x + C,$$

$$\int \sin(2x)\, dx = -\frac{1}{2}\cos 2x + C.$$

The last one can be checked by differentiating, since the chain rule returns $\sin 2x$.

**(b)**

$$\int_0^3 (2x + 1)\, dx = \left[x^2 + x\right]_0^3 = 12.$$

Geometrically this is a trapezoid with parallel sides $f(0)=1$ and $f(3)=7$ and width 3, giving $\frac{1+7}{2}\cdot 3 = 12$. Furthermore,

$$\int_0^{\pi} \sin x\,dx = [-\cos x]_0^{\pi} = 1+1 = 2, \qquad \int_{-1}^{1} x^3\,dx = \left[\frac{x^4}{4}\right]_{-1}^{1} = \frac{1}{4}-\frac{1}{4} = 0.$$

The last integral vanishes because the negative area on $[-1,0]$ cancels the positive area on $[0,1]$; the definite integral gives a signed area.

**(c)**

$$v(t) = \int 6t\,dt = 3t^2 + C_1, \qquad v(0)=2 \implies C_1 = 2, \qquad v(t) = 3t^2+2,$$

$$x(t) = \int \left(3t^2+2\right) dt = t^3 + 2t + C_2, \qquad x(0)=0 \implies C_2 = 0,$$

$$\Delta x = \int_0^{2} v\,dt = \left[t^3+2t\right]_0^{2} = 8+4 = 12 \text{ m}.$$

The integration constants are fixed by the initial conditions, and this problem is the inverse of Example 5(b).

**(d)**

$$W = \int_0^{x_0} kx\,dx = \frac{1}{2}kx_0^2 = \frac{1}{2}(200)(0.1)^2 = 1 \text{ J}.$$

The force is not constant, so $W = Fd$ cannot be used; the integral gives the area under the $F$ versus $x$ line.

### 1.6.8 Example 8: Taylor Series and Physical Approximations

**(a)** Expand $e^x$ and $\sin x$ about $x_0 = 0$ up to third order.

**(b)** Derive the binomial series $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2}x^2$ and use it to estimate $\sqrt{1.02}$.

**(c)** Show that the relativistic energy $E = mc^2/\sqrt{1 - v^2/c^2}$ reduces to $mc^2 + \frac{1}{2}mv^2$ for $v \ll c$.

**(d)** Show that the gravitational potential energy $U = -GMm/(R+h)$ becomes a constant plus $mgh$ for $h \ll R$.

**Solution**

**(a)** For $f(x) = e^x$ we have $f^{(n)}(0) = 1$ for every $n$, so

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

For $f(x) = \sin x$ we have $f(0) = 0$, $f'(0) = 1$, $f''(0) = 0$ and $f'''(0) = -1$, so

$$\sin x = x - \frac{x^3}{3!} + \cdots \qquad \Longrightarrow \qquad \sin\theta \approx \theta,$$

which is the small-angle approximation used for the pendulum. Similarly $\cos x = 1 - \frac{x^2}{2} + \cdots$

**(b)** For $f(x) = (1+x)^n$ we have $f(0) = 1$, $f'(0) = n$ and $f''(0) = n(n-1)$, hence

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2}x^2 + \cdots \qquad \text{for } |x| \ll 1.$$

With $n = \frac{1}{2}$ and $x = 0.02$,

$$\sqrt{1.02} \approx 1 + 0.01 - \frac{1}{8}(0.0004) = 1.00995,$$

while the exact value is $1.009950\ldots$

**(c)** Setting $x = -v^2/c^2$ and $n = -\frac{1}{2}$,

$$E = mc^2\left(1 - \frac{v^2}{c^2}\right)^{-1/2} \approx mc^2\left(1 + \frac{1}{2}\frac{v^2}{c^2} + \frac{3}{8}\frac{v^4}{c^4}\right) = mc^2 + \frac{1}{2}mv^2 + \cdots$$

The first term is the rest energy and the second is the Newtonian kinetic energy.

**(d)**

$$U = -\frac{GMm}{R\,(1+h/R)} = -\frac{GMm}{R}\left(1 + \frac{h}{R}\right)^{-1} \approx -\frac{GMm}{R}\left(1 - \frac{h}{R}\right) = -\frac{GMm}{R} + \frac{GM}{R^2}mh,$$

and since $g = GM/R^2$ we obtain $U \approx \text{constant} + mgh$.

*Closing remark: every simple formula of introductory physics, such as $mgh$, $\frac{1}{2}mv^2$ and $T = 2\pi\sqrt{L/g}$, is the first non-trivial term of a Taylor series.*

## 1.7 Homeworks

1. Compute following indefinite integrals

$$\int \frac{dx}{a+x} \text{ and } \int \left(3x^2\, e^{x^3} + \frac{4x}{1+x^2}\right) dx,$$

where $a$ is a constant.

2. Compute the definite integral

$$\int_1^4 (2x + 3)\, dx.$$

3. Find the derivative of

$$f(x) = \frac{x^3 \ln(x)}{e^{2x}}, \qquad (x > 0).$$

4. Find the local minimum and maximum values of the function

$$f(x) = x^3 - 6x^2 + 9x + 1$$

by using the first and second derivative tests.

5. Find the Taylor series of $\sin x$, $\cos x$, and $e^x$ about $x = 0$ up to and including the term in $x^5$.

6. Compute the partial derivatives of the function

$$f(x, y) = x^2 y + e^{xy}$$

with respect to $x$ and $y$ and write down the total differential $df$.

7. The position of a particle is described by the following vector as a function of time $t$:

$$\mathbf{r}(t) = a\cos(t)\,\hat{\imath} + a\sin(t)\,\hat{\jmath} + ct\,\hat{k}$$

where $a$ and $c$ are constants with units of length and velocity, respectively.

   (a) What kind of trajectory does the particle follow?

   (b) Find the velocity, acceleration and speed of the particle as functions of time.

   (c) When you consider the trajectory of the particle again, are the directions of the velocity and acceleration vectors in the way you expect? Explain.

   (d) Calculate the projection of the acceleration vector onto the velocity vector as a function of time and give a physical interpretation of this result.

8. A flat rectangular surface lies in the $xy$-plane. The surface has sides of length $a = 0.20$ m along the $x$-axis and $b = 0.30$ m along the $y$-axis. A uniform water flow is described by the flux vector field:

$$\mathbf{F} = 200\,\hat{\imath} + 300\,\hat{\jmath} + 400\,\hat{k} \quad \left[\frac{\mathrm{L}}{\mathrm{m}^2 \cdot \mathrm{s}}\right]$$

   (a) Compute the area vector $\mathbf{A}$ of the surface (the vector normal to the surface with magnitude equal to the area of the surface).

   (b) Compute the volume of water per unit time (in liters per second) passing through the surface using:

$$Q = \mathbf{F} \cdot \mathbf{A}.$$

   (c) Discuss how the orientation of the surface affects the water flow through it.

9. A particle moves in a plane such that its position vector at time $t$ is given by

$$\mathbf{r}(t) = at^2\,\hat{\imath} + bt^3\,\hat{\jmath}.$$

   where $a$ and $b$ are constants with appropriate units.

   (a) Find the velocity and acceleration vectors as functions of time.

   (b) Find the angle between the velocity and acceleration vectors at time $t = 1$.

10. Given the vectors

$$\mathbf{A} = 2\,\hat{\imath} - 3\,\hat{\jmath} + \hat{k}, \quad \mathbf{B} = -\hat{\imath} + 4\,\hat{\jmath} + 2\,\hat{k},$$

    compute $\mathbf{A} \times \mathbf{B}$.

# Chapter 2

# Kinematics

In this Chapter, we will study the motion of particles in space without considering the reasons for the motion; that is what *kinematics* is about. We will introduce the concepts of position, displacement, velocity, and acceleration vectors, and explore their relationships. We will also discuss different types of motion, including linear, circular, spiral, oscillatory, and cycloidal motion. Finally, we will work through some examples to illustrate these concepts. We will stick to Cartesian coordinates for simplicity, but the concepts can be extended to other coordinate systems as well.

## 2.1 Notion of a Particle

A *particle* is a body whose size and shape are not important for the problem at hand. In other words, a particle can be treated as a point mass. Its motion is described by its position, velocity, and acceleration as functions of time. Geometrically, a particle is a point in space with an associated mass.

For example, planets can be treated as particles when studying their motion around the Sun, even though they have finite size and shape. However, when studying phenomena such as tidal waves or atmospheric motion, the size and shape of the Earth are important, so the Earth cannot be treated as a particle.

Since a particle's entire mass is idealized as being concentrated at a single point, its mass density is formally singular at that point. This is an idealization and requires some mathematical care; a rigorous treatment uses the Dirac delta distribution, which will be introduced in the Electricity and Magnetism course.

## 2.2 The Position Vector

The position of a particle is described by a vector called the *position vector*:

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}, \tag{2.1}$$

where $x(t)$, $y(t)$, and $z(t)$ are the Cartesian coordinates of the particle as functions of time, and $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are the unit vectors in the $x$-, $y$-, and $z$-directions, respectively.

The position vector points from the origin of the coordinate system to the particle. Its magnitude,

$$|\mathbf{r}(t)| = \sqrt{x^2(t) + y^2(t) + z^2(t)}, \tag{2.2}$$

is the distance of the particle from the origin, while its direction specifies the direction from the origin to the particle.

Because the coordinates can change with time, the position vector $\mathbf{r}(t)$ describes the motion of the particle through space.

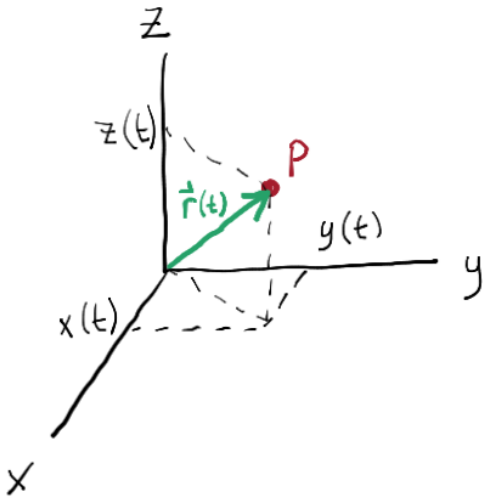



Figure 2.1: The position vector $\mathbf{r}(t)$ of a particle in a three-dimensional Cartesian coordinate system.

## 2.3 The Displacement Vector

Suppose a particle moves from an initial position $\mathbf{r}(t_1)$ to a final position $\mathbf{r}(t_2)$. The *displacement vector* describes the change in the particle's position. It is defined as the difference between the final and initial position vectors:

$$\Delta\mathbf{r} = \mathbf{r}(t_2) - \mathbf{r}(t_1). \tag{2.3}$$

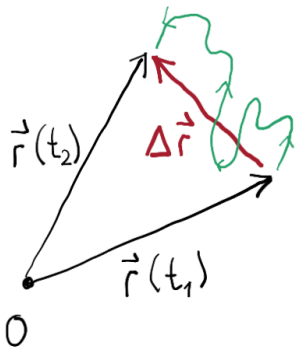



Figure 2.2: Displacement vector of a particle moving from an initial position $\mathbf{r}(t_1)$ to a final position $\mathbf{r}(t_2)$. The actual path, which is shown by the green curve, taken by the particle is not necessarily a straight line.

The displacement vector points from the initial position to the final position. Its magnitude, $|\Delta\mathbf{r}|$, is the straight-line distance between the initial and final positions. In general, it is not equal to the distance traveled by the particle along its trajectory. The displacement depends only on the initial and final positions, not on the path taken between them.

Geometrically, the relation given in Eq. (**??**) follows directly from vector subtraction. Since both position vectors originate at the origin, the vector connecting the initial position $P_1$ to the final position $P_2$ is equal to $\mathbf{r}(t_2) - \mathbf{r}(t_1)$. Thus, the displacement vector is the vector from the initial position to the final position.

## 2.4 The Average Velocity Vector

Velocity describes how the position of a particle changes with time. Suppose the particle moves from an initial position $\mathbf{r}(t_1)$ to a final position $\mathbf{r}(t_2)$ during the time interval $\Delta t = t_2 - t_1$ and let the corresponding displacement be $\Delta\mathbf{r}$. We define *the average velocity vector* as

$$\mathbf{v}_{\text{avg}} = \frac{\Delta\mathbf{r}}{\Delta t}. \tag{2.4}$$

This quantity describes the displacement per unit time during the interval $\Delta t$. Note that the particle may have sped up, slowed down, or even stopped temporarily during the motion from $\mathbf{r}(t_1)$ to $\mathbf{r}(t_2)$. From the initial and

final positions alone, we can therefore determine only the particle's motion *on average* over the time interval as given in Eq. (**??**). In the way the displacement shows where the particle went on average, the average velocity vector shows the direction and magnitude of the particle's motion on average. Consequently, the average velocity vector is in the same direction as the displacement vector, provided $\Delta\mathbf{r} \neq \mathbf{0}$.

Given that the initial and final position vectors be

$$\mathbf{r}(t_1) = x(t_1)\mathbf{i} + y(t_1)\mathbf{j} + z(t_1)\mathbf{k}, \tag{2.5}$$

$$\mathbf{r}(t_2) = x(t_2)\mathbf{i} + y(t_2)\mathbf{j} + z(t_2)\mathbf{k}, \tag{2.6}$$

the displacement can be written as

$$\begin{aligned}\Delta\mathbf{r} &= \mathbf{r}(t_2) - \mathbf{r}(t_1) \\ &= [x(t_2) - x(t_1)]\,\mathbf{i} + [y(t_2) - y(t_1)]\,\mathbf{j} + [z(t_2) - z(t_1)]\,\mathbf{k}.\end{aligned} \tag{2.7}$$

Defining

$$\Delta x = x(t_2) - x(t_1), \qquad \Delta y = y(t_2) - y(t_1), \qquad \Delta z = z(t_2) - z(t_1), \tag{2.8}$$

the average velocity can also be written as

$$\mathbf{v}_{\text{avg}} = \frac{\Delta x}{\Delta t}\mathbf{i} + \frac{\Delta y}{\Delta t}\mathbf{j} + \frac{\Delta z}{\Delta t}\mathbf{k}. \tag{2.9}$$

## 2.5 The Instantaneous Velocity Vector

To determine the velocity at a particular instant, we consider the average velocity over an increasingly small time interval around that instant. Mathematically, considering the limit of the average velocity as $\Delta t \to 0$ leads us to the definition of *instantaneous velocity*

$$\mathbf{v}(t) = \lim_{\Delta t\to 0}\frac{\Delta\mathbf{r}}{\Delta t} = \lim_{\Delta t\to 0}\frac{\Delta x}{\Delta t}\mathbf{i} + \lim_{\Delta t\to 0}\frac{\Delta y}{\Delta t}\mathbf{j} + \lim_{\Delta t\to 0}\frac{\Delta z}{\Delta t}\mathbf{k} \tag{2.10}$$

where we have the definition of derivative for each term. Hence the acceleration of a particle at time $t$ is defined as

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} + \frac{dz}{dt}\mathbf{k}. \tag{2.11}$$

Using Newton's notation,

$$\mathbf{v}(t) = \dot{x}(t)\mathbf{i} + \dot{y}(t)\mathbf{j} + \dot{z}(t)\mathbf{k} = v_x(t)\mathbf{i} + v_y(t)\mathbf{j} + v_z(t)\mathbf{k}. \tag{2.12}$$

The *speed* of the particle is the magnitude of its velocity vector:

$$v(t) = |\mathbf{v}(t)|. \tag{2.13}$$

We use speed if we are only interested in how fast the particle is moving, regardless of its direction of motion. The speed is always a non-negative quantity, while the velocity can be positive or negative depending on the direction of motion.

Compared to the average velocity, calculating the instantaneous velocity requires more detailed information about the motion of the particle. In calculating displacement, we only needed two vectors, the initial and final positions. However, in calculating the instantaneous velocity, we need to know $\mathbf{r}(t)$, i.e., all of the position vectors at each instant in time.

## 2.6 The Average Acceleration Vector

Acceleration is defined as the change in velocity with respect to time. Following the same reasoning as we did for the average velocity in Sec. **??**, we can define the average acceleration vector as follows:

$$\mathbf{a}_{\text{avg}} = \frac{\Delta \mathbf{v}}{\Delta t} \tag{2.14}$$

where $\Delta \mathbf{v} = \mathbf{v}_2 - \mathbf{v}_1$ is the change in velocity vector, and $\Delta t = t_2 - t_1$ is the time interval over which the change in velocity occurs. The average acceleration vector describes how the velocity of the particle changes on average over the time interval $\Delta t$. The average acceleration vector has the same direction as the change in velocity vector. If we are given the initial and final velocity vectors in component form as

$$\mathbf{v}(t_1) = v_x(t_1)\mathbf{i} + v_y(t_1)\mathbf{j} + v_z(t_1)\mathbf{k}$$

$$\mathbf{v}(t_2) = v_x(t_2)\mathbf{i} + v_y(t_2)\mathbf{j} + v_z(t_2)\mathbf{k}$$

Then the change in velocity is

$$\Delta \mathbf{v} = \mathbf{v}(t_2) - \mathbf{v}(t_1) = [v_x(t_2) - v_x(t_1)]\mathbf{i} + [v_y(t_2) - v_y(t_1)]\mathbf{j} + [v_z(t_2) - v_z(t_1)]\mathbf{k}$$

and the average acceleration vector becomes

$$\mathbf{a}_{\text{avg}} = \frac{\Delta v_x}{\Delta t}\mathbf{i} + \frac{\Delta v_y}{\Delta t}\mathbf{j} + \frac{\Delta v_z}{\Delta t}\mathbf{k}$$

## 2.7 The Instantaneous Acceleration Vector

If we are interested in the acceleration of the particle at a given time instead of the average acceleration we should consider the behavior of $\mathbf{a}_{\text{avg}}$ when $\Delta t \to 0$. Therefore,

$$\mathbf{a}(t) = \lim_{\Delta t \to 0} \mathbf{a}_{\text{avg}} = \lim_{\Delta t \to 0} \frac{\Delta \mathbf{v}}{\Delta t} = \frac{d\mathbf{v}}{dt} = \frac{dv_x}{dt}\mathbf{i} + \frac{dv_y}{dt}\mathbf{j} + \frac{dv_z}{dt}\mathbf{k}$$

In Newton's notation

$$\mathbf{a}(t) = \dot{v}_x(t)\mathbf{i} + \dot{v}_y(t)\mathbf{j} + \dot{v}_z(t)\mathbf{k} = a_x(t)\mathbf{i} + a_y(t)\mathbf{j} + a_z(t)\mathbf{k}. \tag{2.15}$$

where $a_x(t)$, $a_y(t)$, and $a_z(t)$ are the components of the acceleration vector in the $x$, $y$, and $z$ directions, respectively.

## 2.8 The Trajectory of a Particle

The *trajectory* of a particle is the path followed by the particle in space as time goes on. The *trajectory equation* of a particle is the equation that describes the trajectory of the particle in space. It can be obtained by eliminating the time variable from the components of the position vector $\mathbf{r}(t)$ and connecting the $x$, $y$, and $z$ coordinates of the particle.

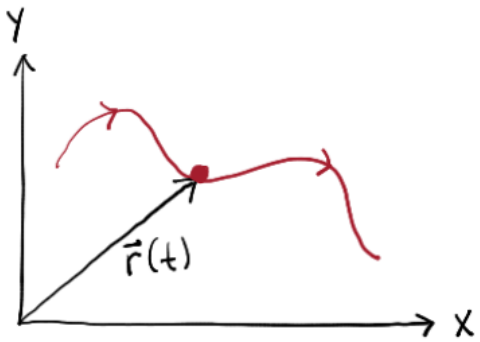



Figure 2.3: The trajectory of a particle in space is shown in red. The trajectory is the path followed by the particle as time goes on and the equation of the trajectory contains the relationship between the $x$, $y$, and $z$ coordinates of the particle.

## 2.9 Connecting position, velocity, and acceleration

We have already seen that once we are given the position vector of a particle, $\mathbf{r}(t)$, we can find the velocity and acceleration vectors by taking the first and second derivatives with respect to time, respectively:

$$\mathbf{r}(t) \xrightarrow{\frac{d}{dt}} \mathbf{v}(t) \xrightarrow{\frac{d}{dt}} \mathbf{a}(t) \tag{2.16}$$

What we really do here is to transform one information into another via a derivative operation. Note that once we know the position vector of a particle, we know everything about the motion of the particle. We can find the velocity and acceleration vectors by taking the first and second derivatives of the position vector with respect to time, respectively.

It is also possible to go the other way around If we are given the acceleration vector of a particle, $\mathbf{a}(t)$, we can find the velocity and position vectors.

For simplicity assume that the motion takes place on $x$-axis only and we know $a_x(t)$ as shown in Fig. **??**(a), which means that we know how the acceleration of the particle is changing with time. We can find how much the velocity of the particle has changed from time $t_1$ to time $t_2$ by integrating the acceleration function with respect to time:

$$\Delta v_x = \int_{t_1}^{t_2} a_x(t)dt = v_x(t_2) - v_x(t_1) \tag{2.17}$$

Rearranging the above equation, we can write:

$$v_x(t_2) = v_x(t_1) + \int_{t_1}^{t_2} a_x(t)dt \tag{2.18}$$

This means that if the only information we have is the form of $a_x(t)$, we can find how much the velocity of the particle has changed from time $t_1$ to time $t_2$. However, we still need to know the initial velocity of the particle at time $t_1$ in order to find the final velocity of the particle at time $t_2$.

If we consider integration as a transformation of information, we can say that we can transform the information about the acceleration of a particle into information about the velocity of the particle. However, we need to know the initial velocity of the particle at time $t_1$ in order to find the final velocity of the particle at time $t_2$.

In terms of indefinite integral, we can write:

$$v_x(t) = \int a_x(t)dt + C \tag{2.19}$$

where $C$ is a constant that represents the initial velocity of the particle at time $t = 0$. The value of $C$ can be determined if we are given the initial velocity of the particle. Like derivative, the integral is also a linear operator, which means that we can integrate each component of the acceleration vector separately to find the corresponding components of the velocity vector:

$$\mathbf{v}(t) = \int \mathbf{a}(t)dt + \mathbf{C} = \int a_x(t)dt\mathbf{i} + \int a_y(t)dt\mathbf{j} + \int a_z(t)dt\mathbf{k} + \mathbf{C} \quad (2.20)$$

where $\mathbf{C}$ is a constant vector that represents the initial velocity of the particle at time $t = 0$. The components of $\mathbf{C}$ can be determined if we are given the initial velocity of the particle. Similarly, we can find the position of the particle by integrating the velocity vector:

$$\mathbf{r}(t) = \int \mathbf{v}(t)dt + \mathbf{C} \quad (2.21)$$

where $\mathbf{C}$ is a constant vector that represents the initial position of the particle at time $t = 0$. The components of $\mathbf{C}$ can be determined if we are given the initial position of the particle.

**Summary:** If we are given how the position vector of a particle is changing with time, it means that we know everything about the motion of the particle. We can find the velocity and acceleration vectors by taking the first and second derivatives of the position vector with respect to time, respectively. If we are given how the acceleration vector of a particle is changing with time, we can find the velocity and position vectors by taking the indefinite integrals of the acceleration vector with respect to time, respectively. However, we need to know the initial conditions of the particle in order to find its exact position and velocity at any given time.

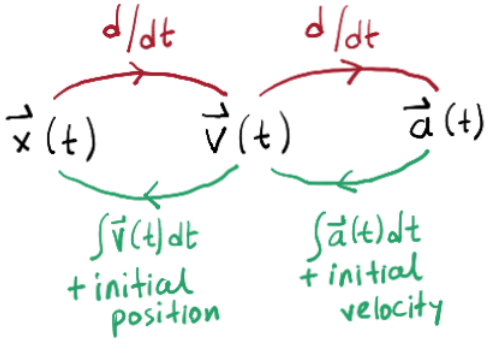



Figure 2.4: Summary of the relationships between position, velocity, and acceleration vectors.

## 2.10 Types of Motion

### 2.10.1 Linear Motion

Motion along a straight line is called *linear motion*. In this case, the position vector of the particle can be written as

$$\mathbf{r}(t) = x(t)\mathbf{i}, \tag{2.22}$$

where the $x$-axis is chosen along the direction of motion. The velocity and acceleration vectors are also directed along this axis:

$$\mathbf{v}(t) = \dot{x}(t)\mathbf{i}, \quad \mathbf{a}(t) = \ddot{x}(t)\mathbf{i}. \tag{2.23}$$

When the motion is restricted to a straight line, or in other words, when the motion is one-dimensional, the position, velocity, and acceleration can therefore be described by the scalar quantities

$$x = x(t), \qquad v = \dot{x}(t), \qquad a = \ddot{x}(t).$$

The sign of each quantity indicates its direction relative to the chosen $x$-axis. In particular, $v > 0$ means that the particle is moving in the positive $x$-direction, while $v < 0$ means that it is moving in the negative $x$-direction. Similarly, $a > 0$ and $a < 0$ indicate acceleration in the positive and negative $x$-directions, respectively. The magnitude $|v|$ is the speed, while $|a|$ is the magnitude of the acceleration.

### 2.10.2 Circular Motion

*Circular motion* is the motion of a particle along a circular path. For a circular path of constant radius $R$ as shown in Fig. **??**, the position vector of the particle can be written as

$$\mathbf{r}(t) = R\cos[\theta(t)]\,\mathbf{i} + R\sin[\theta(t)]\,\mathbf{j}. \tag{2.24}$$

where $R$ is the radius of the circular path and $\theta(t)$ is the angle between the position vector and the positive $x$-axis. Since the particle moves along the circle, this angle generally changes with time:

$$\theta = \theta(t).$$

The *instantaneous angular velocity* is defined as the rate of change of the angle:

$$\omega(t) = \frac{d\theta}{dt}. \tag{2.25}$$

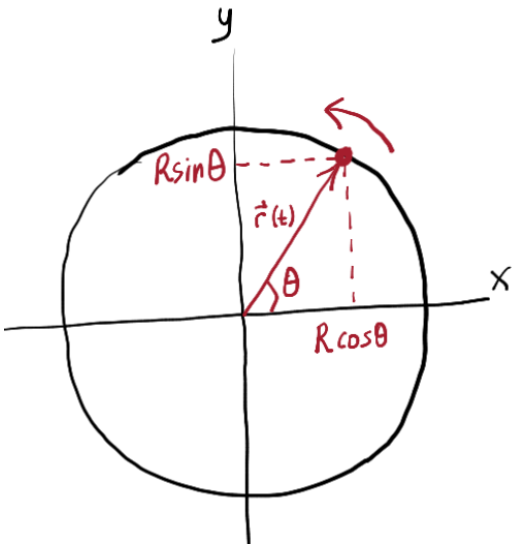



Figure 2.5: Circular motion described by the position vector $\mathbf{r}(t)$.

In *uniform circular motion*, the speed of the particle is constant. In this case, the angular velocity is also constant, so the angle changes linearly with time:

$$\theta(t) = \omega t + \theta_0, \tag{2.26}$$

where $\omega$ is the constant angular velocity and $\theta_0$ is the angle at $t = 0$.

**What is 1 radian?**

A radian is a unit of angular measure. An angle of 1 radian is defined as the angle at the center of a circle that subtends an arc whose length is equal to the radius of the circle. Thus, if an angle $\theta$ subtends an arc of length $s$ on a circle of radius $R$, then, when $\theta$ is measured in radians,

$$\theta = \frac{s}{R}. \tag{2.27}$$

For a complete circle, the arc length is the circumference $2\pi R$. Therefore,

$$2\pi \text{ rad} = 360^\circ.$$

The definition of the radian will be used repeatedly in the following sections, so it is important to understand this relationship between angular displacement and arc length. The following figure illustrates the definition of one radian:

We can obtain the velocity vector of a particle moving in a circular path by differentiating the position vector with respect to time:

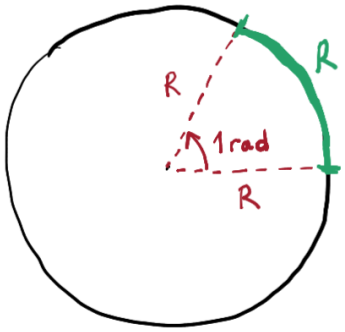



Figure 2.6: Definition of one radian: an angle of 1 radian subtends an arc whose length equals the radius of the circle.

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \frac{d}{dt}\left[R\cos\theta\mathbf{i} + R\sin\theta\mathbf{j}\right] = -R\dot{\theta}\sin\theta\mathbf{i} + R\dot{\theta}\cos\theta\mathbf{j}. \tag{2.28}$$

Since $\dot{\theta} = \omega$ for uniform circular motion,

$$\mathbf{v}(t) = R\omega\left[-\sin\theta\mathbf{i} + \cos\theta\mathbf{j}\right]. \tag{2.29}$$

The velocity vector is tangent to the circular path and therefore perpendicular to the position vector. Its magnitude is

$$|\mathbf{v}| = R|\omega|.$$

Thus, for uniform circular motion, the speed is constant and is related to the angular velocity by

$$v = R|\omega|.$$

Similarly, we can find the acceleration vector by differentiating the velocity vector with respect to time:

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d}{dt}\left[R\omega\left(-\sin\theta\mathbf{i} + \cos\theta\mathbf{j}\right)\right] = -R\omega^2\cos\theta\mathbf{i} - R\omega^2\sin\theta\mathbf{j}. \tag{2.30}$$

Thus, for uniform circular motion, the acceleration vector points towards the center of the circular path and is therefore directed radially inward. Its magnitude is

$$a = R|\omega|^2 \tag{2.31}$$

We can also express the acceleration in terms of the speed $v$:

$$a = \frac{R^2}{R}\left(\frac{d\theta}{dt}\right)^2 = \frac{1}{R}\left(\frac{ds}{dt}\right)^2 = \frac{v^2}{R} \tag{2.32}$$

### 2.10.3 Spiral Motion

It is a composite motion of linear and circular motions. The particle moves along a spiral path, which can be described by a combination of linear and circular motion equations. The position vector of a particle moving in a spiral path can be expressed as:

$$\mathbf{r}(t) = R\cos[\theta(t)]\mathbf{i} + R\sin[\theta(t)]\mathbf{j} + z(t)\mathbf{k}$$

which describes a circular motion in the $xy$-plane with radius $R$ and a linear motion along the $z$-axis. The angle $\theta(t)$ and the height $z(t)$ can be functions of time, allowing for various types of spiral motion, such as helical or conical spirals.

### 2.10.4 Oscillatory Motion

Oscillations are periodic motions; they repeat themselves after a certain interval of time. Complex numbers together with the exponential function provide a powerful and elegant framework for describing and analyzing oscillatory phenomena.

#### Complex Numbers

Mathematicians in the 16th century discovered that cubic formulas sometimes yield three real solutions while still forcing square roots of negative numbers to appear in the intermediate algebra. Bombelli (1572) was the first to state systematic rules for operating with $\sqrt{-1}$, making these calculations consistent and allowing the real results to be recovered, marking the first systematic use of complex numbers. Descartes later coined the term "imaginary" for these numbers, reflecting the skepticism of the time regarding their existence, and Euler adopted the notation $i$ for $\sqrt{-1}$, making the calculations routine; this notation has since become standard in mathematics.

*Complex numbers* are numbers that can be expressed in the form

$$z = x + iy, \tag{2.33}$$

where $x$ and $y$ are real numbers and $i$ is the imaginary unit satisfying $i^2 = -1$. We call them "numbers" because they obey the same algebraic rules as real numbers, with the additional rule that $i^2 = -1$, allowing addition, subtraction, multiplication, and division.

The real part of $z$ is denoted $\mathrm{Re}(z) = x$ and the imaginary part $\mathrm{Im}(z) = y$. Complex numbers can be represented graphically on the *complex plane*, where the horizontal axis represents the real part and the vertical axis the imaginary part, as shown in Fig. **??**.

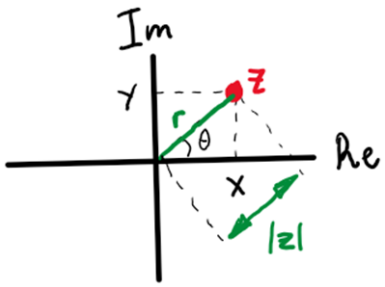



Figure 2.7: The complex plane.

By the Pythagorean theorem, the magnitude (or modulus) of $z = x + iy$ is

$$|z| = \sqrt{x^2 + y^2}, \tag{2.34}$$

the distance from the origin to the point $(x, y)$ in the complex plane. The argument (or angle) of $z$ is the angle $\theta$ that the line from the origin to $z$ makes with the positive real axis. For $x > 0$ it reduces to

$$\theta = \tan^{-1}\left(\frac{y}{x}\right); \tag{2.35}$$

in general the correct quadrant must be taken into account, and the two-argument function $\mathrm{atan2}(y, x)$ returns the correct angle for all $(x, y) \neq (0, 0)$.

It is often useful to write $z$ in *polar form* (see Fig. **??**), which expresses $z$ in terms of its magnitude and argument:

$$z = r\,(\cos\theta + i\sin\theta), \tag{2.36}$$

where $r = |z|$ is the magnitude and $\theta = \arg(z)$ is the argument.

### The connection between complex numbers, circular motion and oscillations

When $r$ is held constant and $\theta$ changes linearly with time, $z$ traces a circle in the complex plane, representing circular motion. Setting $\theta = \omega t + \phi$, where

$\omega$ is the angular velocity (in units $[\mathrm{T}^{-1}]$) as we defined it in Sec. **??**, and $\phi$ is the phase constant, gives

$$z(t) = r\left[\cos(\omega t + \phi) + i\sin(\omega t + \phi)\right]. \tag{2.37}$$

If we take the real part of $z(t)$, we obtain the familiar equation for simple harmonic oscillation which will be examined in more detail in Sec. **??**:

$$x(t) = r\cos(\omega t + \phi), \tag{2.38}$$

which describes the motion of a particle oscillating back and forth along a line. It is customary to call $\omega$ in this case the *angular frequency* since it refers to the rate of change of the phase of an oscillation. However, both angular velocity and angular frequency have the same units and the same numerical value. The imaginary part

$$y(t) = r\sin(\omega t + \phi) \tag{2.39}$$

describes a particle oscillating along a line perpendicular to the $x$-axis (imaginary axis). The combination of these two perpendicular oscillations results in circular motion, as the particle moves around the circle in the complex plane. The angular velocity $\omega$ determines how fast the particle moves around the circle, while the phase constant $\phi$ determines the initial position of the particle on the circle at $t = 0$.

### The Exponential Function

The exponential function $e^x$ describes growth or decay processes, as shown in Fig. **??**.

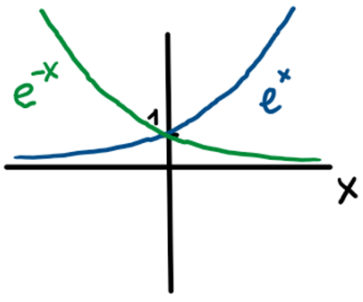



Figure 2.8: The exponential function $e^x$: growth for positive $x$, decay for negative $x$.

It is defined by its Taylor series:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \tag{2.40}$$

A key property is that $e^x$ is the *unique* function satisfying $f' = f$ with $f(0) = 1$, meaning its rate of change at every point equals its value there:

$$\frac{d}{dx}\, e^x = e^x. \tag{2.41}$$

More generally, for a constant rate $\lambda$, where $\lambda$ is a real number:

$$\frac{d}{dt}\, e^{\lambda t} = \lambda\, e^{\lambda t}, \tag{2.42}$$

so the velocity of a particle whose position is $x(t) = e^{\lambda t}$ is always proportional to its position by the factor $\lambda$: exponential growth for $\lambda > 0$ and exponential decay for $\lambda < 0$.

## Complex Exponential Function

Substituting $x = i\theta$ (with $\theta$ real) into the Taylor series of $e^x$ and using $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, ... yields

$$e^{i\theta} = 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \cdots \tag{2.43}$$

Separating real and imaginary parts:

$$e^{i\theta} = \underbrace{\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right)}_{\cos\theta} + i\,\underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right)}_{\sin\theta}, \tag{2.44}$$

which is *Euler's formula*:

$$e^{i\theta} = \cos\theta + i\sin\theta. \tag{2.45}$$

Applying this to the polar form, the time-dependent complex number becomes simply

$$z(t) = r\, e^{i(\omega t + \phi)}. \tag{2.46}$$

While a cosine or sine alone can describe a single oscillation, the complex exponential handles multiple oscillations, phase relationships, and operations

such as differentiation and integration more efficiently, making it the preferred tool in signal processing and wave mechanics.

Compared to Eq. (**??**), replacing the real constant $\lambda$ with the imaginary constant $i\omega$ gives

$$\frac{d}{dt}\, e^{i\omega t} = i\omega\, e^{i\omega t}, \tag{2.47}$$

so the velocity of a particle whose position is $x(t) = e^{i\omega t}$ is always proportional to its position by the factor $i\omega$. The factor $i$ indicates that the velocity is perpendicular to the position, as in circular motion. The magnitude of the velocity is $|\mathbf{v}| = \omega|x|$, as we found in Sec. **??**.

The coefficient of the time parameter $t$ in the exponent can be chosen as a general complex number. In this way, we can describe a combination of oscillatory and exponential growth or decay behavior. For example, if we let $\lambda = \alpha + i\omega$, where $\alpha$ and $\omega$ are real numbers, then

$$\frac{d}{dt}\, e^{(\alpha+i\omega)t} = (\alpha + i\omega)\, e^{(\alpha+i\omega)t}, \tag{2.48}$$

so the velocity of a particle whose position is $x(t) = e^{(\alpha+i\omega)t}$ is always proportional to its position by the factor $\alpha + i\omega$. The real part $\alpha$ indicates that the particle's speed is increasing or decreasing exponentially, while the imaginary part $i\omega$ indicates that the particle is oscillating. The magnitude of the velocity is

$$|\mathbf{v}| = \sqrt{\alpha^2 + \omega^2}\, |x|, \tag{2.49}$$

which combines the effects of exponential growth or decay and oscillation.

### 2.10.5 Cycloidal Motion —*Optional*

It is the path traced by a point on the circumference of a circle as it rolls along straight line without slipping. Imagine drawing a chalk dot on a bicycle wheel and then rolling the wheel along a flat surface. The path traced by the chalk dot is a cycloid.

The position vector of a particle moving along a cycloidal path can be expressed as:

$$\mathbf{r}(t) = R(t - \sin t)\mathbf{i} + R(1 - \cos t)\mathbf{j}$$

# 2.11 Worked Examples

## 2.11.1 Collision of Two Particles

Two particles are moving on a plane according to the following position vectors:

$$\mathbf{r}_1(t) = t\mathbf{i} + t\mathbf{j}$$

$$\mathbf{r}_2(t) = (-t + 4)\mathbf{i} + (2t - 2)\mathbf{j}$$

What time $t$ do the two particles collide? What is the position vector of the collision point? **Solution 1:** The two particles collide when their position vectors are equal:

$$\mathbf{r}_1(t) = \mathbf{r}_2(t)$$

This gives us two equations:

$$t = -t + 4 \quad \text{(i)}$$

$$t = 2t - 2 \quad \text{(ii)}$$

From equation (i), we get:

$$2t = 4 \implies t = 2$$

From equation (ii), we get:

$$t = 2t - 2 \implies t = 2$$

Both equations give us the same time $t = 2$. Now we can find the position vector of the collision point by substituting $t = 2$ into either $\mathbf{r}_1(t)$ or $\mathbf{r}_2(t)$:

$$\mathbf{r}_1(2) = 2\mathbf{i} + 2\mathbf{j} = 2\mathbf{i} + 2\mathbf{j}$$

**Solution 2:** Define a *seperation vector* $\Delta\mathbf{r}$:

$$\Delta\mathbf{r} = \mathbf{r}_2(t) - \mathbf{r}_1(t)$$

The two particles collide when $|\Delta\mathbf{r}| = 0$:

$$|(-2t + 4)\mathbf{i} + (t - 2)\mathbf{j}| = 0$$

This is equivalent to:

$$\sqrt{(-2t + 4)^2 + (t - 2)^2} = 0$$

Squaring both sides:

$$(-2t + 4)^2 + (t - 2)^2 = 0$$

Expanding:

$$4t^2 - 16t + 16 + t^2 - 4t + 4 = 0$$

Combining like terms:

$$5t^2 - 20t + 20 = 0$$

Dividing by 5:

$$t^2 - 4t + 4 = 0$$

This factors as:

$$(t - 2)^2 = 0$$

So $t = 2$.

### 2.11.2 Displacement, Average Velocity, Instantaneous Velocity, and Collision

Two particles move in the $xy$-plane according to

$$\mathbf{r}_1(t) = (t^2 + 1)\mathbf{i} + 2t\mathbf{j},$$

$$\mathbf{r}_2(t) = (5 - t)\mathbf{i} + (t^2 - 1)\mathbf{j}.$$

1. Find the position of each particle at $t = 0$ and $t = 2$.
2. Find the displacement vectors of both particles between $t = 0$ and $t = 2$.
3. Find their average velocity vectors over this time interval.
4. Find $\mathbf{v}_1(t)$ and $\mathbf{v}_2(t)$.
5. Determine whether the particles collide. If they do, find the collision time and collision position.

**Solution:**
**(a) Positions at $t = 0$ and $t = 2$**
For particle 1,

$$\mathbf{r}_1(0) = (0^2 + 1)\mathbf{i} + 2(0)\mathbf{j} = \mathbf{i},$$

and

$$\mathbf{r}_1(2) = (2^2 + 1)\mathbf{i} + 2(2)\mathbf{j} = 5\mathbf{i} + 4\mathbf{j}.$$

For particle 2,

$$\mathbf{r}_2(0) = (5 - 0)\mathbf{i} + (0^2 - 1)\mathbf{j} = 5\mathbf{i} - \mathbf{j},$$

and

$$\mathbf{r}_2(2) = (5 - 2)\mathbf{i} + (2^2 - 1)\mathbf{j} = 3\mathbf{i} + 3\mathbf{j}.$$

Therefore,

$$\boxed{\mathbf{r}_1(0) = \mathbf{i}, \qquad \mathbf{r}_1(2) = 5\mathbf{i} + 4\mathbf{j}}$$

and

$$\boxed{\mathbf{r}_2(0) = 5\mathbf{i} - \mathbf{j}, \qquad \mathbf{r}_2(2) = 3\mathbf{i} + 3\mathbf{j}.}$$

**(b) Displacement vectors**
The displacement is

$$\Delta\mathbf{r} = \mathbf{r}(t_2) - \mathbf{r}(t_1).$$

Thus, for particle 1,

$$\Delta\mathbf{r}_1 = \mathbf{r}_1(2) - \mathbf{r}_1(0) = (5\mathbf{i} + 4\mathbf{j}) - \mathbf{i},$$

so

$$\boxed{\Delta\mathbf{r}_1 = 4\mathbf{i} + 4\mathbf{j}.}$$

For particle 2,

$$\Delta\mathbf{r}_2 = \mathbf{r}_2(2) - \mathbf{r}_2(0) = (3\mathbf{i} + 3\mathbf{j}) - (5\mathbf{i} - \mathbf{j}),$$

hence

$$\boxed{\Delta\mathbf{r}_2 = -2\mathbf{i} + 4\mathbf{j}.}$$

**(c) Average velocity vectors**
The average velocity is

$$\mathbf{v}_{\text{avg}} = \frac{\Delta\mathbf{r}}{\Delta t}.$$

Since

$$\Delta t = 2 - 0 = 2,$$

we obtain

$$\mathbf{v}_{\text{avg},1} = \frac{4\mathbf{i} + 4\mathbf{j}}{2} = 2\mathbf{i} + 2\mathbf{j},$$

and therefore

$$\boxed{\mathbf{v}_{\text{avg},1} = 2\mathbf{i} + 2\mathbf{j}.}$$

Similarly,

$$\mathbf{v}_{\text{avg},2} = \frac{-2\mathbf{i} + 4\mathbf{j}}{2} = -\mathbf{i} + 2\mathbf{j},$$

so

$$\boxed{\mathbf{v}_{\text{avg},2} = -\mathbf{i} + 2\mathbf{j}.}$$

**(d) Instantaneous velocity**

The instantaneous velocity is obtained from

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}.$$

For particle 1,

$$\mathbf{v}_1(t) = \frac{d}{dt}\left[(t^2 + 1)\mathbf{i} + 2t\mathbf{j}\right],$$

giving

$$\boxed{\mathbf{v}_1(t) = 2t\mathbf{i} + 2\mathbf{j}.}$$

For particle 2,

$$\mathbf{v}_2(t) = \frac{d}{dt}\left[(5 - t)\mathbf{i} + (t^2 - 1)\mathbf{j}\right],$$

so

$$\boxed{\mathbf{v}_2(t) = -\mathbf{i} + 2t\mathbf{j}.}$$

**(e) Collision**

The particles collide when their position vectors are equal at the same time:

$$\mathbf{r}_1(t) = \mathbf{r}_2(t).$$

This gives the two equations

$$t^2 + 1 = 5 - t,$$

and

$$2t = t^2 - 1.$$

The first equation gives

$$t^2 + t - 4 = 0,$$

so

$$t = \frac{-1 \pm \sqrt{17}}{2}.$$

The second equation gives

$$t^2 - 2t - 1 = 0,$$

so

$$t = 1 \pm \sqrt{2}.$$

There is no time that satisfies both equations simultaneously. Therefore, the particles do not collide.

$$\boxed{\text{The two particles never collide.}}$$

This illustrates that equality of one coordinate is not sufficient for a collision; all spatial coordinates must be equal at the same instant.

### 2.11.3 Velocity, Acceleration, and the Interpretation of Signs

A particle moves along the $x$-axis according to

$$x(t) = t^3 - 6t^2 + 9t.$$

1. Find $v(t)$ and $a(t)$.

2. Determine the times at which the particle is at rest.

3. For each time interval separated by these instants, determine whether the particle is moving in the $+x$ or $-x$ direction.

4. Determine where the particle is speeding up and where it is slowing down.

5. Find the total distance traveled during $0 \leq t \leq 4$.

6. Compare the total distance traveled with the magnitude of the net displacement.

**Solution:**
**(a) Velocity and acceleration**
The velocity is the derivative of position:

$$v(t) = \frac{dx}{dt} = 3t^2 - 12t + 9.$$

Factoring,

$$v(t) = 3(t^2 - 4t + 3) = 3(t-1)(t-3).$$

Therefore,

$$\boxed{v(t) = 3(t-1)(t-3).}$$

The acceleration is

$$a(t) = \frac{dv}{dt} = 6t - 12,$$

so

$$\boxed{a(t) = 6(t-2).}$$

**(b) Times at which the particle is at rest**
The particle is at rest when

$$v(t) = 0.$$

Therefore,

$$3(t-1)(t-3) = 0,$$

which gives

$$\boxed{t = 1, \qquad t = 3.}$$

**(c) Direction of motion**

The sign of $v(t)$ determines the direction of motion.
For $0 < t < 1$,

$$(t-1) < 0, \qquad (t-3) < 0,$$

so

$$v(t) > 0.$$

Thus the particle moves in the $+x$ direction.
For $1 < t < 3$,

$$(t-1) > 0, \qquad (t-3) < 0,$$

so

$$v(t) < 0.$$

Thus the particle moves in the $-x$ direction.
For $t > 3$,

$$(t-1) > 0, \qquad (t-3) > 0,$$

so

$$v(t) > 0.$$

Thus the particle again moves in the $+x$ direction.
Therefore,

| Time interval | Direction |
|---|---|
| $0 < t < 1$ | $+x$ |
| $1 < t < 3$ | $-x$ |
| $t > 3$ | $+x$ |

**(d) Speeding up and slowing down**

A particle speeds up when the velocity and acceleration have the same sign, and slows down when they have opposite signs.

We have

$$a(t) = 6(t-2).$$

Hence,

$$a(t) < 0 \quad \text{for } t < 2,$$

and

$$a(t) > 0 \quad \text{for } t > 2.$$

Combining this with the sign of $v(t)$:

| Interval | $v$ | $a$ | Behavior |
|---|---|---|---|
| $0 < t < 1$ | $+$ | $-$ | slowing down |
| $1 < t < 2$ | $-$ | $-$ | speeding up |
| $2 < t < 3$ | $-$ | $+$ | slowing down |
| $t > 3$ | $+$ | $+$ | speeding up |

Thus,

$$\boxed{\begin{array}{l}\text{slowing down: } 0 < t < 1, \quad 2 < t < 3, \\ \text{speeding up: } 1 < t < 2, \quad t > 3.\end{array}}$$

**(e) Total distance traveled for** $0 \le t \le 4$

Because the particle changes direction at $t = 1$ and $t = 3$, we must divide the motion into separate intervals.

First calculate the positions:

$$x(0) = 0,$$

$$x(1) = 1 - 6 + 9 = 4,$$

$$x(3) = 27 - 54 + 27 = 0,$$

$$x(4) = 64 - 96 + 36 = 4.$$

Therefore, the distances traveled are

$$d_1 = |x(1) - x(0)| = |4 - 0| = 4,$$

$$d_2 = |x(3) - x(1)| = |0 - 4| = 4,$$

$$d_3 = |x(4) - x(3)| = |4 - 0| = 4.$$

Hence the total distance is

$$\boxed{d_{\text{total}} = 4 + 4 + 4 = 12.}$$

**(f) Distance versus displacement**
The net displacement is

$$\Delta x = x(4) - x(0) = 4 - 0 = 4.$$

Its magnitude is therefore

$$|\Delta x| = 4.$$

Thus,

$$\boxed{\text{total distance} = 12, \qquad |\text{displacement}| = 4.}$$

The difference occurs because the particle changes direction during its motion.

### 2.11.4 Reconstructing Position and Velocity from Acceleration

A particle moves in one dimension with acceleration

$$a(t) = 6t - 4.$$

At $t = 0$,

$$x(0) = 2, \qquad v(0) = -3.$$

1. Find $v(t)$.
2. Find $x(t)$.
3. Determine the time at which the particle changes direction.
4. Find the position of the particle at that instant.
5. Verify the result by differentiating $x(t)$.

**Solution:**
**(a) Finding the velocity**
Since

$$a(t) = \frac{dv}{dt},$$

we integrate:

$$v(t) = \int (6t - 4)\, dt.$$

Therefore,

$$v(t) = 3t^2 - 4t + C_1.$$

Using the initial condition $v(0) = -3$,

$$-3 = C_1.$$

Hence,

$$\boxed{v(t) = 3t^2 - 4t - 3.}$$

**(b) Finding the position**

Since

$$v(t) = \frac{dx}{dt},$$

we integrate again:

$$x(t) = \int (3t^2 - 4t - 3)\, dt.$$

Thus,

$$x(t) = t^3 - 2t^2 - 3t + C_2.$$

Using $x(0) = 2$,

$$2 = C_2.$$

Therefore,

$$\boxed{x(t) = t^3 - 2t^2 - 3t + 2.}$$

This demonstrates why the initial conditions are necessary: one integration introduces one unknown constant, and a second integration introduces another.

**(c) Time at which the particle changes direction**

The particle changes direction when its velocity changes sign. First find when it is at rest:

$$v(t) = 0.$$

Thus,

$$3t^2 - 4t - 3 = 0.$$

Using the quadratic formula,

$$t = \frac{4 \pm \sqrt{16 + 36}}{6} = \frac{4 \pm 2\sqrt{13}}{6}.$$

Therefore,

$$t = \frac{2 \pm \sqrt{13}}{3}.$$

Since physical time in this problem is taken to begin at $t = 0$, the relevant solution is

$$\boxed{t_{\text{turn}} = \frac{2 + \sqrt{13}}{3}.}$$

**(d) Position at the turning point**
Substituting

$$t_{\text{turn}} = \frac{2 + \sqrt{13}}{3}$$

into

$$x(t) = t^3 - 2t^2 - 3t + 2$$

gives the turning-point position

$$\boxed{x_{\text{turn}} = -\frac{52 + 26\sqrt{13}}{27}.}$$

Numerically,

$$x_{\text{turn}} \approx -5.412.$$

**(e) Verification**
Differentiating the position function,

$$x(t) = t^3 - 2t^2 - 3t + 2,$$

gives

$$\frac{dx}{dt} = 3t^2 - 4t - 3 = v(t),$$

and differentiating once more,

$$\frac{d^2x}{dt^2} = 6t - 4 = a(t).$$

Therefore,

$$\boxed{\frac{dx}{dt} = v(t), \qquad \frac{d^2x}{dt^2} = a(t),}$$

as required.

### 2.11.5 Uniform Circular Motion

A particle moves on a circle of radius $R$ according to

$$\mathbf{r}(t) = R\cos(\omega t)\mathbf{i} + R\sin(\omega t)\mathbf{j}.$$

1. Find $\mathbf{v}(t)$.
2. Find the speed $|\mathbf{v}(t)|$.
3. Find $\mathbf{a}(t)$.
4. Show explicitly that
$$\mathbf{r}(t) \cdot \mathbf{v}(t) = 0.$$
5. Show that the acceleration always points toward the center of the circle.
6. Find the magnitude of the acceleration in terms of $R$ and $\omega$.
7. Explain geometrically why the velocity is tangent to the trajectory while the acceleration is directed toward the center.

**Solution:**
**(a) Velocity**
The velocity is

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}.$$

Therefore,

$$\mathbf{v}(t) = -R\omega \sin(\omega t)\mathbf{i} + R\omega \cos(\omega t)\mathbf{j}.$$

Hence,

$$\boxed{\mathbf{v}(t) = -R\omega \sin(\omega t)\mathbf{i} + R\omega \cos(\omega t)\mathbf{j}.}$$

**(b) Speed**
The speed is the magnitude of the velocity:

$$|\mathbf{v}(t)| = \sqrt{R^2\omega^2 \sin^2(\omega t) + R^2\omega^2 \cos^2(\omega t)}.$$

Using

$$\sin^2\theta + \cos^2\theta = 1,$$

we obtain

$$|\mathbf{v}(t)| = R|\omega|.$$

For $\omega > 0$,

$$\boxed{v = R\omega.}$$

Thus the speed is constant.
**(c) Acceleration**
Differentiate the velocity:

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt}.$$

Hence,

$$\mathbf{a}(t) = -R\omega^2 \cos(\omega t)\mathbf{i} - R\omega^2 \sin(\omega t)\mathbf{j}.$$

Factorizing,

$$\boxed{\mathbf{a}(t) = -\omega^2 \left[R\cos(\omega t)\mathbf{i} + R\sin(\omega t)\mathbf{j}\right].}$$

Since the expression in brackets is precisely $\mathbf{r}(t)$,

$$\boxed{\mathbf{a}(t) = -\omega^2 \mathbf{r}(t).}$$

**(d) Orthogonality of position and velocity**
Using the expressions above,

$$\mathbf{r}(t) \cdot \mathbf{v}(t) = [R\cos(\omega t)\mathbf{i} + R\sin(\omega t)\mathbf{j}] \cdot [-R\omega\sin(\omega t)\mathbf{i} + R\omega\cos(\omega t)\mathbf{j}] \quad (2.50)$$
$$= -R^2\omega\cos(\omega t)\sin(\omega t) + R^2\omega\sin(\omega t)\cos(\omega t) \quad (2.51)$$
$$= 0. \quad (2.52)$$

Therefore,

$$\boxed{\mathbf{r}(t) \perp \mathbf{v}(t).}$$

Since the position vector points from the center to the particle, the velocity is perpendicular to the radius and therefore tangent to the circle.

**(e) Direction of the acceleration**

We found that

$$\mathbf{a}(t) = -\omega^2\mathbf{r}(t).$$

The factor $-\omega^2$ is negative, so $\mathbf{a}(t)$ points in the direction opposite to $\mathbf{r}(t)$.

Since $\mathbf{r}(t)$ points from the center toward the particle, the acceleration points from the particle toward the center.

Thus,

$$\boxed{\text{the acceleration is always directed toward the center.}}$$

**(f) Magnitude of the acceleration**

Since

$$\mathbf{a}(t) = -\omega^2\mathbf{r}(t),$$

we have

$$|\mathbf{a}(t)| = \omega^2|\mathbf{r}(t)|.$$

Because

$$|\mathbf{r}(t)| = R,$$

it follows that

$$\boxed{a = R\omega^2.}$$

Using $v = R\omega$, this can also be written as

$$\boxed{a = \frac{v^2}{R}.}$$

**(g) Geometrical interpretation**

The particle remains at a fixed distance $R$ from the center. Therefore, its position vector continuously rotates.

The velocity is the rate of change of the position vector. An infinitesimal change in the position vector along a circular path is tangent to the circle, so the instantaneous velocity is tangent to the trajectory.

The velocity vector itself changes direction continuously. Its change is directed toward the center of the circle, producing an inward acceleration.

Thus,

| **v** is tangent to the trajectory, **a** points toward the center. |
|---|

### 2.11.6 Oscillatory Motion and the Complex Exponential

Consider the complex function

$$z(t) = Ae^{i(\omega t+\phi)},$$

where $A$, $\omega$, and $\phi$ are constants.

1. Using Euler's formula, write $z(t)$ in terms of its real and imaginary parts.

2. What trajectory does the corresponding point follow in the complex plane?

3. Calculate $\dot{z}(t)$ and $\ddot{z}(t)$.

4. Show that
$$\ddot{z} + \omega^2 z = 0.$$

5. What are the real and imaginary parts of this equation?

6. Explain why the complex exponential is useful for describing oscillatory motion.

**Solution:**

**(a) Real and imaginary parts**

Using Euler's formula,

$$e^{i\theta} = \cos\theta + i\sin\theta,$$

we obtain

$$z(t) = A\left[\cos(\omega t + \phi) + i\sin(\omega t + \phi)\right].$$

Therefore,

$$\boxed{\mathrm{Re}[z(t)] = A\cos(\omega t + \phi)}$$

and

$$\boxed{\mathrm{Im}[z(t)] = A\sin(\omega t + \phi).}$$

Thus, the same complex function contains both sine and cosine oscillations.

**(b) Trajectory in the complex plane**

The real and imaginary coordinates are

$$x(t) = A\cos(\omega t + \phi),$$

$$y(t) = A\sin(\omega t + \phi).$$

Squaring and adding,

$$x^2 + y^2 = A^2\cos^2(\omega t + \phi) + A^2\sin^2(\omega t + \phi).$$

Hence,

$$x^2 + y^2 = A^2.$$

Therefore, the point follows a circle of radius $A$ centered at the origin:

$$\boxed{x^2 + y^2 = A^2.}$$

**(c) First and second derivatives**

Differentiating,

$$\dot{z}(t) = i\omega A e^{i(\omega t+\phi)}.$$

Therefore,

$$\boxed{\dot{z}(t) = i\omega z(t).}$$

Differentiating again,

$$\ddot{z}(t) = (i\omega)^2 A e^{i(\omega t+\phi)}.$$

Since

$$i^2 = -1,$$

we have

$$\boxed{\ddot{z}(t) = -\omega^2 z(t).}$$

**(d) Differential equation**
From the previous result,

$$\ddot{z} = -\omega^2 z.$$

Therefore,

$$\boxed{\ddot{z} + \omega^2 z = 0.}$$

Thus the complex exponential automatically satisfies the differential equation of simple harmonic motion.

**(e) Real and imaginary parts**
Since

$$z(t) = x(t) + iy(t),$$

we have

$$\ddot{z} + \omega^2 z = \left(\ddot{x} + \omega^2 x\right) + i\left(\ddot{y} + \omega^2 y\right).$$

For this expression to vanish, both the real and imaginary parts must vanish:

$$\boxed{\ddot{x} + \omega^2 x = 0}$$

and

$$\boxed{\ddot{y} + \omega^2 y = 0.}$$

Thus both the real and imaginary components independently describe simple harmonic motion.

**(f) Why use complex exponentials?**

The complex exponential provides a compact representation of two real oscillatory functions:

$$e^{i\omega t} = \cos(\omega t) + i\sin(\omega t).$$

Differentiation becomes particularly simple:

$$\frac{d}{dt}e^{i\omega t} = i\omega e^{i\omega t}.$$

Similarly,

$$\frac{d^2}{dt^2}e^{i\omega t} = -\omega^2 e^{i\omega t}.$$

Thus, instead of carrying separate sine and cosine functions through every calculation, we can perform the calculation using a single complex exponential and take the real or imaginary part at the end.

The complex representation therefore provides a convenient mathematical tool for analyzing oscillatory motion, while the underlying physical motion can remain completely real.

## 2.12 Homework

1. A particle is under uniform circular motion (counter-clockwise) with a radius of 5 m and a constant speed of 10 m/s. The center of the circle is at point (3m, 4m) in the xy-plane.

   (a) Write the position vector of the particle as a function of time, assuming it starts at the point (8m, 4m) at time t = 0.

   (b) Calculate the velocity and acceleration vectors at $t = \pi/2$ seconds and show that they are perpendicular to each other by considering their dot product. Is this true for all time?

2. The position vectors of two particles are given by:

$$\mathbf{r}_1(t) = (1 + 2t)\,\hat{\imath} + (3 - t)\,\hat{\jmath}$$

$$\mathbf{r}_2(t) = (4 - t)\,\hat{\imath} + (1 + 3t)\,\hat{\jmath}$$

   (a) Find the velocity and acceleration of each particle.

   (b) Find the time at which the two particles are closest to each other and determine that minimum distance.

3. A stationary body of total mass $M = 13$ kg suddenly disintegrates into three fragments with masses

$$m_1 = 3 \text{ kg}, \qquad m_2 = 4 \text{ kg}, \qquad m_3 = 6 \text{ kg}.$$

Right after the break-up:

- fragment $m_1$ moves due east with speed $v_1 = 4$ m/s,
- fragment $m_2$ moves due north with speed $v_2 = 3$ m/s.

Using only conservation of linear momentum, determine the velocity (both speed and direction) of fragment $m_3$ immediately after the break-up.

4. A block of mass $m = 4$ kg rests on a horizontal surface. The coefficient of static friction between the block and the surface is $\mu_s = 0.5$, and the coefficient of kinetic friction is $\mu_k = 0.3$.

A horizontal force $F(t)$ is applied to the block such that it increases linearly with time:

$$F(t) = 2t \quad \text{(in N, with } t \text{ in seconds)}.$$

(a) Determine the time at which the block starts to move.

(b) Once the block starts moving, find its acceleration immediately after it starts moving.

(c) Find the velocity of the block 5 seconds after it starts moving. *(Assume $g = 10$ m/s$^2$.)*

# Chapter 3

# Dynamics

In this chapter, we turn our attention to the causes of motion: What makes an object change its motion? For centuries, philosophers believed that an object must be continuously acted upon in order to remain in motion. This view was consistent with everyday experience. For example, when you slide a book across a table, it eventually comes to rest. It therefore seems natural to conclude that a "push" is required to keep the book moving.

Galileo Galilei challenged this idea by introducing the principle of inertia. He argued that an object does not need a cause to maintain a constant velocity. In the absence of external influences, an object at rest remains at rest, while an object in motion continues to move in a straight line at constant speed. The apparent need for a cause to maintain motion in everyday situations arises because friction and other external influences are present.

Isaac Newton later formulated this idea and developed a quantitative theory of dynamics. In particular, Newton's asserted that there must be a cause for objects to accelerate. He called this cause *force*. Thus, the fundamental question of dynamics is not why an object has a velocity, but why its velocity changes.

In this Chapter, we will restrict our attention to the motion of a single particle. We will call this particle a *system* to make our terminology consistent with the rest of the text where we will be revisiting the dynamics for system of particles in Chapter **??**.

## 3.1 Newton's Axioms of Motion

**Axiom 1:** A body remains at rest or continues to move in a straight line with constant speed unless acted upon by a *force*.

**Axiom 2:** A body acted upon a *force* moves in such a manner that the

time rate of change of its *momentum* equals the net force acting on it.

**Axiom 3:** To every action, there is an equal and opposite reaction.

It looks like the 1st axiom is a special case of the 2nd axiom, where the net force is zero. However, the 1st axiom is important because Newton defines a special class of reference frames called *inertial frames* where his axioms of motion hold true. Geometrically, a *reference frame* is the coordinate system where we make measurements. For example, if you are in a car, the car itself is your reference frame; you measure all lengths relative to the car. When the car accelerates, you feel a force pushing you back into your seat. This force is not due to any physical interaction with the car; rather, it arises because you are in a non-inertial frame of reference. Newton's axioms are valid only in intertial frames, which are reference frames that are not accelerating. Inertial frames are special because they allow us to describe the motion of objects in a simple and consistent way.

The second axiom mentions the quantity *momentum*, which is defined as

$$\mathbf{p} = m\mathbf{v}, \tag{3.1}$$

where $\mathbf{p}$ is the momentum of a body, $m$ is its mass, and $\mathbf{v}$ is its velocity. The units of momentum are kilogram meters per second (kg·m/s). The second axiom states that if the momentum changes in time it is due to a net force acting on the body. Mathematically, this can be expressed as

$$\mathbf{F} = \frac{d\mathbf{p}}{dt}, \tag{3.2}$$

where $\mathbf{F}$ is the net force acting on the body. The units of force are newtons (N), which are defined as one kilogram meter per second squared (kg·m/s$^2$). In general, both mass and velocity can change with time resulting in a change in momentum. If the mass of the body is constant, then this equation can be rewritten as

$$\mathbf{F} = m\frac{d\mathbf{v}}{dt} = m\mathbf{a}, \tag{3.3}$$

where $\mathbf{a}$ is the acceleration of the body. This equation is often referred to as Newton's second law of motion. It states that the acceleration of a body is directly proportional to the net force acting on it and inversely proportional to its mass. In other words,

$$\mathbf{a} = \frac{\mathbf{F}}{m}. \tag{3.4}$$

The third axiom states the nature of forces that they never exist in isolation, but always come in pairs. If body A exerts a force on body B, then body B exerts an equal and opposite force on body A. This is known as the

*action-reaction* principle. For example, when you push against a wall, the wall pushes back against you with an equal and opposite force. If you push a book across a table, the book pushes back against your hand with an equal and opposite force.

These three axioms explain a wide range of phenomena, from the motion of planets to the behavior of everyday objects. They form the foundation of classical mechanics and are still used today to analyze the motion of objects in a variety of contexts.

## 3.2 Galilean Invariance and Inertial Frames

All reference frames that move with a constant velocity relative to an inertial reference frame are also inertial reference frames. The laws of mechanics have the same form in all inertial reference frames. This is known as *Galilean invariance*. For example, if you are in a car moving at constant velocity along a straight road, you can perform mechanical experiments and obtain the same results as if you were at rest. The outcome of a mechanical experiment does not depend on the uniform motion of the reference frame.

To see this, consider two inertial reference frames, $S$ and $S'$, where $S'$ moves with a constant velocity $\mathbf{v}_0$ relative to $S$. If their origins coincide at $t = 0$, the position of a body in the two reference frames is related by

$$\mathbf{r}' = \mathbf{r} - \mathbf{v}_0 t, \tag{3.5}$$

where $\mathbf{r}$ is the position of the body in $S$ and $\mathbf{r}'$ is its position in $S'$. Differentiating with respect to time gives the relation between the velocities:

$$\mathbf{v}' = \frac{d\mathbf{r}'}{dt} = \frac{d\mathbf{r}}{dt} - \mathbf{v}_0 = \mathbf{v} - \mathbf{v}_0, \tag{3.6}$$

where $\mathbf{v}$ and $\mathbf{v}'$ are the velocities of the body in $S$ and $S'$, respectively. Differentiating once more gives

$$\mathbf{a}' = \frac{d\mathbf{v}'}{dt} = \frac{d\mathbf{v}}{dt} = \mathbf{a}. \tag{3.7}$$

Thus, the acceleration of a body is the same in all inertial reference frames. Consequently, Newton's second law,

$$\mathbf{F} = m\mathbf{a}, \tag{3.8}$$

has the same form in both frames:

$$\mathbf{F}' = m\mathbf{a}' = m\mathbf{a} = \mathbf{F}. \tag{3.9}$$

This invariance of Newton's laws under Galilean transformations is the basis of Galilean invariance in classical mechanics.

An important feature of Galilean transformations is that time is absolute. The time interval between two events is the same in all inertial reference frames:

$$t' = t. \tag{3.10}$$

Therefore, if two events occur simultaneously in one inertial reference frame, they occur simultaneously in every other inertial reference frame. This absolute nature of time is a fundamental assumption of classical mechanics.

## 3.3 Validity of Newton's Axioms

The validity of Newton's axioms has been extensively tested and confirmed by experiments. Within the domain of classical mechanics, they provide an accurate description of the motion of macroscopic bodies when their speeds are much smaller than the speed of light and quantum effects can be neglected.

The first and second axioms form the basis for relating forces to the motion of a particle, while the third axiom expresses a fundamental property of interactions between two bodies. In its usual form,

$$\mathbf{F}_{12} = -\mathbf{F}_{21},$$

the third axiom holds for central interactions, for which the force between two particles acts along the line joining them and depends only on their separation. Newtonian gravitational and electrostatic forces are important examples. By contrast, forces such as magnetic forces, friction, and fluid drag are non-central: their direction is not determined solely by the line joining the interacting bodies. Nevertheless, non-central forces do not necessarily violate the third axiom; in interactions involving fields, the momentum of the field must also be taken into account.

The axioms therefore constitute the postulates of Newtonian mechanics: their validity is established not by derivation from more fundamental principles within classical mechanics, but by their agreement with experimental observations within their range of applicability.

## 3.4 Measurement of Mass

An *isolated system* is a system on which the net external force is zero. Internal forces may still act between the bodies within the system. Therefore, for an isolated system consisting of $N$ bodies, the sum of the internal forces acting on all bodies is zero:

$$\sum_{i=1}^{N} \mathbf{F}_i = 0. \tag{3.11}$$

For a system consisting of two bodies, Newton's third axiom gives

$$\mathbf{F}_1 = -\mathbf{F}_2. \tag{3.12}$$

Using Newton's second axiom,

$$m_1\mathbf{a}_1 = -m_2\mathbf{a}_2, \tag{3.13}$$

and therefore, taking magnitudes,

$$\frac{m_2}{m_1} = \frac{|\mathbf{a}_1|}{|\mathbf{a}_2|}. \tag{3.14}$$

Thus, if one of the masses is chosen as a reference mass, the other mass can be determined by measuring the accelerations of the two bodies.

There is another way to measure mass. Near the surface of the Earth, an object is acted upon by the gravitational force, called its *weight*. Neglecting air resistance, all objects at the same location have the same gravitational acceleration, denoted by $\mathbf{g}$. Newton's second axiom therefore gives

$$\mathbf{W} = m\mathbf{g}, \tag{3.15}$$

where $\mathbf{W}$ is the weight of the object. Taking magnitudes,

$$|\mathbf{W}| = m|\mathbf{g}|. \tag{3.16}$$

Thus, by comparing the weights of two bodies at the same location, their masses are related by

$$\frac{m_2}{m_1} = \frac{|\mathbf{W}_2|}{|\mathbf{W}_1|}, \tag{3.17}$$

or equivalently,

$$m_2 = \frac{|\mathbf{W}_2|}{|\mathbf{W}_1|}m_1. \tag{3.18}$$

The equality of inertial and gravitational mass is an experimental fact in Newtonian mechanics and is known as the *equivalence of inertial and gravitational mass*. This equivalence plays a central role in Einstein's theory of gravity, where it is elevated to the *equivalence principle*.

## 3.5 Equations of Motion

The 2nd axiom provides the *equation of motion* of a body once the force acting on it is specified. It is a differential equation that determines how the position of the body changes with time under the influence of a force. In general,

$$\dot{\mathbf{p}} = \mathbf{F}(\mathbf{r}, \mathbf{v}, t), \tag{3.19}$$

where $\mathbf{r}$ is the position vector, $\mathbf{v}$ is the velocity vector, and $t$ is time. The force may depend on position, velocity, and time.

For a body of constant mass, $\mathbf{p} = m\mathbf{v}$, so the equation of motion takes the familiar form

$$m\ddot{\mathbf{r}} = \mathbf{F}(\mathbf{r}, \dot{\mathbf{r}}, t). \tag{3.20}$$

This is a second-order differential equation for the position vector $\mathbf{r}(t)$. Together with the initial position and velocity, its solution determines the motion of the body.

### 3.5.1 Angular-Momentum Form of the 2nd Axiom

The 2nd axiom also has a useful form associated with rotational motion. In translational motion, the momentum $\mathbf{p}$ characterizes the motion of a body, while the force $\mathbf{F}$ determines how its momentum changes. There is a corresponding description of motion about a reference point: the angular momentum $\mathbf{L}$ characterizes the motion of a body about that point, while the torque $\boldsymbol{\tau}$ determines how its angular momentum changes. Thus, angular momentum and torque play roles in rotational motion analogous to those of momentum and force in translational motion.

The angular momentum of a particle about a reference point $O$ is defined as

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}, \tag{3.21}$$

where $\mathbf{r}$ is the position vector of the particle relative to $O$. The units of angular momentum are kilogram meter squared per second (kg·m$^2$/s). The torque of a force about the same point is defined as

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}. \tag{3.22}$$

These definitions reflect the fact that a force can affect not only the translational motion of a body but also its tendency to rotate about a reference point. The torque depends on both the magnitude and direction of the force and on its position relative to the reference point.

Taking the time derivative of the angular momentum and using the 2nd axiom gives

$$\frac{d\mathbf{L}}{dt} = \frac{d\mathbf{r}}{dt} \times \mathbf{p} + \mathbf{r} \times \frac{d\mathbf{p}}{dt}. \tag{3.23}$$

For a particle of constant mass, $\mathbf{p} = m\mathbf{v}$, so $\mathbf{v}$ and $\mathbf{p}$ are parallel and therefore

$$\mathbf{v} \times \mathbf{p} = 0. \tag{3.24}$$

Hence,

$$\boxed{\frac{d\mathbf{L}}{dt} = \boldsymbol{\tau}}. \tag{3.25}$$

Thus, the 2nd axiom can be written in two analogous forms,

$$\boxed{\frac{d\mathbf{p}}{dt} = \mathbf{F}}, \qquad \boxed{\frac{d\mathbf{L}}{dt} = \boldsymbol{\tau}}. \tag{3.26}$$

The first describes the change of linear momentum caused by a force, while the second describes the change of angular momentum caused by a torque. These relations will be used in the following section to formulate the conservation laws.

## 3.6 Conservation Theorems

In this section, we will discuss the direct consequences of Newton's axioms of motion, which are the conservation theorems. These theorems state that certain physical quantities remain constant in time for an isolated system; this is what we mean by *conservation*. The conservation theorems are important because they provide a powerful tool for analyzing the motion of systems without having to solve the equations of motion directly. The conservation

theorems are derived from Newton's axioms and are valid for all isolated systems, regardless of the specific forces acting within the system.

### 3.6.1 Conservation of Linear Momentum

The linear momentum of a system is conserved if the net external force acting on the system is zero:

$$\dot{\mathbf{p}} = \mathbf{F}_{\text{ext}} = 0 \implies \mathbf{p} = \text{constant}, \tag{3.27}$$

The system here can be a single particle or a collection of particles. The conservation of linear momentum say that it does not matter how asingle particle moves in space, or how the particles move in space and interact with each other, as long as there is no net external force acting on the system the total linear momentum of the system remains constant.

A corrolary of the conservation of linear momentum is that if the it is conserved in a system, so all components of linear momentum vector are conserved. This means that if the system is isolated, the total linear momentum in the $x$-direction, $y$-direction, and $z$-direction are all conserved separately. To see this, let $s$ be any arbitrary direction in space, and let $\hat{\mathbf{s}}$ be the unit vector in that direction. The component of the linear momentum along this direction is given by

$$p_s = \mathbf{p} \cdot \hat{\mathbf{s}}. \tag{3.28}$$

Taking the time derivative gives

$$\dot{p}_s = \dot{\mathbf{p}} \cdot \hat{\mathbf{s}} = \mathbf{F}_{\text{ext}} \cdot \hat{\mathbf{s}} = F_{\text{ext},s}, \tag{3.29}$$

where $F_{\text{ext},s}$ is the component of the net external force along the direction $\hat{\mathbf{s}}$. If the net external force is zero, then $F_{\text{ext},s} = 0$, and therefore $\dot{p}_s = 0$. This means that the component of the linear momentum along any arbitrary direction is conserved. Since the $x$-direction, $y$-direction, and $z$-direction are just three specific directions in space, it follows that the components of linear momentum along these directions are also conserved separately.

### 3.6.2 Conservation of Angular Momentum

The angular momentum of a system is conserved if the net external torque acting on the system is zero.

$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\tau}_{\text{ext}} = 0 \implies \mathbf{L} = \text{constant}, \tag{3.30}$$

### 3.6.3 Work and Kinetic Energy

When a particle is acted upon by a force, $\mathbf{F}$, the *work* done by the force as the particle moves from position $\mathbf{r}_1$ to position $\mathbf{r}_2$ is defined as

$$W_{12} = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r}. \tag{3.31}$$

where $d\mathbf{r}$ is an infinitesimal displacement vector along the path of the particle from $\mathbf{r}_1$ to $\mathbf{r}_2$ and it can be expressed as

$$d\mathbf{r} = dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k} = \mathbf{v}dt, \tag{3.32}$$

If $\mathbf{F}$ is the only force acting on the particle, then we can write the integrant of the work integral as

$$\mathbf{F}\cdot d\mathbf{r} = \frac{d\mathbf{p}}{dt}\cdot d\mathbf{r} = \frac{d\mathbf{p}}{dt}\cdot\mathbf{v}dt = \frac{d\mathbf{p}}{dt}\cdot\frac{\mathbf{p}}{m}dt = \frac{1}{m}\mathbf{p}\cdot d\mathbf{p} = \frac{1}{2m}d(\mathbf{p}\cdot\mathbf{p}) = d\left(\frac{1}{2}mv^2\right) = dK, \tag{3.33}$$

where $K = \frac{1}{2}mv^2$ is defined as the *kinetic energy* of the particle. Therefore, the work done by the force as the particle moves from $\mathbf{r}_1$ to $\mathbf{r}_2$ can also be expressed as the change in kinetic energy of the particle:

$$W_{12} = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r} = \int_{K_1}^{K_2} dK = K_2 - K_1 = \Delta K. \tag{3.34}$$

From the differential point of view, if a force $\mathbf{F}$ acts on a particle along an infinitesimal path $d\mathbf{r}$, it changes its kinetic energy by an infinitesimal amount $dK$:

$$dK = \mathbf{F} \cdot d\mathbf{r}. \tag{3.35}$$

Both equations, Eq. (**??**) and Eq. (**??**), are known as the *work-energy theorem*. They state that the work done by a force on a particle is equal to the change in its kinetic energy. This theorem provides a powerful tool for analyzing the motion of particles, especially when the forces involved are complex or difficult to describe in terms of acceleration.

It is worth empasizing that a positive work done on a particle by a force increases its kinetic energy, while a negative work done by a force decreases its kinetic energy. If the work done by the force is zero, then the kinetic energy of the particle remains constant. This can happen if the force is perpendicular to the displacement of the particle, as in uniform circular motion, or if there is no net force acting on the particle.

The unit of work and kinetic energy is the joule (J), which is defined as one newton-meter (N·m). In terms of base SI units, one joule is equal to one kilogram meter squared per second squared ($\mathrm{kg{\cdot}m^2/s^2}$).

### 3.6.4 Conservative Forces and Potential Energy

A *conservative force* is a force for which the work done in moving a particle between two points is independent of the path taken. Thus, the work depends only on the initial and final positions of the particle. Equivalently, the work done by a conservative force around any closed path is zero:

$$\oint \mathbf{F} \cdot d\mathbf{r} = 0. \tag{3.36}$$

This property allows us to define a scalar function of position called the *potential energy*, $U(\mathbf{r})$, such that the work done by the conservative force in moving the particle from point $A$ to point $B$ is equal to the negative change in potential energy:

$$W_{AB} = \int_{\mathbf{r_A}}^{\mathbf{r_B}} \mathbf{F} \cdot d\mathbf{r} = -\Delta U = U(\mathbf{r}_A) - U(\mathbf{r}_B). \tag{3.37}$$

For simplicity, consider motion in the $x$ direction only. Then,

$$W_{AB} = \int_{x_A}^{x_B} F_x dx = U(x_A) - U(x_B). \tag{3.38}$$

Since this relation must hold for arbitrary $x_A$ and $x_B$, differentiating with respect to the upper limit gives

$$F_x = -\frac{dU}{dx}. \tag{3.39}$$

In three dimensions, this generalizes to

$$\mathbf{F} = -\nabla U, \tag{3.40}$$

where $\nabla U$ is the gradient of the potential energy:

$$\nabla U = \frac{\partial U}{\partial x}\mathbf{i} + \frac{\partial U}{\partial y}\mathbf{j} + \frac{\partial U}{\partial z}\mathbf{k}. \tag{3.41}$$

Thus, the force is determined by how the potential energy varies in space, while the work between two points depends only on the values of $U$ at those points. The zero of potential energy is arbitrary: adding a constant to $U$ does not affect the force or the work:

$$\mathbf{F} = -\nabla(U + C) = -\nabla U \tag{3.42}$$

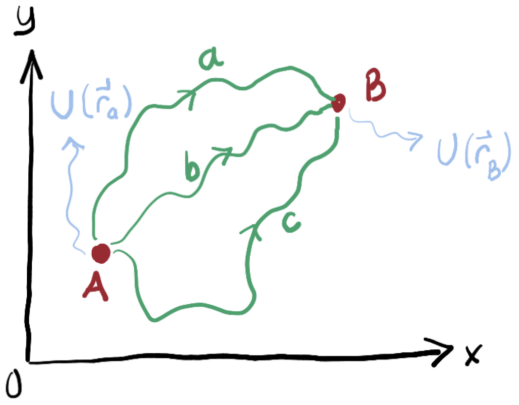



Figure 3.1: For a conservative force, the work done in moving a particle between two points is independent of the path taken; it depends only on the initial and final positions. Therefore, the work done around any closed path is zero.

### 3.6.5 Conservation of Mechanical Energy

The *mechanical energy* of a particle is defined as the sum of its kinetic energy and potential energy:

$$E = K + U, \tag{3.43}$$

where $K$ is the kinetic energy and $U$ is the potential energy. A very important consequence of Newton's axioms is that the mechanical energy of a particle is conserved if the only forces acting on it are conservative forces. In other words, if a particle is acted upon only by conservative forces, then its total mechanical energy remains constant. We will show this mathematically as follows:

$$\frac{dE}{dt} = \frac{dK}{dt} + \frac{dU}{dt} \tag{3.44}$$

The first term in the summation can be written as

$$\frac{dK}{dt} = \frac{d}{dt}\left(\frac{1}{2}m\mathbf{v}\cdot\mathbf{v}\right) = m\mathbf{v}\cdot\dot{\mathbf{v}} = m\mathbf{v}\cdot\mathbf{a} = \mathbf{F}\cdot\mathbf{v} = \mathbf{F}\cdot\dot{\mathbf{r}} \tag{3.45}$$

For the second term, we will consider the most general case where the potential energy is a function of position and time, $U(\mathbf{r}, t)$. Then, using the chain rule, we can write

$$\frac{dU}{dt} = \sum_{i=1}^{3}\frac{\partial U}{\partial x_i}\frac{dx_i}{dt} + \frac{\partial U}{\partial t} = \sum_{i=1}^{3}\frac{\partial U}{\partial x_i}\dot{x}_i + \frac{\partial U}{\partial t} = \nabla U\cdot\dot{\mathbf{r}} + \frac{\partial U}{\partial t}. \tag{3.46}$$

Therefore, we can write the time derivative of the mechanical energy as

$$\frac{dE}{dt} = \mathbf{F} \cdot \dot{\mathbf{r}} + \nabla U \cdot \dot{\mathbf{r}} + \frac{\partial U}{\partial t} = (\mathbf{F} + \nabla U) \cdot \dot{\mathbf{r}} + \frac{\partial U}{\partial t}. \tag{3.47}$$

If the force is conservative, then $\mathbf{F} = -\nabla U$, so the first term in the summation vanishes, and we are left with

$$\frac{dE}{dt} = \frac{\partial U}{\partial t}. \tag{3.48}$$

If the potential energy is independent of time, then $\frac{\partial U}{\partial t} = 0$, and we have

$$\frac{dE}{dt} = 0 \implies E = \text{constant}. \tag{3.49}$$

Thus, the mechanical energy of a particle is conserved if the only forces acting on it are conservative forces and the potential energy is independent of time. This is known as the *conservation of mechanical energy*. It is a powerful tool for analyzing the motion of particles, especially in situations where the forces involved are complex or difficult to describe in terms of acceleration.

## 3.7 More on the Potential Energy

Knowing the potential energy function $U(\mathbf{r})$ together with the total mechanical energy $E$ of a particle provides us with a powerful tool for analyzing its motion without solving the equations of motion directly.

Let us consider the following example. Suppose a particle of mass $m$ is moving in one dimension under the influence of a conservative force with potential energy $U(x) = k(x^3 - 3x)$ as given in Fig. **??**, where $k$ is a positive constant.

First, note that the kinetic energy of a particle, $K = \frac{1}{2}mv^2$, is always non-negative, since the mass $m$ is positive and the square of the speed $v^2$ is non-negative. Therefore, for a particle with conserved total mechanical energy $E$, we must have

$$K = E - U(x) \geq 0 \implies E \geq U(x). \tag{3.50}$$

Thus, the motion of a particle with conserved total mechanical energy $E$ is restricted to regions where $E \geq U(x)$. Regions where $U(x) > E$ are called *classically forbidden regions*, since the particle would need negative kinetic energy to reach them, which is impossible in classical mechanics. Therefore, $E_1$ is the minimum energy for a particle to have in existence of the potential $U(x)$ as seen in Fig. **??**.

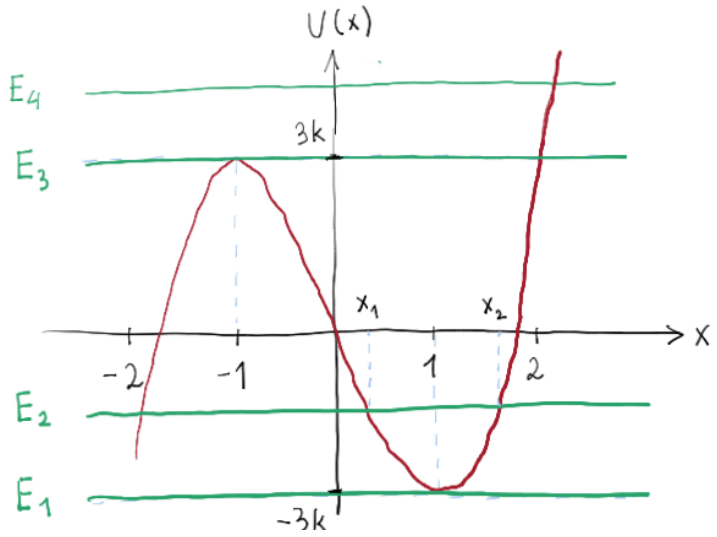



Figure 3.2: Potential energy function $U(x) = k(x^3 - 3x)$ for a particle in one dimension.

In general, at positions where $E = U(x)$, the kinetic energy vanishes, and the particle is momentarily at rest. If the force at such a position is nonzero, the particle reverses its direction of motion, and the position is called a *turning point*. On the other hand, if the force vanishes, the particle is at an *equilibrium position* and can remain at rest.

If the total energy of the particle is $E_1$, then the particle is at rest at the equilibrium position $x = 1$. For $E_2$, the particle can move between the two turning points $x_1$ and $x_2$, where $E_2 = U(x_1) = U(x_2)$. Even if we do not know the exact function $x(t)$, wecan still say that the particle oscillates between $x_1$ and $x_2$ around the equilibrium position $x = 1$. For energy $E_3$, the particle's motion if bounded by point $x = 2$ on the right. As particle moves to the left to reach point $x = -1$, it's velocity decreases with an ever decreasing rate so that it takes an infinite amount of time to reach $x = -1$ under ideal conditions. Practically, we would see it still at point $x = -1$ after a long time, but it would never reach that point. For energy $E_4$, even though the motion of the particle is bounded from the right, it can move freely to the left forever without any restrictions.

## 3.8 Stable and Unstable Equilibrium

A point of equilibrium is said to be *stable* if a small displacement from the equilibrium position results in a restoring force that tends to bring the par-

ticle back to the equilibrium position. Conversely, an equilibrium point is *unstable* if a small displacement from the equilibrium position results in a force that tends to move the particle further away from the equilibrium position.

Mathematically, a point of equilibrium $x_0$ is stable if the potential energy function $U(x)$ has a local minimum at that point, and it is unstable if $U(x)$ has a local maximum at that point. In other words, if $U''(x_0) > 0$, the equilibrium is stable, and if $U''(x_0) < 0$, the equilibrium is unstable.



Figure 3.3: Stable and unstable equilibrium points. We may think of a ball placed in a concave-up potential well as being in stable equilibrium, while a ball placed on top of a concave-down potential hill is in unstable equilibrium. However, this analogy is not perfect, since the ball can roll away from the equilibrium point in any direction, while a particle in one-dimensional motion can only move along a line as given in Fig. **??**.

## 3.9 Impulse and Power

The *impulse* of a force $\mathbf{F}$ acting on a particle over a time interval from $t_1$ to $t_2$ is defined as

$$\mathbf{J} = \int_{t_1}^{t_2} \mathbf{F} dt. \tag{3.51}$$

The impulse is a vector quantity that has the same direction as the force. It represents the change in momentum of the particle due to the force acting over the time interval. From Newton's second axiom, we have

$$\mathbf{J} = \int_{t_1}^{t_2} \frac{d\mathbf{p}}{dt} dt = \int_{t_1}^{t_2} d\mathbf{p} = \mathbf{p}_2 - \mathbf{p}_1 = \Delta\mathbf{p}, \tag{3.52}$$

where $\mathbf{p}_1$ and $\mathbf{p}_2$ are the momenta of the particle at times $t_1$ and $t_2$, respectively. Thus, the impulse of a force is equal to the change in momentum of the particle. This relationship is known as the *impulse-momentum theorem*. It provides a useful way to analyze the effects of forces that act over short time intervals, such as collisions or explosions.

If we know the change in momentum of a particle and the time interval over which the change occurs, we can calculate the average force acting on the particle during that time interval. As it is geometrically shown in Fig. **??**, the average force is can be written as

$$\mathbf{F}_{\text{avg}} = \frac{\Delta \mathbf{p}}{\Delta t} = \frac{\mathbf{p}_2 - \mathbf{p}_1}{t_2 - t_1}. \tag{3.53}$$

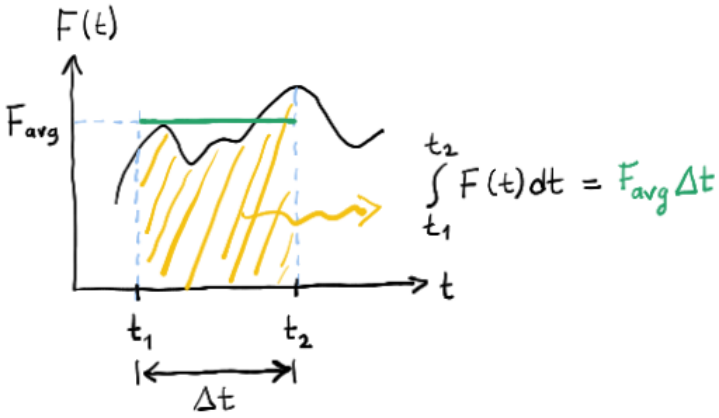



Figure 3.4: The average force acting on a particle over a time interval $\Delta t$ is equal to the change in momentum (area under the force-time curve) of the particle divided by the time interval.

The *power* is defined as the rate at which work is done by a force on a particle. According to the situation considered, it can also be interpreted as the rate at which energy is transferred/transformed to or from a particle/system. It is measured in watts (W), where one watt is equal to one joule per second (J/s).

If a force $\mathbf{F}$ acts on a particle that moves with velocity $\mathbf{v}$, the instantaneous power delivered to the particle is given by

$$P = \frac{dW}{dt} = \mathbf{F} \cdot \frac{d\mathbf{r}}{dt} = \mathbf{F} \cdot \mathbf{v}. \tag{3.54}$$

The power can be positive or negative, depending on the direction of the force relative to the velocity of the particle. If the force is in the same direction as the velocity, the power is positive, indicating that energy is being transferred to the particle. If the force is in the opposite direction, the power is negative, indicating that energy is being taken away from the particle.

## 3.10 Mechanical Energy as a First Integral of Motion

The conservation of mechanical energy can be used to obtain a first-order equation governing the motion of a particle. If the mechanical energy of a particle is conserved, then

$$E = K + U = \frac{1}{2}mv^2 + U(x) = \text{constant}. \tag{3.55}$$

This equation is called a *first integral of motion* because it is obtained by integrating the second-order equation of motion once with respect to time. In this sense, energy conservation provides a first-order form of the equation of motion. This equation can be rearranged to give the speed of the particle as a function of position:

$$v(x) = \sqrt{\frac{2}{m}\left[E - U(x)\right]}. \tag{3.56}$$

For one-dimensional motion, we have

$$v = \frac{dx}{dt}, \tag{3.57}$$

so that

$$dt = \frac{dx}{\pm\sqrt{\frac{2}{m}\left[E - U(x)\right]}}. \tag{3.58}$$

Integrating from $(x_0, t_0)$ to $(x, t)$ gives

$$t - t_0 = \int_{x_0}^{x} \frac{dx}{\pm\sqrt{\frac{2}{m}\left[E - U(x)\right]}}. \tag{3.59}$$

Thus, if the potential energy $U(x)$ is known, this equation can, in principle, be used to determine the motion $x(t)$. The integral may not have an analytical solution for an arbitrary potential, but it can be evaluated numerically.

For motion in more than one dimension, conservation of energy determines the speed of the particle but not the direction of its velocity. Therefore, the energy equation alone is not sufficient to determine the full trajectory $\mathbf{r}(t)$.

# 3.11 Worked Examples

## 3.11.1 Motion of a Particle under a Constant Force

Assume that a particle of mass $m$ is acted upon by a constant force in $\mathbf{F}$. Write down the equation of motion and solve it to find the position and velocity of the particle as functions of time.

**Solution:** If the force is constant, then motion takes place in a straight line. The equation of motion is given by Newton's second law can be written in a scalar form as

$$m\ddot{x} = F, \tag{3.60}$$

where $x$ is the position of the particle along the direction of the force. It can be solved for velocity as

$$d\dot{x} = \frac{F}{m}dt \implies \int d\dot{x} = \int \frac{F}{m}dt + C \implies \dot{x} = \frac{F}{m}t + C, \tag{3.61}$$

where $C$ is the constant of integration. If the initial velocity of the particle at $t = 0$ is $v_0$, then $C = \dot{x}_0$, and we have

$$\dot{x}(t) \equiv v_x(t) = \frac{F}{m}t + v_0. \tag{3.62}$$

Integrating once more gives the position of the particle as a function of time:

$$dx = \left(\frac{F}{m}t + v_0\right)dt \implies \int dx = \int \left(\frac{F}{m}t + v_0\right)dt \implies x(t) = \frac{F}{2m}t^2 + v_0t + x_0, \tag{3.63}$$

where $x_0$ is the initial position of the particle at $t = 0$. Thus, the position, velocity and acceleration of the particle as functions of time are given by

$$\boxed{x(t) = \frac{F}{2m}t^2 + v_0t + x_0, \quad v_x(t) = \frac{F}{m}t + v_0, \quad a_x(t) = \frac{F}{m} = \text{constant}} \tag{3.64}$$

which are plotted in Fig. **??**.

## 3.11.2 Free fall Motion

A particle is released from a height $h$ and falls under the influence of gravity freely from rest, then $F = mg$, $v_0 = 0$, and $x_0 = h$.

(a) Write down the equation of motion and its solution to find the position and velocity of the particle as functions of time.

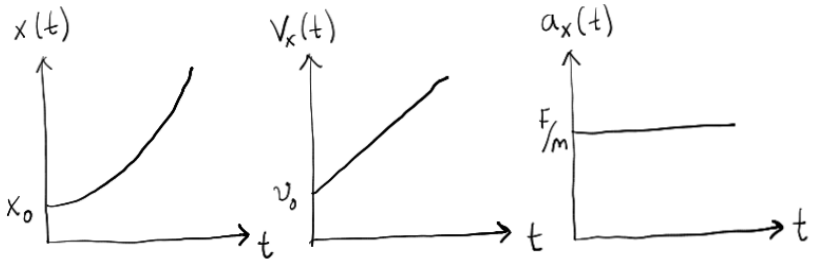



Figure 3.5: Motion of a particle under a constant force. The position, velocity, and acceleration of the particle as functions of time.

(b) Find the speed of the particle when it crushes into the ground.

**Solution:**

(a) The equation of motion is given by Newton's second law can be written in a scalar form as since it takes place in a straight line along the vertical direction, we have

$$m\ddot{y} = -mg \implies \ddot{y} = -g, \tag{3.65}$$

where $y$ is the position of the particle along the vertical direction. This problem is the same as Example **??**, with $F = -mg$, $v_0 = 0$, and $x_0 = h$. Therefore, the position and velocity of the particle as functions of time are given by

$$\boxed{y(t) = -\frac{1}{2}gt^2 + h, \quad v_y(t) = -gt, \quad a_y(t) = -g = \text{constant}} \tag{3.66}$$

(b) The particle crushes into the ground when $y(t) = 0$, which gives

$$0 = -\frac{1}{2}gt^2 + h \implies t = \sqrt{\frac{2h}{g}}. \tag{3.67}$$

The speed of the particle when it crushes into the ground is given by

$$v_y\left(\sqrt{\frac{2h}{g}}\right) = -g\sqrt{\frac{2h}{g}} = -\sqrt{2gh} \implies \boxed{v = \sqrt{2gh}} \tag{3.68}$$

### 3.11.3 Motion on an Inclined Plane and Friction Forces

A block is placed on a frictionless inclined plane that makes an angle $\theta$ with the horizontal. The block is released from rest at the top of the incline as shown in Fig. **??**. Write down the equation of motion, and solve it to find the position and velocity of the block for the following cases:

(a) The block slides down the frictionless incline.

(b) The block slides down the incline on a rough surface with a coefficient of kinetic friction $\mu_k$.

(c) Assume that we can increase the angle of the incline $\theta$ until the block starts sliding down the incline. Find the critical angle $\theta_c$ at which the block starts sliding down the incline given the coefficient of static friction $\mu_s$.

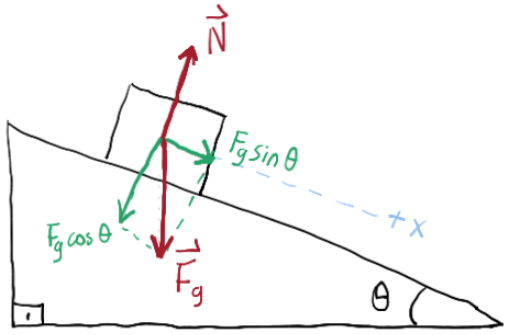



Figure 3.6: A block on a frictionless inclined plane.

**Solution:**

(a) For a frictionless incline, the only force acting on the block along the incline is the component of its weight parallel to the incline, which is $mg\sin\theta$ as seen in Fig. **??**. This force causes the block to accelerate down the incline (in $+x$ direction) and does not change in time. Therefore, this problem can be considered as a motion of a particle under a constant force as considered in Example **??**. The acceleration of the block along the incline is given by

$$a_x = \frac{F}{m} = \frac{mg\sin\theta}{m} = g\sin\theta. \tag{3.69}$$

Therefore, the position and velocity of the block as functions of time are given by

$$\boxed{x(t) = \frac{1}{2}g\sin\theta t^2, \quad v_x(t) = g\sin\theta t, \quad a_x(t) = g\sin\theta = \text{constant}} \tag{3.70}$$

Note that the acceleration of the block along the incline is less than $g$ because only a component of the weight acts along the incline. The acceleration is maximum when $\theta = 90°$, in which case $a_x = g$. The form of the

equations of motion is the same as that for free fall, but the acceleration is reduced by a factor of $\sin\theta$. Therefore, the motion on an inclined plane can be considered as a "slowed-down version of free fall".

(b) When the surfaces are rough, the block experiences a *kinetic friction force* because the block is sliding relative to the surface. Kinetic friction acts opposite to the direction of sliding. Thus, since the block moves down the incline, the kinetic friction force acts up the incline.

The magnitude of the kinetic friction force is described by the empirical relation

$$f_k = \mu_k N, \tag{3.71}$$

where $N$ is the normal force exerted by the surface on the block. The dimensionless quantity $\mu_k$ is called the *coefficient of kinetic friction*. It characterizes the strength of the frictional interaction between the two surfaces: for a given normal force, a larger $\mu_k$ corresponds to a larger kinetic friction force.

The relation $f_k = \mu_k N$ means that, within the kinetic-friction model, the magnitude of the friction force is proportional to the normal force. The normal force measures how strongly the two surfaces press against each other. It is important to note that $\mu_k$ is a property of the pair of contacting surfaces and is not itself a force. The normal force is equal to the component of the weight perpendicular to the incline, which is $mg\cos\theta$. Therefore, the force of friction is $f_k = \mu_k mg\cos\theta$. The net force along the incline is then

$$F_{net} = mg\sin\theta - \mu_k mg\cos\theta. \tag{3.72}$$

The acceleration of the block along the incline is given by

$$a_x = \frac{F_{net}}{m} = g(\sin\theta - \mu_k\cos\theta). \tag{3.73}$$

This is a constant acceleration, so the position and velocity of the block as functions of time can be found by using Eq. (**??**) with $a_x = g(\sin\theta - \mu_k\cos\theta)$.

(c) Before the block starts sliding, it is at rest relative to the incline. The friction force is therefore *static friction*. Static friction acts along the surface and prevents relative motion between the two surfaces. Unlike kinetic friction, its magnitude is not generally fixed at a particular value; it adjusts to balance the other forces, up to a maximum value given by

$$f_s \leq f_{s,\max} = \mu_s N, \tag{3.74}$$

where $\mu_s$ is the *coefficient of static friction* and $N$ is the normal force. The coefficient $\mu_s$ is dimensionless and characterizes the strength of static friction between the two surfaces.

As the angle $\theta$ of the incline is increased, the component of the gravitational force parallel to the incline,

$$mg\sin\theta, \tag{3.75}$$

also increases. The block remains at rest as long as static friction can balance this component. At the critical angle $\theta_c$, the block is just about to start sliding down the incline. At this point, static friction has reached its maximum value:

$$f_s = f_{s,\max} = \mu_s N. \tag{3.76}$$

The normal force is determined by the component of the gravitational force perpendicular to the incline. Since there is no acceleration perpendicular to the surface,

$$N = mg\cos\theta_c. \tag{3.77}$$

At the critical angle, the component of gravity down the incline is exactly balanced by the maximum static friction acting up the incline:

$$mg\sin\theta_c = \mu_s N. \tag{3.78}$$

Substituting $N = mg\cos\theta_c$ gives

$$mg\sin\theta_c = \mu_s mg\cos\theta_c \implies \tan\theta_c = \mu_s. \tag{3.79}$$

Therefore, the critical angle at which the block starts sliding down the incline is

$$\boxed{\theta_c = \arctan(\mu_s)}. \tag{3.80}$$

For $\theta < \theta_c$, static friction is sufficient to keep the block at rest. At $\theta = \theta_c$, static friction reaches its maximum value and the block is on the verge of sliding. For $\theta > \theta_c$, the block starts sliding down the incline, and the friction force becomes kinetic friction.

### 3.11.4 Mass-Pulley System

Two blocks of unequal masses are connected by a string over a smooth pulley as shown in Fig. **??**. If the coefficient of kinetic friction is $\mu_k$ between the heavier block and the surface, write down the equations of motion and solve them to find the acceleration of the blocks and the tension in the string.

**Solution:**

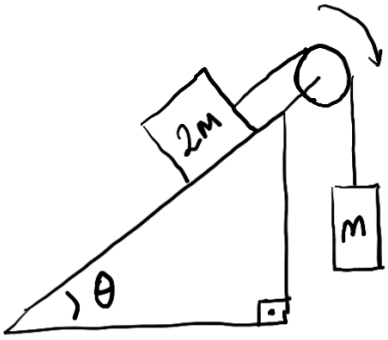



Figure 3.7: Two blocks of unequal masses connected by a string over a smooth pulley. The heavier block is on a horizontal surface with kinetic friction, while the lighter block hangs vertically.

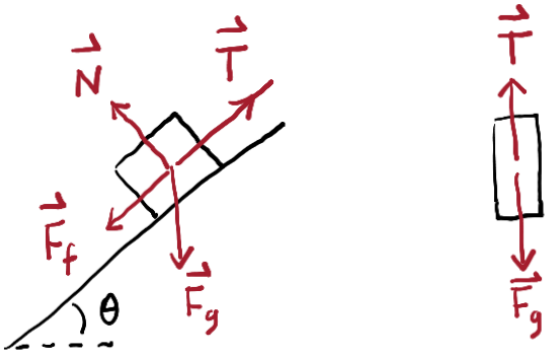



Figure 3.8: Free-body diagrams of the two blocks in the mass-pulley system. The forces acting on each block are shown, including the tension in the string, the weight of each block, and the kinetic friction acting on the heavier block.

### 3.11.5 A "More Realistic" Free Fall and Terminal Speed

A particle is released from a height $h$ with initial velocity $v_0 = 0$ and falls under the influence of gravity. The particle experiences a resistive force due to air resistance, which is proportional to its velocity: $F_r = -kv$, where $k$ is a positive constant that characterizes the strength of the resistive force.

(a) Write down the equation of motion and solve it to find the velocity of the particle as function of time.

(b) Determine the terminal speed of the particle.

**Solution:**

(a) The equation of motion is given by Newton's second law can be written in a scalar form as

$$m\ddot{y} = -mg - kv \implies m\frac{dv}{dt} = -mg - kv, \tag{3.81}$$

where $y$ is the position of the particle along the vertical direction. This is a first-order linear ordinary differential equation for the velocity $v(t)$. It can be solved using an integrating factor or by separation of variables. Rearranging gives

$$\frac{dv}{dt} + \frac{k}{m}v = -g. \tag{3.82}$$

The integrating factor is $e^{(k/m)t}$, so multiplying both sides gives

$$e^{(k/m)t}\frac{dv}{dt} + \frac{k}{m}e^{(k/m)t}v = -ge^{(k/m)t}. \tag{3.83}$$

The left-hand side can be recognized as the derivative of $ve^{(k/m)t}$, so we have

$$\frac{d}{dt}\left(ve^{(k/m)t}\right) = -ge^{(k/m)t}. \tag{3.84}$$

Integrating both sides with respect to $t$ gives

$$ve^{(k/m)t} = -\frac{mg}{k}e^{(k/m)t} + C, \tag{3.85}$$

where $C$ is the constant of integration. Solving for $v(t)$ gives

$$v(t) = -\frac{mg}{k} + Ce^{-(k/m)t}. \tag{3.86}$$

Using the initial condition $v(0) = 0$, we find $C = \frac{mg}{k}$, so the solution is

$$v(t) = -\frac{mg}{k}\left(1 - e^{-(k/m)t}\right). \tag{3.87}$$

The plot of $v(t)$ is shown in Fig. **??**.

(b) The terminal speed is the speed at which the net force on the particle becomes zero, meaning that the resistive force balances the gravitational force. Setting $m\frac{dv}{dt} = 0$ gives

$$0 = -mg - kv_{\text{terminal}} \implies v_{\text{terminal}} = -\frac{mg}{k}. \tag{3.88}$$

The negative sign indicates that the terminal speed is directed downward. The magnitude of the terminal speed is

$$\boxed{v_{\text{terminal}} = \frac{mg}{k}}, \tag{3.89}$$

which agrees with the long-time limit of Eq. (**??**) as $t \to \infty$.

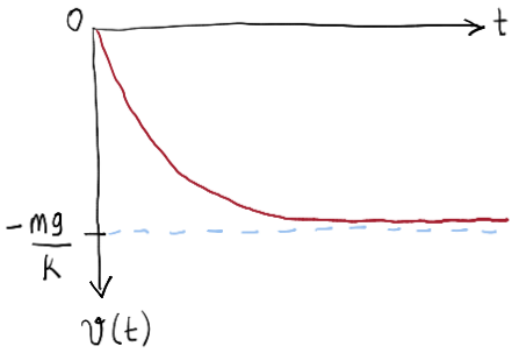



Figure 3.9: Velocity of a particle falling under gravity with air resistance proportional to its velocity. The velocity approaches the terminal speed $v_{\text{terminal}} = \frac{mg}{k}$ as $t \to \infty$.

### 3.11.6 Projectile Motion

A particle is projected from the ground with an initial speed $v_0$ at an angle $\theta$ above the horizontal.

(a) Find the time of flight ($t_{\text{flight}}$) of the particle.

(b) Find the maximum height ($y_{\max}$) reached by the particle.

(c) Find the horizontal range ($x_{\max}$) of the particle.

(d) Find the trajectory equation $y(x)$ of the particle.

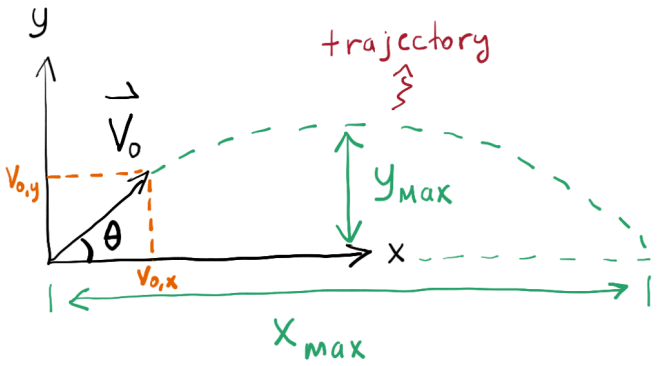



Figure 3.10: Projectile motion of a particle launched with initial speed $v_0$ at an angle $\theta$ above the horizontal. The motion can be analyzed by decomposing the initial velocity into horizontal and vertical components.

**Solution:**

The motion can be divided into horizontal and vertical components. The horizontal motion is uniform, i.e., the horizontal velocity is constant, while the vertical motion is uniformly accelerated due to gravity. The initial velocity can be decomposed into horizontal and vertical components:

$$v_{0x} = v_0 \cos\theta, \quad v_{0y} = v_0 \sin\theta. \tag{3.90}$$

(a) To determine the time of flight, we analyze the vertical motion. The particle is acted upon only by the force of gravity; yet again, this is a constant force problem as considered in Ex. **??**. Therefore, the vertical acceleration is $a_y = -g$. The vertical velocity as a function of time is given by

$$v_y(t) = v_{0y} - gt. \tag{3.91}$$

To find the time of flight, we can find the time where the velocity becomes zero at the maximum height, and then double it to account for the descent back to the ground. Setting $v_y(t) = 0$ gives

$$0 = v_{0y} - gt_{\text{up}} \implies t_{\text{up}} = \frac{v_{0y}}{g} = \frac{v_0 \sin\theta}{g}. \tag{3.92}$$

The total time of flight is then

$$\boxed{t_{\text{flight}} = 2t_{\text{up}} = \frac{2v_0 \sin\theta}{g}}. \tag{3.93}$$

(b) We know that the maximum height is reached when the vertical velocity becomes zero. The height at that time can be found using the vertical motion equation by using Eq. (**??**) with $a_y = -g$, $v_{0y} = v_0 \sin\theta$, and $y_0 = 0$:

$$y(t) = v_{0y}t - \frac{1}{2}gt^2. \tag{3.94}$$

Using the vertical motion equation, we can find the maximum height:

$$y_{\max} = v_{0y}t_{\text{up}} - \frac{1}{2}gt_{\text{up}}^2 = v_0 \sin\theta \cdot \frac{v_0 \sin\theta}{g} - \frac{1}{2}g\left(\frac{v_0 \sin\theta}{g}\right)^2. \tag{3.95}$$

$$\boxed{y_{\max} = \frac{v_0^2 \sin^2\theta}{2g}}. \tag{3.96}$$

(c) The horizontal range can be found by determining the horizontal distance traveled during the total time of flight. The horizontal motion is uniform, so the horizontal velocity is constant:

$$v_x = v_{0x} = v_0 \cos\theta. \tag{3.97}$$

The horizontal range is then

$$x_{\max} = v_{x0} t_{\text{flight}} = v_0 \cos\theta \cdot \frac{2v_0 \sin\theta}{g}. \tag{3.98}$$

Using the double angle identity $\sin 2\theta = 2\sin\theta\cos\theta$, we can simplify the expression:

$$\boxed{x_{\max} = \frac{v_0^2 \sin 2\theta}{g}}. \tag{3.99}$$

(d) The trajectory equation $y(x)$ can be found by eliminating time $t$ from the horizontal and vertical motion equations. From the horizontal motion, we have

$$x(t) = v_{0x} t \implies t = \frac{x}{v_{0x}} = \frac{x}{v_0 \cos\theta}. \tag{3.100}$$

Substituting this expression for $t$ into the vertical motion equation gives

$$y(x) = v_{0y}\left(\frac{x}{v_0\cos\theta}\right) - \frac{1}{2}g\left(\frac{x}{v_0\cos\theta}\right)^2 = x\tan\theta - \frac{gx^2}{2v_0^2\cos^2\theta}. \tag{3.101}$$

Thus, the trajectory equation is

$$\boxed{y(x) = x\tan\theta - \frac{gx^2}{2v_0^2\cos^2\theta}}. \tag{3.102}$$

Assigning $A = \frac{g}{2v_0^2\cos^2\theta}$, and $B = \tan\theta$, we can write the trajectory equation as

$$\boxed{y(x) = Bx - Ax^2} \tag{3.103}$$

which is a quadratic equation in $x$, representing a parabolic trajectory.

### 3.11.7 Force law from a given speed function

A particle of mass $m$ has speed $v(x) = \alpha/x$ where $\alpha$ is a positive constant and $x$ is the position of the particle along a straight line. Find the force responsible for this motion as a function of position $x$.

**Solution:**

$$F = ma = m\frac{dv}{dt} = m\frac{dv}{dx}\frac{dx}{dt} = mv\frac{dv}{dx} = m\left(\frac{\alpha}{x}\right)\left(-\frac{\alpha}{x^2}\right) = -\frac{m\alpha^2}{x^3}. \tag{3.104}$$

### 3.11.8 Circular motion

A particle of mass $m$ moves in a circular path of radius $r$ with constant speed $v$. The motion is uniform circular motion.

(a) Write down the acceleration of the particle and its direction. What is the corresponding force acting on the particle?

(b) Calculate the acceleration of the particle by using geometry; that is, by using the change in the velocity vector over a small time interval $dt$.

(c) Instead of considering two very close velocity vectors seperated by the time interval $dt$, consider two velocity vectors seperated by a finite time interval so that the velocity vectors will be at the opposite ends of the diameter of the circle. Calculate the change in the velocity vector and the corresponding acceleration. Compare it with the result obtained in part (b).

(c) What is the kinetic energy of the particle?

(d) Assume that we have $N$ particles of different masses $m_i$ $(i = 1, 2, \ldots, N)$ moving in circular paths of different radii $r_i$ with different constant speeds $v_i$. Write down the total kinetic energy of the system of particles.

(e) What if the particles have the same angular frequency $\omega$? Write down the total kinetic energy of the system of particles in terms of $\omega$ and $r_i$.

(f) What is the angular momentum of the particle? Is there any torque acting on the particle? Express the angular momentum in terms of the moment of inertia $I$ and angular frequency $\omega$.

**Solution:** (a) As we have seen in Kinematics, the position vector of a particle in circular motion is given by

$$\mathbf{r}(t) = r\cos(\omega t)\hat{i} + r\sin(\omega t)\hat{j}, \tag{3.105}$$

where $r$ is the radius of the circle and $\omega$ is the angular frequency. The second derivative of the position vector with respect to time gives the acceleration:

$$\mathbf{a}(t) = -\omega^2 r\cos(\omega t)\hat{i} - \omega^2 r\sin(\omega t)\hat{j} = -\omega^2\mathbf{r}(t). \tag{3.106}$$

The acceleration is directed toward the center of the circle, which is called centripetal acceleration, and its magnitude is given by

$$a = \omega^2 r = \frac{v^2}{r}, \tag{3.107}$$

where $v = \omega r$ is the speed of the particle. The direction of the acceleration is always toward the center of the circle, which is perpendicular to the velocity vector.

The corresponding force acting on the particle is given by Newton's second law:

$$\mathbf{F} = m\mathbf{a} = -m\omega^2\mathbf{r}(t) = -\frac{mv^2}{r}\hat{r}, \tag{3.108}$$

where $\hat{r}$ is the unit vector pointing from the particle toward the center of the circle. The force is called the *centripetal force*, and it is responsible for keeping the particle in circular motion. The magnitude of the centripetal force is given by

$$F = \frac{mv^2}{r}. \tag{3.109}$$

(b) Let us consider two velocity vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ at two different points on the circle separated by a small angle $d\theta$ as shown in the Fig. **??**. Since we consider a very small angle $d\theta$, the arc length is approximately equal to the length of $|d\mathbf{v}|$. The change in the velocity vector is given by

$$dv \approx vd\theta \implies a = \frac{dv}{dt} \approx \frac{vd\theta}{dt} = \frac{v}{R}\frac{(Rd\theta)}{dt} = \frac{v}{R}\frac{ds}{dt} = \frac{v^2}{R}, \tag{3.110}$$

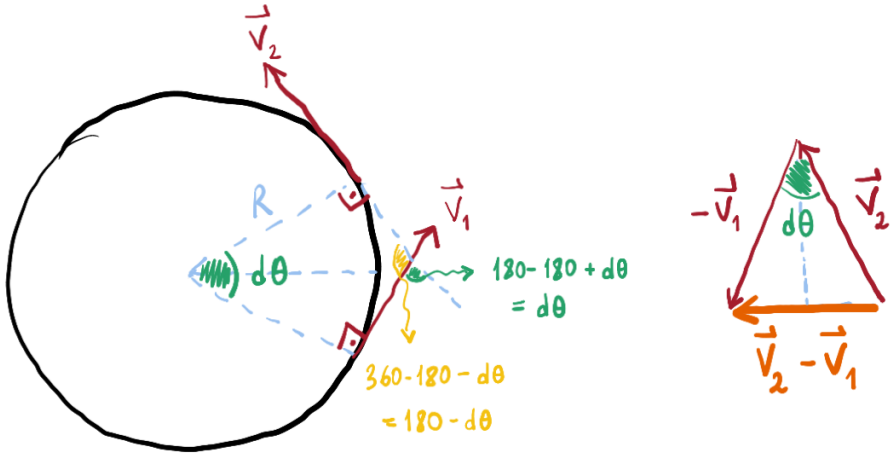



Figure 3.11: Velocity vectors of a particle in circular motion at two different points on the circle separated by a small angle $d\theta$. The change in the velocity vector $\Delta\mathbf{v}$ is directed toward the center of the circle.

(c) Let us consider two velocity vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ at two different points on the circle separated by an angle of 180°. The change in the velocity vector is given by

$$\Delta\mathbf{v} = \mathbf{v}_2 - \mathbf{v}_1 = -\mathbf{v}_1 - \mathbf{v}_1 = -2\mathbf{v}_1. \tag{3.111}$$

The magnitude of the change in the velocity vector is given by

$$|\Delta \mathbf{v}| = 2v. \tag{3.112}$$

The time interval between the two velocity vectors is given by

$$\Delta t = \frac{T}{2} = \frac{\pi}{\omega}, \tag{3.113}$$

where $T = 2\pi/\omega$ is the period of the circular motion. Therefore, the magnitude of the acceleration is given by

$$a = \frac{|\Delta \mathbf{v}|}{\Delta t} = \frac{2v}{\pi/\omega} = \frac{2\omega v}{\pi} = \frac{2\omega^2 r}{\pi}. \tag{3.114}$$

This is the average acceleration over the time interval $\Delta t$. The direction of the acceleration is toward the center of the circle, which is consistent with the result obtained in part (b). The magnitude of the acceleration obtained in part (b) is $a = \omega^2 r$, which is the instantaneous acceleration at any point on the circle. The average acceleration over a finite time interval is less than the instantaneous acceleration, but the direction is the same. This is something that we would expect since some acceleration vectors are cancelled out during the averaging.

(b) The kinetic energy of the particle is given by

$$K = \frac{1}{2} m v^2. \tag{3.115}$$

which can also be expressed in terms of the angular frequency as

$$K = \frac{1}{2} m (\omega r)^2 = \frac{1}{2} m \omega^2 r^2. \tag{3.116}$$

(c) The total kinetic energy of the system of $N$ particles is given by the sum of the kinetic energies of each particle:

$$K_{\text{total}} = \sum_{i=1}^{N} \frac{1}{2} m_i v_i^2 = \sum_{i=1}^{N} \frac{1}{2} m_i (\omega_i r_i)^2 = \sum_{i=1}^{N} \frac{1}{2} m_i \omega_i^2 r_i^2. \tag{3.117}$$

(d) If the particles have the same angular frequency $\omega$, then the total kinetic energy of the system can be expressed as

$$K_{\text{total}} = \sum_{i=1}^{N} \frac{1}{2} m_i \omega^2 r_i^2 = \frac{1}{2} \omega^2 \left( \sum_{i=1}^{N} m_i r_i^2 \right). \tag{3.118}$$

The term in the parantheses is called the *moment of inertia* of the system of particles about the axis of rotation, denoted by $I$:

$$I = \sum_{i=1}^{N} m_i r_i^2. \tag{3.119}$$

It is a measure of how the mass is distributed with respect to the axis of rotation; the larger the scattering of the masses from the axis, the larger the moment of inertia. The assumption of the same angular frequency $\omega$ is equivalent to assuming that the system of particles is rigidly rotating about the axis of rotation. Therefore, the total kinetic energy of a rigidly rotating system of particles can be expressed in terms of its moment of inertia and angular frequency as

$$K_{\text{total}} = \frac{1}{2} I\omega^2. \tag{3.120}$$

This is the rotational analog of the kinetic energy of a particle moving in a straight line, $K = \frac{1}{2}mv^2$ where $I$ plays the role of mass $m$ and $\omega$ plays the role of speed $v$. It directly follows from Eq. **??** that the moment of inertia of a single particle of mass $m$ moving in a circular path of radius $r$ is given by

$$I = mr^2. \tag{3.121}$$

(e) The angular momentum of the particle about the center of the circle is given by

$$\mathbf{L} = \mathbf{r} \times m\mathbf{v} = mr^2\omega\mathbf{k}, \tag{3.122}$$

where $\mathbf{k}$ is the unit vector perpendicular to the plane of motion. The torque acting on the particle is given by

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = \mathbf{r} \times (-m\omega^2\mathbf{r}) = 0, \tag{3.123}$$

since the force is directed toward the center of the circle and is therefore parallel to the position vector. Thus, there is no torque acting on the particle, and the angular momentum is conserved.

We can express the angular momentum of the particle undergoing a uniform circular motion in terms of the moment of inertia $I$ and angular frequency $\omega$,

$$\mathbf{L} = I\omega\mathbf{k}. \tag{3.124}$$

This form of angular momentum is the rotational analog of the linear momentum of a particle moving in a straight line, $\mathbf{p} = m\mathbf{v}$, where $I$ plays the role of mass $m$ and $\omega$ plays the role of speed $v$.

### 3.11.9 Elastic collision of two hockey pucks – 1

Two hockey pucks with identical masses are on a flat, horizontal ice hockey rink. Black puck is motionless and white is moving 2.5 m/s toward the black puck. The two pucks collide. After the collision, the black puck moves at 2.5 m/s in the same direction as the white puck was moving.

(a) What is the speed of the white puck after the collision?

(b) Is the energy conserved in this collision? If so, what is the total kinetic energy before and after the collision?

**Solution:**

(a) Let $m$ be the mass of each puck, $v_{1i}$ and $v_{2i}$ be the initial velocities of the white and black pucks, respectively, and $v_{1f}$ and $v_{2f}$ be their final velocities after the collision. The conservation of momentum gives us:

$$mv_{1i} + mv_{2i} = mv_{1f} + mv_{2f}. \tag{3.125}$$

Substituting the known values:

$$m(2.5) + m(0) = mv_{1f} + m(2.5). \tag{3.126}$$

Dividing through by $m$ and simplifying:

$$2.5 = v_{1f} + 2.5 \implies v_{1f} = 0. \tag{3.127}$$

Thus, the speed of the white puck after the collision is $\boxed{0\text{m/s}}$.

(b) The total kinetic energy before the collision is:

$$K_i = \frac{1}{2}mv_{1i}^2 + \frac{1}{2}mv_{2i}^2 = \frac{1}{2}m(2.5)^2 + \frac{1}{2}m(0)^2 = \frac{1}{2}m(6.25) = 3.125m. \tag{3.128}$$

The total kinetic energy after the collision is:

$$K_f = \frac{1}{2}mv_{1f}^2 + \frac{1}{2}mv_{2f}^2 = \frac{1}{2}m(0)^2 + \frac{1}{2}m(2.5)^2 = \frac{1}{2}m(6.25) = 3.125m. \tag{3.129}$$

Since $K_i = K_f$, the energy is conserved in this collision, and the total kinetic energy before and after the collision is $\boxed{3.125m}$. If the energy is conserved before and after the collision, then the collision is called an *elastic collision* meaning that no energy was lost to heat, sound, or deformation of the pucks.

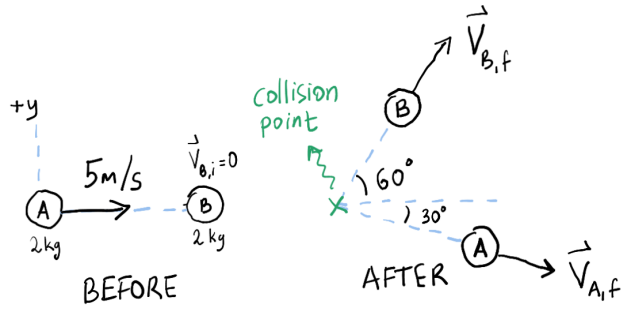



Figure 3.12: aa

### 3.11.10 Elastic collision of two hockey pucks – 2

Two hockey pucks with identical masses collide as shown in Fig. **??**. Find the final velocities of the two pucks after the collision.

**Solution:** The momentum vector has to stay the same before and after the collision. Before the collision, the momentum of the system is

$$\mathbf{p}_i = 2\text{kg} \times 5\mathbf{i}\frac{\text{m}}{\text{s}} = 10\mathbf{i}\frac{\text{kg m}}{\text{s}} \implies \mathbf{p}_f = 10\mathbf{i}\frac{\text{kg m}}{\text{s}}. \tag{3.130}$$

We know the directions of the two pucks after the collision, so we can decompose the $\mathbf{p}_f$ into its components along these directions as shown in Fig. **??**. Therefore, we have

$$|\mathbf{p}_{A,f}| = 10\cos(30°) = 10\frac{\sqrt{3}}{2} = 5\sqrt{3}\frac{\text{kg m}}{\text{s}}. \tag{3.131}$$

$$|\mathbf{p}_{B,f}| = 10\sin(30°) = 10\frac{1}{2} = 5\frac{\text{kg m}}{\text{s}}. \tag{3.132}$$

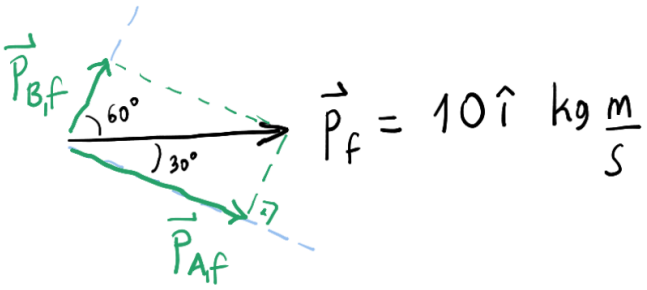



Figure 3.13: aa

### 3.11.11 Inelastic collision of two hockey pucks

A hockey puck of mass $m_1 = 2\,\text{kg}$ moves with velocity $v_1 = 5\,\text{m/s}$ toward a stationary hockey puck of mass $m_2 = 3\,\text{kg}$. Assume that the collision is one-dimensional and that the external impulse during the collision is negligible. The two pucks collide and stick together.

(a) Find the final velocity of the combined pucks.

(b) Determine the mechanical energy lost in the collision.

(c) If instead the pucks separate after the collision, and half of the initial kinetic energy is lost, find their final velocities.

**Solution:**

**(a)** When two objects collide and stick together, the collision is called a *perfectly inelastic collision*. The total momentum is conserved when the external impulse is negligible. Since there is no defined potential energy, the mechanical energy of the system is solely kinetic energy. Kinetic energy is not conserved: some of it is converted into internal energy, such as heating and deformation.

Choose the positive $x$-direction along the initial motion of the first puck. Conservation of momentum gives

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f. \tag{3.133}$$

Since $v_2 = 0$, the common final velocity is

$$v_f = \frac{m_1 v_1}{m_1 + m_2} = \frac{(2\,\text{kg})(5\,\text{m/s})}{2\,\text{kg} + 3\,\text{kg}} = \boxed{2\,\text{m/s}}. \tag{3.134}$$

The combined pucks move in the initial direction of the first puck.

**(b)** The initial kinetic energy is

$$K_i = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 = \frac{1}{2}(2\,\text{kg})(5\,\text{m/s})^2 = 25\,\text{J}. \tag{3.135}$$

The final kinetic energy is

$$K_f = \frac{1}{2}(m_1 + m_2) v_f^2 = \frac{1}{2}(5\,\text{kg})(2\,\text{m/s})^2 = 10\,\text{J}. \tag{3.136}$$

Therefore, the kinetic energy lost is

$$K_{\text{lost}} = K_i - K_f = 25\,\text{J} - 10\,\text{J} = \boxed{15\,\text{J}}. \tag{3.137}$$

This energy is converted into other forms; it is not destroyed. The change in kinetic energy is $\Delta K = K_f - K_i = -15\,\mathrm{J}$.

**(c)** Half of the initial kinetic energy is lost, so

$$K_f = \frac{1}{2}K_i = 12.5\,\mathrm{J}. \tag{3.138}$$

Let $v_{1f}$ and $v_{2f}$ be the signed final velocities. Conservation of momentum and the specified final kinetic energy give

$$m_1 v_{1f} + m_2 v_{2f} = m_1 v_1, \tag{3.139}$$

$$\frac{1}{2}m_1 v_{1f}^2 + \frac{1}{2}m_2 v_{2f}^2 = 12.5\,\mathrm{J}. \tag{3.140}$$

From the momentum equation,

$$v_{2f} = \frac{m_1 v_1 - m_1 v_{1f}}{m_2} = \frac{10\,\mathrm{m/s} - 2v_{1f}}{3}. \tag{3.141}$$

Substituting this expression into the kinetic energy equation gives

$$(1\,\mathrm{kg})v_{1f}^2 + \frac{3}{2}\,\mathrm{kg}\left(\frac{10\,\mathrm{m/s} - 2v_{1f}}{3}\right)^2 = 12.5\,\mathrm{J}. \tag{3.142}$$

Completing the square yields

$$10\,\mathrm{J} + \frac{5}{3}\,\mathrm{kg}\,(v_{1f} - 2\,\mathrm{m/s})^2 = 12.5\,\mathrm{J}. \tag{3.143}$$

Therefore,

$$v_{1f} = \left(2 \pm \frac{\sqrt{6}}{2}\right)\mathrm{m/s}, \qquad v_{2f} = \left(2 \mp \frac{\sqrt{6}}{3}\right)\mathrm{m/s}. \tag{3.144}$$

To select the physically relevant solution, note that the first puck approaches the second puck from behind. After the collision, the pucks must separate, which requires

$$v_{2f} > v_{1f}. \tag{3.145}$$

Thus,

$$\boxed{v_{1f} = \left(2 - \frac{\sqrt{6}}{2}\right)\mathrm{m/s} \approx 0.775\,\mathrm{m/s}} \tag{3.146}$$

and

$$\boxed{v_{2f} = \left(2 + \frac{\sqrt{6}}{3}\right)\mathrm{m/s} \approx 2.816\,\mathrm{m/s}}. \tag{3.147}$$

The other algebraic solution has $v_{1f} > v_{2f}$ and corresponds to approaching rather than separating pucks, so it cannot describe the state immediately after this collision.

Both final velocities are positive. The first puck slows down, while the second puck moves forward faster than the first.

### 3.11.12 Conservation of angular momentum

A ceiling fan with moment of inertia $I$ and radius $R$ is freely rotating with angular velocity $\omega_i$. A mouse of mass $m$ jumps onto the outer edge of the fan with zero tangential velocity relative to the ground and remains there. Assuming that no external torque acts about the fan's rotation axis, determine the ratio of the final angular velocity to the initial angular velocity of the fan.

**Solution:** Since no external torque acts about the rotation axis, the total angular momentum of the system is conserved. Initially, only the fan contributes to the angular momentum because the mouse has zero tangential velocity. Therefore,

$$L_i = I\omega_i. \tag{3.148}$$

After the mouse lands on the fan, they rotate together with angular velocity $\omega_f$. Since the mouse can be treated as a point mass at a distance $R$ from the rotation axis, its moment of inertia is $mR^2$. The final angular momentum is therefore

$$L_f = (I + mR^2)\omega_f. \tag{3.149}$$

From conservation of angular momentum,

$$L_i = L_f \implies I\omega_i = (I + mR^2)\omega_f. \tag{3.150}$$

Solving for the ratio of the final to the initial angular velocity, we obtain

$$\boxed{\frac{\omega_f}{\omega_i} = \frac{I}{I + mR^2}}. \tag{3.151}$$

Since $I/(I + mR^2) < 1$, the angular velocity decreases after the mouse lands on the fan. This is because the total moment of inertia increases while the total angular momentum remains constant.

### 3.11.13 Energy and Conservation of Angular Momentum – 1

An ice skater with moment of inertia $I$ and angular velocity $\omega_i$ is spinning with her arms extended. The skater then pulls her arms in, reducing her

moment of inertia to $I/2$. Assuming that no external torque acts about her rotation axis,

(a) Determine the final angular velocity of the skater.

(b) Determine whether the rotational kinetic energy is conserved. If not, calculate the change in rotational kinetic energy and explain its physical origin.

**Solution:**

(a) Since no external torque acts about the rotation axis, the angular momentum of the skater is conserved. Initially, the skater has moment of inertia $I$ and angular velocity $\omega_i$, so the initial angular momentum is $L_i = I\omega_i$. After pulling her arms in, her moment of inertia becomes $I/2$, and her final angular velocity is $\omega_f$. The final angular momentum is therefore $L_f = (I/2)\omega_f$.

From conservation of angular momentum, we have

$$L_i = L_f \implies I\omega_i = \frac{I}{2}\omega_f. \tag{3.152}$$

Solving for the final angular velocity, we obtain

$$\boxed{\omega_f = 2\omega_i}. \tag{3.153}$$

Therefore, the skater's angular velocity doubles when her moment of inertia is reduced by half.

(b) The initial rotational kinetic energy of the skater is

$$K_i = \frac{1}{2}I\omega_i^2. \tag{3.154}$$

After pulling her arms in, her moment of inertia is reduced to $I/2$, while her angular velocity increases to $\omega_f = 2\omega_i$. Therefore, the final rotational kinetic energy is

$$K_f = \frac{1}{2}\left(\frac{I}{2}\right)\omega_f^2 = \frac{1}{2}\left(\frac{I}{2}\right)(2\omega_i)^2 = I\omega_i^2. \tag{3.155}$$

Comparing the initial and final kinetic energies, we find

$$\boxed{K_f = 2K_i}. \tag{3.156}$$

Thus, the rotational kinetic energy is not conserved; it doubles. The change in rotational kinetic energy is

$$\Delta K = K_f - K_i = I\omega_i^2 - \frac{1}{2}I\omega_i^2 = \boxed{\frac{1}{2}I\omega_i^2}. \tag{3.157}$$

The increase in rotational kinetic energy comes from the work performed by the skater's muscles while pulling her arms inward. Therefore, although angular momentum is conserved in the absence of external torque, rotational kinetic energy need not be conserved when the moment of inertia changes.

### 3.11.14 Energy and Conservation of Angular Momentum – 2

A uniform thin rod of mass $M = 3\,\text{kg}$ and length $L = 2\,\text{m}$ rotates counterclockwise in a horizontal plane about a frictionless vertical axis passing through its center. Its initial angular velocity is $\omega_i = 4\,\text{rad/s}$. A small lump of clay of mass $m = 1\,\text{kg}$ moves horizontally with a speed of $v = 6\,\text{m/s}$ and strikes one end of the rod. At the moment of impact, the clay moves tangentially to the circular path of the rod's end, in the clockwise direction, and sticks to the rod.

The moment of inertia of a uniform thin rod about an axis passing through its center and perpendicular to its length is $I = ML^2/12$.

(a) Determine the angular velocity of the rod immediately after the collision, including its direction of rotation.

(b) Calculate the mechanical energy lost during the collision.

(c) Determine the minimum speed of the clay required to reverse the direction of rotation of the rod.

**Solution:**

(a) Since no external torque acts about the rotation axis, the total angular momentum of the rod and clay is conserved during the collision.

We choose the counterclockwise direction as positive. Initially, the angular momentum of the system is the sum of the angular momentum of the rod and the clay:

$$L_i = I\omega_i - mv\frac{L}{2}. \tag{3.158}$$

The minus sign appears because the clay moves in the clockwise direction, opposite to the initial rotation of the rod.

The moment of inertia of the rod is

$$I = \frac{ML^2}{12} = \frac{3 \cdot 2^2}{12} = 1\,\text{kg}\,\text{m}^2. \tag{3.159}$$

After the collision, the clay remains attached to the end of the rod. Therefore, the total moment of inertia becomes

$$I_f = I + m\left(\frac{L}{2}\right)^2. \tag{3.160}$$

Since the rod and clay rotate together with angular velocity $\omega_f$, the final angular momentum is

$$L_f = \left(I + \frac{mL^2}{4}\right)\omega_f. \tag{3.161}$$

Applying conservation of angular momentum,

$$I\omega_i - \frac{mvL}{2} = \left(I + \frac{mL^2}{4}\right)\omega_f. \tag{3.162}$$

Solving for the final angular velocity,

$$\omega_f = \frac{I\omega_i - mvL/2}{I + mL^2/4}. \tag{3.163}$$

Substituting the numerical values,

$$\omega_f = \frac{1\cdot 4 - 1\cdot 6\cdot 1}{1 + 1\cdot 1^2} = \boxed{-1\,\text{rad/s}}. \tag{3.164}$$

The negative sign indicates that the rod reverses its direction of rotation and rotates clockwise after the collision.

(b) The initial kinetic energy of the system is the sum of the rotational kinetic energy of the rod and the translational kinetic energy of the clay:

$$K_i = \frac{1}{2}I\omega_i^2 + \frac{1}{2}mv^2. \tag{3.165}$$

Substituting the numerical values,

$$K_i = \frac{1}{2}(1)(4^2) + \frac{1}{2}(1)(6^2) = 8 + 18 = 26\,\text{J}. \tag{3.166}$$

After the collision, the rod and clay rotate together. Their total kinetic energy is

$$K_f = \frac{1}{2}\left(I + \frac{mL^2}{4}\right)\omega_f^2. \tag{3.167}$$

Substituting the numerical values,

$$K_f = \frac{1}{2}(1+1)(-1)^2 = 1\,\text{J}. \tag{3.168}$$

The mechanical energy lost during the collision is therefore

$$\Delta K = K_i - K_f = 26 - 1 = \boxed{25\,\mathrm{J}}. \tag{3.169}$$

Although angular momentum is conserved, kinetic energy is not conserved because the collision is perfectly inelastic. Part of the initial mechanical energy is converted into other forms, such as internal energy and deformation.

(c) To reverse the direction of rotation, the final angular velocity must be negative:

$$\omega_f < 0. \tag{3.170}$$

Since the denominator in the expression for $\omega_f$ is always positive, this condition requires

$$I\omega_i - \frac{mvL}{2} < 0. \tag{3.171}$$

Rearranging, we obtain

$$v > \frac{2I\omega_i}{mL}. \tag{3.172}$$

Using $I = ML^2/12$, this becomes

$$v > \frac{ML\omega_i}{6m}. \tag{3.173}$$

Substituting the numerical values,

$$v > \frac{3 \cdot 2 \cdot 4}{6 \cdot 1} = 4\,\mathrm{m/s}. \tag{3.174}$$

Therefore, the threshold speed is

$$\boxed{v_c = 4\,\mathrm{m/s}}. \tag{3.175}$$

At exactly $v = v_c$, the total initial angular momentum vanishes, and the rod comes to rest immediately after the collision. For $v > v_c$, the rod reverses its direction of rotation.

### 3.11.15 Scalar fields and vector fields

Given the potential $U(x) = ax^2 + by^2$, find the corresponding force field $\mathbf{F}(x, y)$.

**Solution:** The force field is given by $\mathbf{F}(x, y) = -\nabla U(x, y)$. Computing the gradient, we get

$$\mathbf{F}(x, y) = -\left(\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j}\right) = -\left(2ax\hat{i} + 2by\hat{j}\right). \tag{3.176}$$

Thus, the force field is

$$\boxed{\mathbf{F}(x, y) = -2ax\hat{i} - 2by\hat{j}}. \tag{3.177}$$

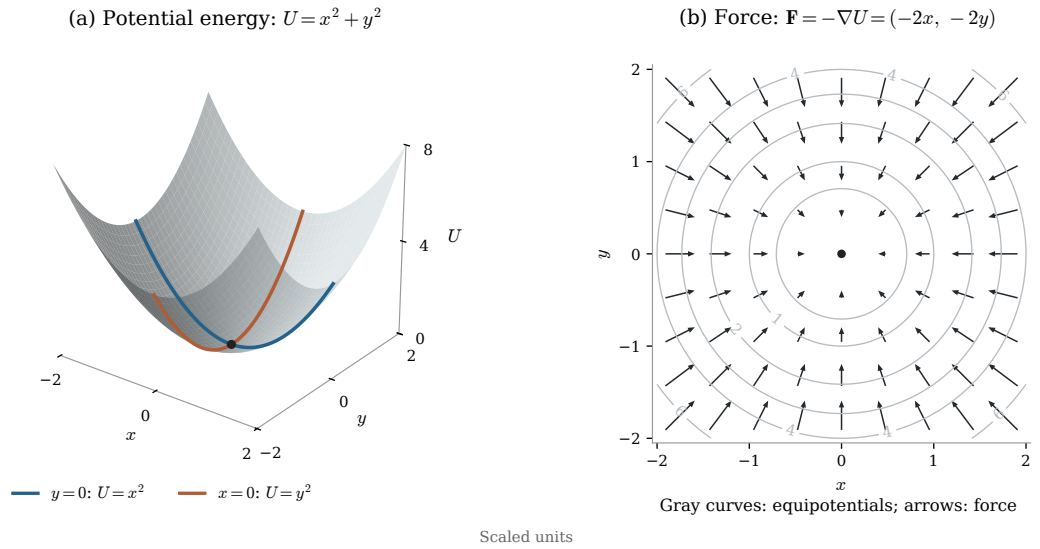



Figure 3.14: Potential energy $U = x^2 + y^2$ and its force field $\mathbf{F} = -\nabla U = (-2x, -2y)$, in scaled units. The force points inward, perpendicular to the circular equipotential curves, and its magnitude increases linearly with distance from the origin. The origin is a stable equilibrium.

### 3.11.16 Potential energy for the motion close to the surface of the earth

A particle of mass $m$ is raised to a height $h$ above the ground. Assume that the gravitational force is constant near the surface of the earth.

(a) Show that gravitational force is conservative.

(b) Determine the work done by the gravitational force on the particle as it moves from the ground to a height $h$.

(c) How much work is done by us in raising the particle to a height $h$?

**Solution:**

(a) When the distance from the Earth's surface is much smaller than the Earth's radius, the gravitational field can be approximated as uniform. In this approximation, the gravitational force acting on a particle of mass $m$ is given by

$$\mathbf{F} = -mg\hat{\mathbf{z}}, \tag{3.178}$$

where $g$ is the gravitational acceleration and $\hat{\mathbf{z}}$ is the unit vector pointing vertically upward. Recall that a force is conservative if the work done in

moving a particle between two points is independent of the path taken. To verify this property for gravity, consider a particle moving from an initial position $A$ to a final position $B$ along an arbitrary path $C$.

The work done by the gravitational force is

$$W_{AB} = \int_C \mathbf{F} \cdot d\mathbf{r}, \tag{3.179}$$

where the infinitesimal displacement vector is

$$d\mathbf{r} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}}. \tag{3.180}$$

Substituting the gravitational force into the work integral, we obtain

$$\begin{aligned} W_{AB} &= \int_C (-mg\hat{\mathbf{z}}) \cdot d\mathbf{r} \\ &= -mg \int_C \hat{\mathbf{z}} \cdot d\mathbf{r} \\ &= -mg \int_C dz \\ &= -mg \int_{z_A}^{z_B} dz. \end{aligned} \tag{3.181}$$

Evaluating the integral gives

$$\boxed{W_{AB} = -mg(z_B - z_A).} \tag{3.182}$$

Notice that the work done by gravity depends only on the initial and final heights, $z_A$ and $z_B$, and not on the particular path followed by the particle. Therefore, the gravitational force is conservative.

In particular, if the particle moves along a closed path, its initial and final positions coincide, so $z_A = z_B$. Consequently,

$$\boxed{\oint_C \mathbf{F} \cdot d\mathbf{r} = 0,} \tag{3.183}$$

which is another defining property of a conservative force. Note that the assumption of a uniform gravitational field is an approximation valid near the Earth's surface. We will see that the full Newtonian gravitational force, $\mathbf{F} = -GMm\hat{\mathbf{r}}/r^2$, is also conservative.

(b) The work done by the gravitational force on the particle as it moves from the ground ($z_A = 0$) to a height $h$ ($z_B = h$) is

$$W_{AB} = -mg(h - 0) = -mgh. \tag{3.184}$$

(c) The work done by us in raising the particle to a height $h$ is the negative of the work done by gravity:

$$W_{us} = -W_{AB} = mgh. \tag{3.185}$$

### 3.11.17 Horsepower

In British engineering units, the unit of power is the horsepower (hp). One horsepower is defined as the power required to lift 550 pounds by 1 foot in 1 second. In SI units, it is approximately (according to DIN; German Institute for Standardization) defined as the power required to lift 75 kg by 1 meter in 1 second.

(a) Write 1 horsepower in SI units.

(b) If a car has a 200 hp engine, what is the maximum power output of the engine in watts?

**Solution:** (a) The work done in lifting 75 kg by 1 meter is

$$W = mgh = (75 \text{ kg})(9.8 \text{ m/s}^2)(1 \text{ m}) = 735 \text{ J}. \tag{3.186}$$

Since this work is done in 1 second, the power is

$$P = \frac{W}{t} = \frac{735 \text{ J}}{1 \text{ s}} = 735 \text{ W}. \tag{3.187}$$

Therefore, 1 horsepower is approximately $\boxed{735 \text{ W}}$.

(b) The maximum power output of a 200 hp engine is

$$P_{\max} = 200 \text{ hp} \times 735 \text{ W/hp} = 147,000 \text{ W} = 147\text{kW}. \tag{3.188}$$

Therefore, the maximum power output of the engine is $\boxed{147\text{kW}}$. In other words, a 200 hp engine can deliver a maximum energy of 147 kJ in 1 second.

### 3.11.18 Impulse

A block of mass crashes into a wall blabla...

### 3.11.19 Infinite time required to reach point $x = -1$

### 3.11.20 Stable and Unstable Equilibrium Points – 1

Find the stable and unstable equilibrium points of the potential energy function $U(x) = k(x^3 - 3x)$ given in Sec. **??** and justify your answers mathematically using the second derivative of the potential energy.

**Solution:** The equilibrium points are found by setting the force, $F(x) = -dU/dx$, equal to zero. Equivalently, we set the first derivative of the potential energy to zero:

$$\frac{dU}{dx} = k(3x^2 - 3) = 3k(x^2 - 1) = 3k(x - 1)(x + 1) = 0. \tag{3.189}$$

This gives us two equilibrium points: $x = -1$ and $x = 1$.

To determine the stability of each equilibrium point, we compute the second derivative of the potential energy:

$$U''(x) = \frac{d^2U}{dx^2} = 6kx. \tag{3.190}$$

An equilibrium point is stable if $U''(x) > 0$, corresponding to a local minimum of the potential energy, and unstable if $U''(x) < 0$, corresponding to a local maximum.

At $x = -1$, we have

$$U''(-1) = -6k < 0, \tag{3.191}$$

so the equilibrium at $x = -1$ is unstable.

At $x = 1$, we have

$$U''(1) = 6k > 0, \tag{3.192}$$

so the equilibrium at $x = 1$ is stable.

### 3.11.21 Stable and Unstable Equilibrium Points – 2

A particle of mass $m$ moves along the $x$-axis under the influence of a conservative force described by the potential energy function

$$U(x) = U_0 \left[ \left(\frac{x}{a}\right)^4 - 2\left(\frac{x}{a}\right)^2 \right], \tag{3.193}$$

where $U_0 > 0$ and $a > 0$ are constants.

(a) Determine the force $F(x)$ acting on the particle as a function of its position.

(b) Find all equilibrium positions of the particle, i.e., the positions at which the net force vanishes.

(c) Determine which equilibrium positions are stable and which are unstable. Justify your answers mathematically using the second derivative of the potential energy.

**Solution:** (a) The force is given by $F(x) = -\frac{dU}{dx}$. Taking the derivative, we get

$$F(x) = -U_0 \left[ 4\left(\frac{x}{a}\right)^3 - 4\left(\frac{x}{a}\right) \right] = 4U_0 \left(\frac{x}{a}\right) \left[ \left(\frac{x}{a}\right)^2 - 1 \right]. \tag{3.194}$$

(b) The equilibrium positions are found by setting $F(x) = 0$:

$$4U_0 \left(\frac{x}{a}\right) \left[\left(\frac{x}{a}\right)^2 - 1\right] = 0. \tag{3.195}$$

This gives us three equilibrium positions: $x = 0$, $x = a$, and $x = -a$.

(c) To determine the stability of each equilibrium position, we compute the second derivative of the potential energy:

$$U''(x) = \frac{d^2U}{dx^2} = \frac{4U_0}{a^2} \left[3\left(\frac{x}{a}\right)^2 - 1\right]. \tag{3.196}$$

At $x = 0$, we have $U''(0) = -\frac{4U_0}{a^2} < 0$, so the equilibrium at $x = 0$ is unstable. At $x = a$ and $x = -a$, we have $U''(a) = U''(-a) = \frac{8U_0}{a^2} > 0$, so the equilibria at these points are stable.

### 3.11.22 Mechanical energy as the first integral of Newton's second axiom

Show that the mechanical energy $E = K + U$ can be derived by integrating Newton's second law for a particle moving in one dimension under the influence of a conservative force $F(x) = -dU/dx$.

**Solution:** For a particle of mass $m$ moving in one dimension, Newton's second axiom:

$$F(x) = m\frac{dv}{dt}, \tag{3.197}$$

where $v = dx/dt$ is the velocity of the particle. Using the chain rule, we can express the acceleration as

$$\frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx}. \tag{3.198}$$

Substituting this into Newton's second law, we have

$$F(x) = mv\frac{dv}{dx}. \tag{3.199}$$

Rearranging, we obtain

$$F(x)dx = mvdv. \tag{3.200}$$

Integrating both sides, we get

$$\int F(x)dx = \int mvdv. \tag{3.201}$$

The left-hand side can be expressed in terms of the potential energy $U(x)$, since

$$F(x) = -\frac{dU}{dx} \implies \int F(x)dx = -U(x) + C, \tag{3.202}$$

where $C$ is an integration constant. The right-hand side evaluates to

$$\int mvdv = \frac{1}{2}mv^2 + C'. \tag{3.203}$$

Combining these results, we have

$$-U(x) + C = \frac{1}{2}mv^2 + C'. \tag{3.204}$$

Rearranging, we find

$$\frac{1}{2}mv^2 + U(x) = C - C'E, \tag{3.205}$$

where $E$ is the total mechanical energy of the particle, which is conserved in the absence of non-conservative forces. Thus, we have derived the expression for the mechanical energy as the first integral of Newton's second law for a particle moving under a conservative force.

### 3.11.23 Using Total Energy as the Equation of Motion for a Particle Under a Constant Force

Starting from $E = K + U$, derive $x(t)$ for a particle under a constant force $F$ and show that it is equivalent to the solution obtained from Newton's second law.

**Solution:** The total energy of the particle is given by

$$E = \frac{1}{2}mv^2 + U(x), \tag{3.206}$$

where $m$ is the mass of the particle, $v$ is its velocity, and $U(x)$ is the potential energy. For a constant force $F$,

$$F = -\frac{dU}{dx} \implies dU = -F\,dx \implies U(x) = -Fx + C, \tag{3.207}$$

where $C$ is a constant. Since the potential energy is defined up to an arbitrary constant, we can set $C = 0$ without loss of generality. The total energy becomes

$$E = \frac{1}{2}mv^2 - Fx. \tag{3.208}$$

Solving for the velocity, we obtain

$$\frac{1}{2}mv^2 = E + Fx \implies v = \frac{dx}{dt} = \pm\sqrt{\frac{2}{m}(E + Fx)}. \tag{3.209}$$

The sign depends on the direction of motion. Let us first consider a particle with a positive initial velocity, $v_0 > 0$, and assume that its velocity remains positive over the time interval considered. We then have

$$\frac{dx}{dt} = \sqrt{\frac{2}{m}(E + Fx)}. \tag{3.210}$$

Separating the variables gives

$$\frac{dx}{\sqrt{E + Fx}} = \sqrt{\frac{2}{m}}\, dt. \tag{3.211}$$

Integrating both sides from the initial position $x_0$ at time $t_0$ to the position $x(t)$ at time $t$, we obtain

$$\int_{x_0}^{x(t)} \frac{dx}{\sqrt{E + Fx}} = \sqrt{\frac{2}{m}} \int_{t_0}^{t} dt. \tag{3.212}$$

Assuming $F \neq 0$, and using the substitution $u = E + Fx$, we have $du = F\, dx$, and the integral on the left-hand side is

$$\frac{1}{F} \int u^{-1/2} du = \frac{1}{F} \cdot 2u^{1/2} = \frac{2}{F}\sqrt{u} = \frac{2}{F}\sqrt{E + Fx}. \tag{3.213}$$

Therefore,

$$\frac{2}{F}\left[\sqrt{E + Fx(t)} - \sqrt{E + Fx_0}\right] = \sqrt{\frac{2}{m}}(t - t_0). \tag{3.214}$$

Rearranging, we find

$$\sqrt{E + Fx(t)} = \sqrt{E + Fx_0} + \frac{F}{\sqrt{2m}}(t - t_0). \tag{3.215}$$

At the initial position $x_0$, the total energy is

$$E = \frac{1}{2}mv_0^2 - Fx_0, \tag{3.216}$$

which implies

$$\sqrt{E + Fx_0} = \sqrt{\frac{m}{2}}\, v_0, \tag{3.217}$$

where we have used $v_0 > 0$. Substituting this into Eq. (**??**) expression gives

$$\sqrt{E + Fx(t)} = \sqrt{\frac{m}{2}}\, v_0 + \frac{F}{\sqrt{2m}}(t - t_0). \tag{3.218}$$

Squaring both sides, we obtain

$$E + Fx(t) = \frac{m}{2}v_0^2 + Fv_0(t - t_0) + \frac{F^2}{2m}(t - t_0)^2. \tag{3.219}$$

Using the initial energy $E = \frac{1}{2}mv_0^2 - Fx_0$, we can simplify this expression to

$$Fx(t) = Fx_0 + Fv_0(t - t_0) + \frac{F^2}{2m}(t - t_0)^2. \tag{3.220}$$

Finally, dividing both sides by $F$, we obtain

$$\boxed{x(t) = x_0 + v_0(t - t_0) + \frac{F}{2m}(t - t_0)^2}. \tag{3.221}$$

This is precisely the familiar equation of motion for a particle under constant acceleration,

$$x(t) = x_0 + v_0(t - t_0) + \frac{1}{2}a(t - t_0)^2, \tag{3.222}$$

where $a = F/m$, according to Newton's second law.

The same result is obtained for a negative initial velocity by choosing the negative square-root branch. If the particle changes direction, the appropriate branch must be changed at the turning point. For $F = 0$, the energy equation gives constant speed, and the initial direction determines the uniform motion $x(t) = x_0 + v_0(t - t_0)$.

**Remark:** The conservation of mechanical energy determines the magnitude of the velocity but not its direction.

1. A child of mass $m_c$ is sitting in a boat of mass $m_b$, which is initially moving with velocity $\vec{v}_0$ on calm water. The child throws a package of mass $m_p$ with velocity $\vec{v}_p$ at an angle $\theta$ above the horizontal, measured relative to the water (i.e., an inertial frame). Assuming that no external horizontal forces act on the system and neglecting any water resistance, determine the velocity $\vec{v}_b$ of the boat (including both $x$ and $y$ components) immediately after the throw.

2. A particle of mass $m$ moves along the $x$-axis under the influence of a position-dependent force given by

$$F(x) = -\frac{kx}{(x^2 + a^2)^{3/2}},$$

where $k > 0$ and $a > 0$ are constants. Initially, the particle is at rest at $x = x_0 > 0$.

(a) Calculate the work done by the force as the particle moves from $x = x_0$ to a general position $x$.

(b) Using the work-energy theorem, find the velocity of the particle as a function of $x$.

(c) Determine the maximum speed of the particle.

3. A solid disk of mass $M$ and radius $R$ is free to rotate about its central axis (perpendicular to its plane) and is initially at rest. A small particle of mass $m$ is moving with speed $v_0$ in a direction tangential to the edge of the disk and lands at the rim of the disk, sticking to it (i.e., the particle's velocity is perpendicular to the radius vector connecting the disk's center to the point of contact). The collision is perfectly inelastic. The moment of inertia of a solid disk about its central axis is $I = \frac{1}{2}MR^2$.

   (a) Determine the angular velocity $\omega$ of the disk+particle system immediately after the particle sticks.

   (b) Determine the rotational kinetic energy $T_{\text{rot}}$ of the disk+particle system after the collision.

4. A particle of mass $m$ moves along the $x$-axis under the influence of a conservative force derived from the potential

$$U(x) = U_0 \left( \frac{x^4}{a^4} - 2\frac{x^2}{a^2} \right),$$

   where $U_0 > 0$ and $a$ are constants.

   (a) Determine the force acting on the particle as a function of $x$.

   (b) Find the positions of stable and unstable equilibrium.

   (c) If the particle is released from rest at $x = 0$, determine its speed when it reaches $x = a$.

   (d) Compute the work done by the force as the particle moves from $x = 0$ to $x = a$.

# Chapter 4

# Small Oscillations

: In this chapter, we will study how physical systems behave in the vicinity of their stable equilibrium points. Such behavior is often referred to as *small oscillations* because the system exhibits oscillatory motion about its equilibrium position, with a displacement that remains small.

We will first examine the simplest case of oscillatory motion, in which a restoring force acts on the system, pushing it back toward its equilibrium position. We will then consider the effects of a retarding force, which dissipates mechanical energy and causes the amplitude of oscillations to decrease over time. Finally, we will study driven oscillations, in which an external periodic force acts on the system, and investigate how the resulting motion depends on the frequency of the driving force, including the phenomenon of resonance. For simplicity, we will restrict our attention to one-dimensional systems throughout this chapter.

## 4.1 Simple Oscillations

We begin by considering the motion of a particle under a conservative force described by a potential energy function $U(x)$. We make the following three assumptions:

(i) $U(x)$ is sufficiently smooth to allow a Taylor expansion about the equilibrium point.

(ii) The potential has a local minimum at $x = x_0$, with $U'(x_0) = 0$ and $U''(x_0) > 0$, so that $x_0$ is a stable equilibrium point.

(iii) The displacement from equilibrium, $|x-x_0|$, remains "sufficiently small".

Assumption (i) allows us to expand the potential energy about the equilibrium point $x_0$ using a Taylor series:

$$U(x) = U(x_0) + U'(x_0)(x - x_0) + \frac{1}{2}U''(x_0)(x - x_0)^2 + \frac{1}{6}U'''(x_0)(x - x_0)^3 + \dots \tag{4.1}$$

According to assumption (ii), the first derivative vanishes at equilibrium, $U'(x_0) = 0$, while the second derivative is positive, $U''(x_0) > 0$. Furthermore, assumption (iii) allows us to neglect the third and higher-order terms compared to the quadratic term. This is what we actually mean when we say that the displacement is "sufficiently small" in assumption (iii). Since the reference level of potential energy can be chosen arbitrarily, we may also set $U(x_0) = 0$ without loss of generality. We therefore obtain the quadratic approximation

$$U(x) \approx \frac{1}{2}k(x - x_0)^2, \tag{4.2}$$

where $k \equiv U''(x_0) > 0$. It is worth emphasizing that the approximation for the potential energy follows directly from our assumptions and is not limited to specific systems such as springs or pendulums. A wide range of physical systems can be described by a quadratic potential near their stable equilibrium points, provided that the second derivative of the potential at equilibrium is positive. This allows us to study the small oscillations of different physical systems using the same mathematical framework.

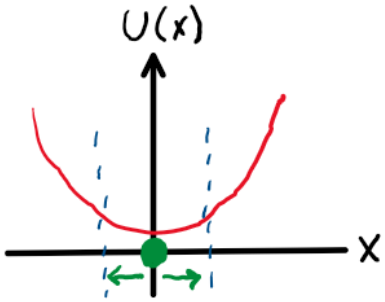



Figure 4.1: Small oscillations of a particle about a stable equilibrium point under potential energy $U(x)$. The particle oscillates about the equilibrium position $x_0$ with a small amplitude.

Using the quadratic approximation, we can derive the force acting on the particle:

$$F(x) = -\frac{dU}{dx} \approx -k(x - x_0). \tag{4.3}$$

This is known as *Hooke's law*. The force is proportional to the displacement from equilibrium and always points toward the equilibrium position $x_0$. For this reason, it is called a *restoring force*. The constant $k$ is called the *force constant* or *spring constant*, and it characterizes the stiffness of the system. A larger value of $k$ corresponds to a stiffer system, which requires more force to achieve the same displacement from equilibrium.

From the experimental point of view, the force constant can be determined by measuring the force required to displace the particle from its equilibrium position and calculating the slope of the force-displacement curve as shown in Fig. **??**. The force constant is a fundamental property of the system and depends on its physical characteristics, such as the material properties and geometry.

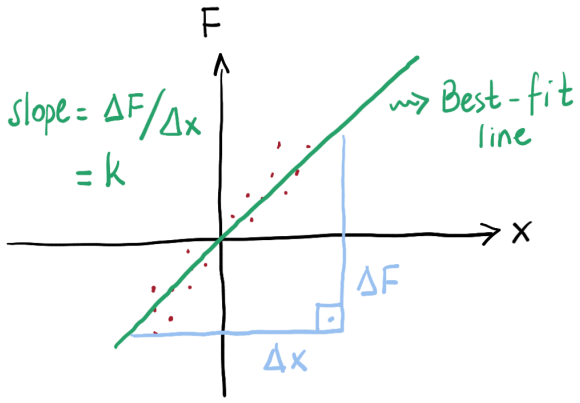



Figure 4.2: Empirical determination of the force constant $k$. The slope of the force-displacement curve gives the value of $k$. Note that the linear best-fit line is a model; the idea is there is always a sufficiently small region around the equilibrium point where the force is linear in the displacement. The data points may scatter according to different models as the displacement increaases.

### 4.1.1 Equation of Motion for Simple Oscillations

The dynamics of the particle are governed by Newton's second law:

$$F_{\text{net}} = m\ddot{x}, \tag{4.4}$$

where $m$ is the mass of the particle. Assuming that the restoring force is the only net force acting on the particle, and choosing the origin at the equilibrium position ($x_0 = 0$), we obtain

$$m\ddot{x} = -kx. \tag{4.5}$$

Rearranging this equation gives

$$\ddot{x} + \omega_0^2 x = 0, \tag{4.6}$$

where we have introduced the positive parameter

$$\omega_0 \equiv \sqrt{\frac{k}{m}}. \tag{4.7}$$

This is a second-order linear homogeneous differential equation describing *simple harmonic motion*. The physical meaning of $\omega_0$ will become clear when we solve this equation and examine the resulting motion.

### 4.1.2 Solution to the Equation of Motion

We know that the particle oscillates back and forth about the equilibrium position in a periodic way. We saw in Chapter 1 that the exponential function $e^{\lambda t}$ exhibits a wide range of different types of behavior depending on how the parameter $\lambda$ is chosen as summarized in the following table:

| $\lambda$ | Behavior of $e^{\lambda t}$ |
|---|---|
| Real and positive ($\lambda > 0$) | Exponential growth |
| Real and negative ($\lambda < 0$) | Exponential decay |
| Purely imaginary ($\lambda = i$) | Sinusoidal oscillation |
| Complex, $\mathrm{Re}(\lambda) < 0$ | Oscillation with exponentially decaying amplitude |
| Complex, $\mathrm{Re}(\lambda) > 0$ | Oscillation with exponentially growing amplitude |

We know that the solution to the differential equation in Eq. **??** must be oscillatory according to our assumptions. Therefore, it is natural[1] to look for a solution of the form $x(t) = e^{\lambda t}$, where $\lambda$ will be determined by substituting this ansatz into the differential equation. This leads to the characteristic equation

$$\lambda^2 + \omega_0^2 = 0, \tag{4.8}$$

which has the two purely imaginary roots

$$\lambda = \pm i\omega_0. \tag{4.9}$$

This leads to two different solutions $x_{1,2}(t) = e^{\pm i\omega_0 t}$. Not surprisingly, the both solutions have a pure complex factor of the time leading to oscillatory behavior.

[1] I admit that this is a handwaving argument. For the complete description, we need to study Laplace transforms which is beyond the scope of this text.

A second-order linear homogeneous differential equation requires two initial conditions to determine a unique solution; therefore, the linear combination of two linearly independent solutions, containing two arbitrary constants, is sufficient to describe all possible solutions. Thus, the general solution can be written as

$$x(t) = Ae^{i\omega_0 t} + Be^{-i\omega_0 t}, \tag{4.10}$$

where $A$ and $B$ are arbitrary constants determined by the initial conditions. In general, these constants can be complex. However, since $x(t)$ represents the position of a particle, it must be a real-valued function. To express the solution in a real form, we use Euler's formula,

$$e^{i\theta} = \cos\theta + i\sin\theta, \tag{4.11}$$

which also implies

$$e^{-i\theta} = \cos\theta - i\sin\theta. \tag{4.12}$$

Substituting these expressions into the general solution gives

$$\begin{aligned} x(t) &= A\left[\cos(\omega_0 t) + i\sin(\omega_0 t)\right] + B\left[\cos(\omega_0 t) - i\sin(\omega_0 t)\right] \\ &= (A+B)\cos(\omega_0 t) + i(A-B)\sin(\omega_0 t). \end{aligned} \tag{4.13}$$

At first glance, the coefficients $A + B$ and $i(A - B)$ may appear to be complex. However, we can show that they must be real by considering the initial conditions. Suppose that the particle has an initial position $x(0) = x_0$ and an initial velocity $\dot{x}(0) = v_0$. Evaluating the position at $t = 0$, we obtain

$$x(0) = A + B = x_0. \tag{4.14}$$

Differentiating the solution with respect to time gives

$$\dot{x}(t) = -\omega_0(A+B)\sin(\omega_0 t) + i\omega_0(A-B)\cos(\omega_0 t). \tag{4.15}$$

Evaluating the velocity at $t = 0$, we find

$$\dot{x}(0) = i\omega_0(A-B) = v_0, \tag{4.16}$$

and therefore

$$A + B = x_0, \qquad i(A-B) = \frac{v_0}{\omega_0}. \tag{4.17}$$

Since both the initial position $x_0$ and the initial velocity $v_0$ are real physical quantities, the combinations $A + B$ and $i(A - B)$ must also be real. Substituting these expressions back into the general solution, we finally obtain

$$x(t) = x_0\cos(\omega_0 t) + \frac{v_0}{\omega_0}\sin(\omega_0 t). \tag{4.18}$$

This expression describes the motion of the particle for any given initial position $x_0$ and initial velocity $v_0$. For convenience, we can introduce two real constants, $C = x_0$ and $D = v_0/\omega_0$, and write the general solution in the more compact form

$$x(t) = C\cos(\omega_0 t) + D\sin(\omega_0 t). \tag{4.19}$$

The constants $C$ and $D$ are determined by the initial conditions. This general solution can also be expressed in terms of the *amplitude* and *phase* of the oscillation. To see this, we start from Eq. **??** and use Euler's formula to rewrite the trigonometric functions in terms of complex exponentials:

$$\begin{aligned} x(t) &= \mathrm{Re}\left[Ce^{i\omega_0 t} - iDe^{i\omega_0 t}\right] \\ &= \mathrm{Re}\left[(C - iD)e^{i\omega_0 t}\right], \end{aligned} \tag{4.20}$$

where Re denotes the real part of a complex number. We now introduce the complex number

$$Z = C - iD, \tag{4.21}$$

which is called a *phasor* or *complex amplitude*. Recall that any nonzero complex number can be expressed in polar form as

$$Z = Re^{i\phi}, \tag{4.22}$$

where $R = |Z|$ is its magnitude and $\phi$ is its argument (the angle it makes with the positive real axis). Since $Z = C - iD$, its magnitude is

$$\boxed{R = \sqrt{C^2 + D^2},} \tag{4.23}$$

and its argument $\phi$ is determined by

$$\arctan\left(-\frac{D}{C}\right) = \phi. \tag{4.24}$$

Substituting the polar form of $Z$ into the expression for the displacement gives

$$\begin{aligned} x(t) &= \mathrm{Re}\left[Ze^{i\omega_0 t}\right] \\ &= \mathrm{Re}\left[Re^{i\phi}e^{i\omega_0 t}\right] \\ &= \mathrm{Re}\left[Re^{i(\omega_0 t+\phi)}\right]. \end{aligned} \tag{4.25}$$

Finally, using Euler's formula, we obtain

$$\boxed{x(t) = R\cos(\omega_0 t + \phi).} \tag{4.26}$$

This is called the *amplitude-phase form* of the general solution. The constant $R \geq 0$ is the *amplitude*, representing the maximum displacement of the particle from its equilibrium position, while $\phi$ is the *initial phase* (or *phase constant*), which determines the position of the particle within its oscillation cycle at $t = 0$.

The phasor representation also provides a geometric interpretation of the motion as shown in Fig. **??**. The complex quantity $Re^{i(\omega_0 t+\phi)}$ represents a vector of constant length $R$ rotating counterclockwise in the complex plane with angular velocity $\omega_0$. The projection of this rotating vector onto the real axis gives the physical displacement $x(t)$ of the particle.

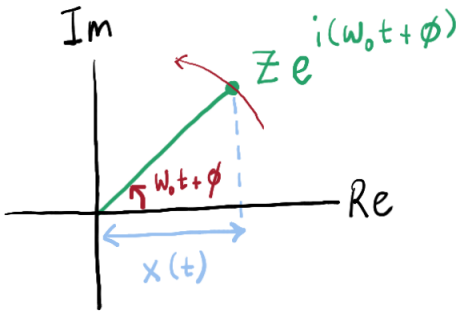



Figure 4.3: Phasor representation of simple harmonic motion. The complex vector $Re^{i(\omega_0 t+\phi)}$ rotates in the complex plane, and its projection onto the real axis gives the physical displacement $x(t)$.

Recalling that $C = x_0$ and $D = v_0/\omega_0$, the amplitude can be expressed in terms of the initial conditions as

$$\boxed{R = \sqrt{x_0^2 + \frac{v_0^2}{\omega_0^2}}.} \tag{4.27}$$

Similarly, the initial phase is determined by

$$\cos\phi = \frac{x_0}{R}, \qquad \sin\phi = -\frac{v_0}{\omega_0 R}. \tag{4.28}$$

Thus, both the amplitude and the initial phase are determined by the initial position and velocity. In the special case $R = 0$, the particle remains at its equilibrium position, $x(t) = 0$, and the phase $\phi$ is undefined.

### 4.1.3 Energy of Simple Oscillations

The mechanical energy of a particle undergoing simple oscillation

$$E = K + U = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2 \tag{4.29}$$

Substituting the general solution $x(t) = R\cos(\omega_0 t + \phi)$ into the expressions for kinetic and potential energies, we find

$$K(t) = \frac{1}{2}m\left[-R\omega_0 \sin(\omega_0 t + \phi)\right]^2 = \frac{1}{2}mR^2\omega_0^2 \sin^2(\omega_0 t + \phi), \tag{4.30}$$

$$U(t) = \frac{1}{2}k\left[R\cos(\omega_0 t + \phi)\right]^2 = \frac{1}{2}kR^2 \cos^2(\omega_0 t + \phi). \tag{4.31}$$

Using the relation $k = m\omega_0^2$, we can express the total energy as

$$E = \frac{1}{2}mR^2\omega_0^2\left[\sin^2(\omega_0 t + \phi) + \cos^2(\omega_0 t + \phi)\right] = \frac{1}{2}mR^2\omega_0^2. \tag{4.32}$$

By substituting $\omega_0^2 = k/m$, we can also express the total energy in terms of the force constant $k$ and the amplitude $R$:

$$E = \frac{1}{2}kR^2. \tag{4.33}$$

Hence, the mechanical energy of a simple oscillator is constant and proportional to the square of the amplitude. This is a characteristic feature of conservative systems, where energy is conserved and oscillates between kinetic and potential forms without any loss as shown in the Fig. **??**, with the maximum kinetic energy occurring when the particle passes through the equilibrium position ($x = 0$) and the maximum potential energy occurring at the turning points of the motion ($x = \pm R$).

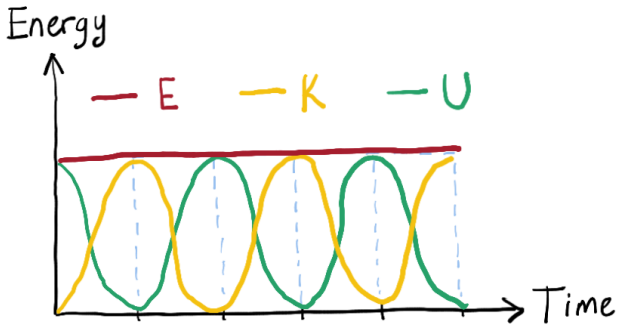



Figure 4.4: Oscillations of kinetic energy $K(t)$, potential energy $U(t)$, and total energy $E$ for a particle undergoing simple harmonic motion. The total energy remains constant, while the kinetic and potential energies oscillate out of phase with each other.

### 4.1.4 Physical Interpretation of the Solution

We have obtained four equivalent forms of the general solution to the equation of motion for simple harmonic oscillations:

$$x(t) = Ae^{i\omega_0 t} + Be^{-i\omega_0 t}, \tag{4.34}$$
$$x(t) = C\cos(\omega_0 t) + D\sin(\omega_0 t), \tag{4.35}$$
$$x(t) = R\cos(\omega_0 t + \phi), \tag{4.36}$$
$$x(t) = \mathrm{Re}\left[Re^{i(\omega_0 t+\phi)}\right]. \tag{4.37}$$

## 4.2 Damped Oscillations

If the oscillating system is not isolated, it may experience a retarding force that dissipates mechanical energy. We will consider a model in which the retarding force is proportional to the velocity of the particle, which is a common situation in many physical systems. The retarding force can be expressed as

$$F_\mathrm{r} = -b\dot{x}, \tag{4.38}$$

where $b$ is a positive constant called the *damping coefficient*. The negative sign indicates that the retarding force opposes the motion of the particle. The damping coefficient has units of force per unit velocity (N·s/m) and depends on the properties of the medium through which the particle moves, such as viscosity or air resistance. The total force acting on the particle is then the sum of the restoring force and the retarding force:

$$F_\mathrm{net} = F_\mathrm{restoring} + F_\mathrm{retarding} = -kx - b\dot{x}. \tag{4.39}$$

This leads to the equation of motion for damped oscillations:

$$m\ddot{x} + b\dot{x} + kx = 0. \tag{4.40}$$

Rearranging this equation gives

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0, \tag{4.41}$$

where we have introduced the positive parameters

$$\gamma\frac{b}{2m}, \qquad \omega_0\sqrt{\frac{k}{m}}. \tag{4.42}$$

We would expect that the damping force would cause the amplitude of oscillations to decrease over time, eventually bringing the particle to rest at

its equilibrium position. The parameter $\gamma$ characterizes the strength of the damping, with larger values corresponding to stronger damping effects. The parameter $\omega_0$ is the natural angular frequency of the undamped system, as we have seen in the previous section.

We know thatthe exponential function $e^{\lambda t}$ exhibits a wide range of different types of behavior depending on how the parameter $\lambda$ is chosen. For the damped oscillation equation, we can look for a solution of the form $x(t) = e^{\lambda t}$, as we did it in Section **??**, which leads to the characteristic equation

$$\lambda^2 + 2\gamma\lambda + \omega_0^2 = 0 \tag{4.43}$$

with the roots

$$\lambda_\pm = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}. \tag{4.44}$$

## 4.3 Phase Space Analysis

## 4.4 Driven Oscillations

## 4.5 Worked Examples

### 4.5.1 Work done by the spring force

### 4.5.2 Simple pendulum

# Chapter 5

# Central Force Motion

In this Chapter, hede hödö hödö...

## 5.1 Definition of Central Force

## 5.2 Direct Consequences of the Definitions

## 5.3 Worked Examples

### 5.3.1 Work and kinetic energy in a rotating system

Two identical small masses, each of mass $m$, are connected to a frictionless vertical rotation axis by two massless rods of adjustable length. The system rotates in a horizontal plane with an initial angular velocity $\omega_i$, and each mass is initially at a distance $r_i$ from the rotation axis. The rods are then shortened, pulling the masses radially inward until their distance from the axis becomes $r_f = r_i/\sqrt{2}$. Assume that no external torque acts about the rotation axis and that the radial velocities of the masses are zero both initially and finally.

(a) Calculate the change in the total kinetic energy of the system using conservation of angular momentum.

(b) Calculate the work done by the pulling forces directly from the definition of work, $W = \int \mathbf{F} \cdot d\mathbf{r}$, and verify that it agrees with the result obtained in part (a).

**Solution:**

(a) Since no external torque acts about the rotation axis, the total angular momentum of the system is conserved. Treating the two masses as point particles, the initial moment of inertia is

$$I_i = 2mr_i^2. \tag{5.1}$$

After the masses are pulled inward to $r_f = r_i/\sqrt{2}$, the final moment of inertia becomes

$$I_f = 2mr_f^2 = 2m\left(\frac{r_i}{\sqrt{2}}\right)^2 = mr_i^2 = \frac{I_i}{2}. \tag{5.2}$$

From conservation of angular momentum, we have

$$I_i\omega_i = I_f\omega_f \implies 2mr_i^2\omega_i = mr_i^2\omega_f. \tag{5.3}$$

Therefore, the final angular velocity is

$$\boxed{\omega_f = 2\omega_i}. \tag{5.4}$$

Since the radial velocities are zero initially and finally, the kinetic energy at these instants is purely rotational. The initial kinetic energy is

$$K_i = \frac{1}{2}I_i\omega_i^2 = mr_i^2\omega_i^2. \tag{5.5}$$

Similarly, the final kinetic energy is

$$K_f = \frac{1}{2}I_f\omega_f^2 = \frac{1}{2}(mr_i^2)(2\omega_i)^2 = 2mr_i^2\omega_i^2. \tag{5.6}$$

Thus, the change in kinetic energy is

$$\Delta K = K_f - K_i = 2mr_i^2\omega_i^2 - mr_i^2\omega_i^2, \tag{5.7}$$

which gives

$$\boxed{\Delta K = mr_i^2\omega_i^2}. \tag{5.8}$$

The kinetic energy of the system increases even though its angular momentum remains constant. This increase must come from the work performed by the forces pulling the masses inward.

(b) To calculate this work directly, we start from the definition of work,

$$W = \int \mathbf{F} \cdot d\mathbf{r}. \tag{5.9}$$

Since the pulling force acts in the radial direction, only the radial component of displacement contributes to the work. For each mass, we can therefore write

$$W_1 = \int_{r_i}^{r_f} F_r \, dr, \tag{5.10}$$

where $F_r$ is the radial component of the pulling force, with the outward radial direction chosen as positive.

From Newton's second law in polar coordinates, the radial acceleration of a particle is

$$a_r = \ddot{r} - r\omega^2. \tag{5.11}$$

Therefore, the radial force acting on each mass is

$$F_r = m(\ddot{r} - r\omega^2). \tag{5.12}$$

Since there are two identical masses, the total work done by the pulling forces is

$$W = 2\int_{r_i}^{r_f} F_r \, dr = 2m\int_{r_i}^{r_f} \ddot{r} \, dr - 2m\int_{r_i}^{r_f} r\omega^2 \, dr. \tag{5.13}$$

Let us first evaluate the integral containing the radial acceleration. Using

$$\ddot{r} = \frac{d\dot{r}}{dt}, \qquad dr = \dot{r} \, dt, \tag{5.14}$$

we obtain

$$\int_{r_i}^{r_f} \ddot{r} \, dr = \int_{\dot{r}_i}^{\dot{r}_f} \dot{r} \, d\dot{r} = \frac{1}{2}\left(\dot{r}_f^2 - \dot{r}_i^2\right). \tag{5.15}$$

Since the radial velocities are zero both initially and finally, this integral vanishes. Hence, the total work reduces to

$$W = -2m\int_{r_i}^{r_f} r\omega^2 \, dr. \tag{5.16}$$

However, the angular velocity is not constant during the inward motion. To express it as a function of the radial position $r$, we use conservation of angular momentum:

$$I(r)\omega(r) = I_i\omega_i. \tag{5.17}$$

Since $I(r) = 2mr^2$, we have

$$2mr^2\omega(r) = 2mr_i^2\omega_i, \tag{5.18}$$

which gives

$$\omega(r) = \frac{r_i^2}{r^2}\omega_i. \tag{5.19}$$

Substituting this expression into the work integral, we obtain

$$W = -2m \int_{r_i}^{r_f} r \left( \frac{r_i^2}{r^2} \omega_i \right)^2 dr \tag{5.20}$$

$$= -2m r_i^4 \omega_i^2 \int_{r_i}^{r_f} \frac{dr}{r^3}. \tag{5.21}$$

Evaluating the integral,

$$W = -2m r_i^4 \omega_i^2 \left[ -\frac{1}{2r^2} \right]_{r_i}^{r_f} \tag{5.22}$$

$$= m r_i^4 \omega_i^2 \left( \frac{1}{r_f^2} - \frac{1}{r_i^2} \right). \tag{5.23}$$

Finally, substituting $r_f = r_i/\sqrt{2}$, we find

$$W = m r_i^4 \omega_i^2 \left( \frac{2}{r_i^2} - \frac{1}{r_i^2} \right) \tag{5.24}$$

$$= m r_i^2 \omega_i^2. \tag{5.25}$$

Therefore,

$$\boxed{W = m r_i^2 \omega_i^2}. \tag{5.26}$$

This result agrees exactly with the change in kinetic energy calculated in part (a):

$$\boxed{W = \Delta K = K_f - K_i}. \tag{5.27}$$

Thus, although the angular momentum of the system is conserved, its kinetic energy increases due to the positive work performed by the pulling forces. This demonstrates that conservation of angular momentum does not necessarily imply conservation of kinetic energy.

## 5.4 Central Motion Force - Homeworks

1. A point mass moves in the plane with polar coordinates $(r, \theta)$ relative to a fixed origin. Let

$$r(t) = r_0 + v_r\, t, \qquad \theta(t) = \Omega\, t + \tfrac{1}{2}\alpha t^2,$$

where $r_0 > 0$, $v_r$, $\Omega$, and $\alpha$ are constants.

(a) Write the general expressions for the velocity and acceleration in polar coordinates.

(b) Find the radial and angular components of the velocity and acceleration as functions of time.

(c) Compute the speed $v(t) = \|\mathbf{v}\|$ and its time derivative $\dot{v}(t)$. State the condition under which $\dot{v}$ is positive, negative, or zero.

(d) Find the time(s) $t^*$ (if any) at which the radial component of acceleration $a_r$ vanishes. State the condition on parameters for such a time to exist (with $t^* \geq 0$).

2. A particle of mass $m$ moves under a central force $f(r)$. Its trajectory in polar coordinates is observed to be

$$r(\theta) = \frac{a}{1 + b\cos(\theta)},$$

where $a > 0$ and $0 < b < 1$.

(a) Determine the central force $f(r)$ as a function of $r$.

(b) State whether this force is attractive or repulsive for $0 < r < a$.

3. (Hard) Perform an explicit calculation of time average of potential energy over a one complete period for a particle moving in an elliptical orbit under a central inverse square law force field. Express the result in terms of the force constant of the field and the semi-major axis of the ellipse. Perform a similar calculation for the time average of kinetic energy. Compare your results and verify that they are consistent with the virial theorem.

4. Two particles are moving under the influence of their mutual gravitational force describe circular orbits about one another with a period $\tau$. If they suddenly stopped in their orbits and allowed to gravitate toward each other, show that they would collide after a time interval $\tau/4\sqrt{2}$.